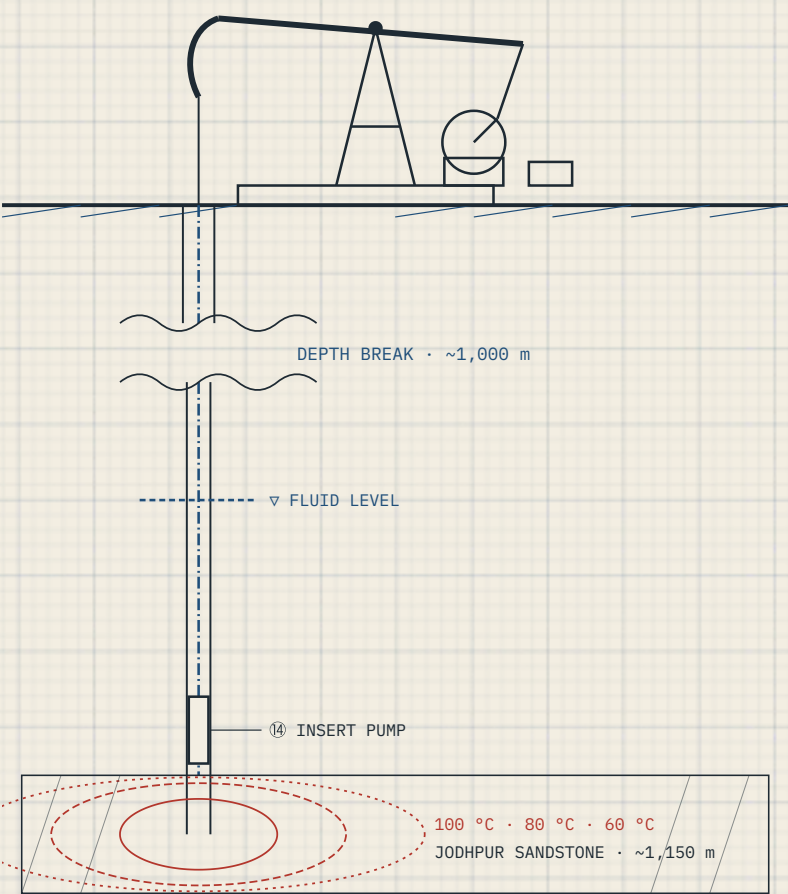


REV ΔB · BUILD SPEC

SIH 2026 · PS SIH26120 · OIL INDIA LIMITED · SMART AUTOMATION

MANTLE

WELL-TO-SURFACE DIGITAL TWIN FOR CSS + SRP
OPTIMIZATION OF BAGHEWALA HEAVY-OIL WELLS: PRODUCT,
TECHNICAL & HACKATHON PLAN



DOCUMENT	PRODUCT, TECHNICAL & HACKATHON PLAN
PROJECT	MANTLE · BAGHEWALA TWIN
PROBLEM STMT.	SIH26120 · OIL INDIA LTD.
DATE · REV.	29 SEP 2026 · REV B
SCALE	NTS · DEPTH PIECEWISE
STATUS	APPROVED SCOPE · PRE-BUILD

MANTLE: PRODUCT, TECHNICAL & HACKATHON PLAN

Team project: Mantle · **Event:** Smart India Hackathon 2026 · **Problem statement:** SIH26120 (Oil India Limited)

Document status: Revision B, 29 Sep 2026. This is the single source of truth for building, testing, demoing and pitching Mantle. Every section is written to be handed off on its own: to a teammate, to the 3D asset agent, or to a coding agent.

Rev B changes: verbatim problem statement up front · Next.js (not Vite) · uv for all Python · single-command Docker · HTML/CSS draft-first workflow · 6 screens with progressive disclosure · full screen, API and database specifications · Strategy Arena (static / heuristic / greedy / RL / ... vs Mantle) · test strategy for everything · hackathon plan · phased implementation plan · pitch & demo scripts · judges' Q&A · new features · diagrams throughout.

No code exists yet. This document describes what we will build.

HOW THIS DOCUMENT IS ORGANISED

PART	SECTIONS	READ IT IF YOU ARE...
I. The problem	0 Problem statement (verbatim) · 1 One-page version · 2 The problem, understood · 3 Editorial decisions	everyone, first
II. The product	4 Information architecture · 5 Design language · 6 Screen specifications · 7 Twin behaviour · 8 3D asset brief · 9 Metrics & labels · 10 Blueprint · 11 Transition · 12 Condition Deck · 13 Feature catalogue	design, frontend, 3D
III. The engine	14 Physics · 15 Estimation, ML & optimisation · 16 Strategy Arena · 17 Data strategy	physics, ML, backend
IV. The build	18 Architecture · 19 API reference · 20 Database schema · 21 Repository & workflow · 22 Docker · 23 Testing · 24 Budgets	engineering
V. Winning	25 Impact framework · 26 Hackathon plan · 27 Implementation plan · 28 Visual evidence map · 29 Scripts · 30 Judges' Q&A · 31 Competitive analysis · 32 Honesty & risks	whole team, pitch
Appendices	A Traceability · B Metric dictionary · C Glossary · D References · E Configuration · F Decision records	reference

Diagrams are Mermaid (they render on GitHub; the PDF uses pre-rendered SVGs) plus hand-drawn wireframes in `docs/diagrams/wireframes/`.

0. THE SIH PROBLEM STATEMENT (VERBATIM)

Reproduced verbatim from the SIH 2026 problem listing, as mirrored in a public copy of the listing (`problem_statement.json`, retrieved 29 Sep 2026). Only line breaks were added, at the original bullet markers (•). Spelling, spacing and punctuation are unchanged. **Before quoting it on slides, cross-check against the official SIH portal.**

FIELD	VALUE
Problem Statement ID	SIH26120
Title	Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.
Organization	Oil India Limited
Department	Oil India Limited
Category	Software
Theme	Smart Automation

Description

- Background Baghewala Field in Rajasthan produces heavy crude oil (17–19° API) from the Jodhpur Sandstone reservoir. The reservoir is characterized by High crude viscosity, High asphaltene content, Low reservoir pressure, Low reservoir temperature (46–48°C) and Poor oil mobility under primary recovery. Consequently, artificial lift and thermal enhanced oil recovery are critical for sustained production. At present, CSS cycle design and SRP operation are optimized separately using historical experience. As reservoir temperature declines after steam injection, crude viscosity increases, leading to reduced pump efficiency, higher energy consumption, rod floating issues, rod failures and lower oil recovery. There is a need for an integrated, data-driven system that continuously optimizes both CSS and artificial lift operations.
- Problem Description Current operations face the following challenges:
 - CSS parameters (steam volume, injection pressure, soak time and production cut-off) are largely based on historical practices.
 - SRP operating parameters (stroke length, SPM and VFD settings) are adjusted manually and reactively.
 - Heavy crude causes rod floating, impact loading, frequent pump unsetting, rod failures and increased maintenance.
 - Reservoir behaviour, wellbore conditions and SRP performance are not optimized together.
 - Lack of predictive analytics results in higher Steam-Oil Ratio (SOR), increased energy consumption and reduced production efficiency.
- Expected Outcome / Solution Develop an AI-enabled Well-to-Surface Digital Twin that integrates reservoir, wellbore and surface production systems to provide real-time monitoring, prediction and optimization. The solution should:
 - Optimize CSS cycle parameters.
 - Predict reservoir heating, cooling and production performance.
 - Continuously optimize SRP operation by adjusting stroke speed and SPM based on well conditions.
 - Detect rod floating and minimize impact loading.
 - Improve pump efficiency and equipment reliability.
 - Optimize steam and energy consumption while reducing operating cost.
- Expected Benefits
 - Increased oil production and recovery.
 - Reduced Steam-Oil Ratio (SOR).
 - Lower energy consumption per barrel.
 - Reduced rod failures and pump unsetting.
 - Improved equipment life and operational reliability.
- Data-driven and predictive decision making.
- Relevant Data Availability The field has sufficient historical and operational data, including:
 - Production history
 - CSS cycle records
 - Steam injection parameters
 - VFD and SRP operating data
 - Rod failure and pump unsetting history

- Well completion and reservoir data
- Fluid properties and pressure data

Dataset: none attached to the public listing. **YouTube:** none.

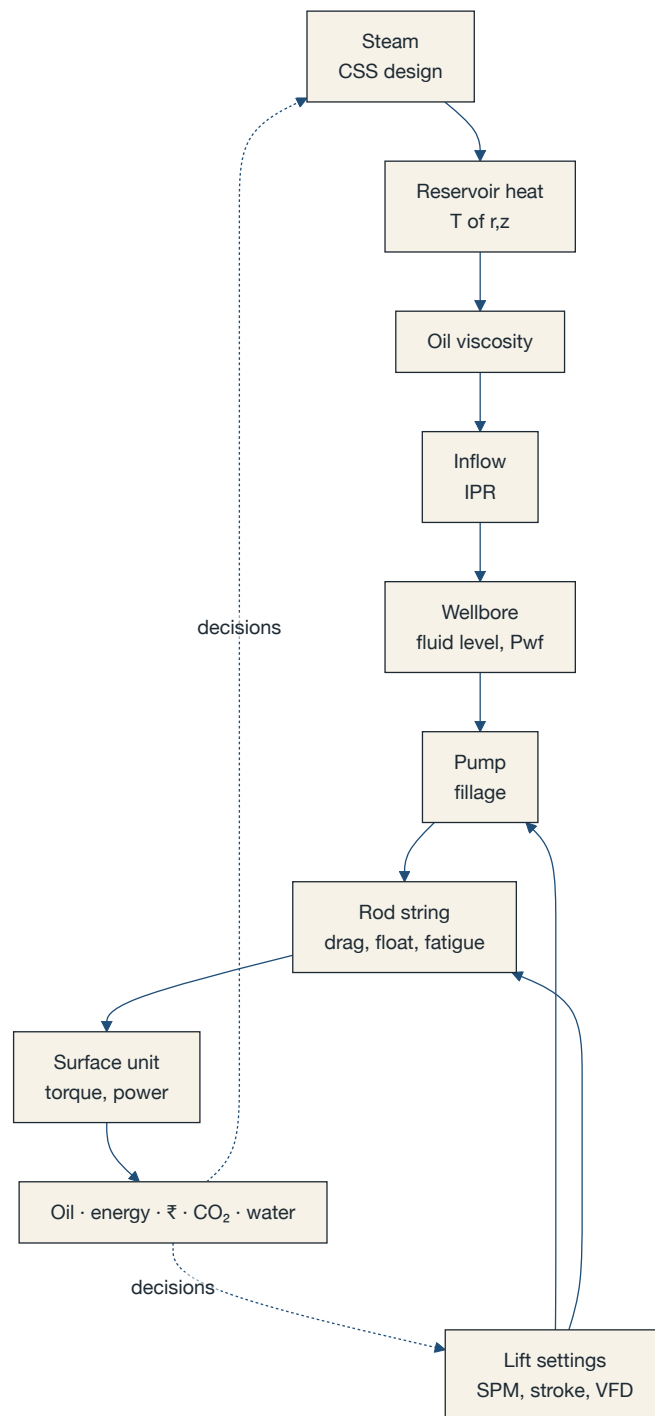
0.1 HOW EVERY LINE OF THE STATEMENT IS ANSWERED

Each line maps to a feature *and to a visual that proves it* (full matrix in Appendix A; visual proof in §28).

STATEMENT LINE	MANTLE ANSWER	WHERE A JUDGE SEES IT
CSS parameters "based on historical practices"	Joint optimiser over steam volume, pressure, soak and cut-off (§15.4); cut-off by the marginal value theorem (§14.9)	Well → Reservoir sheet; Arena race chart
SRP settings "adjusted manually and reactively"	Receding-horizon SPM/stroke/VFD schedule + Stroke Shaper (§15.4–15.5)	Blueprint Plate 3 (safe-SPM envelope)
Rod floating, impact loading, pump unsetting, rod failures	Physics margins, wave-equation dynamics, dynocard classifier, hazard models (§14.7, §15.3)	3D bridle slack + impact flash; Risk margins; Plate 2 card
"Not optimized together"	Coupling Dividend = joint minus sequential optimum, with Shapley decomposition (§15.6)	Twin chip; Arena decomposition
Higher SOR, energy, lower efficiency	Economics/energy/CO ₂ engine; SOR in CWE (§14.9, §25)	Impact screen; carbon receipt
Real-time monitoring, prediction, optimisation	Synthetic SCADA → EnKF twin → forecasts with P10–P90 → optimiser (§15)	Twin live; Timeline fan
Expected benefits (production, SOR, energy/bbl, failures, equipment life, data-driven decisions)	Measured policy-vs-policy in the Strategy Arena (§16); tracked in Ledger (§6.7)	Arena scoreboard; Impact; Ledger calibration
Seven data categories	Canonical schema maps 1:1 (§17.3, §20)	Proof → Real-data readiness; live CSV import

1. THE ONE-PAGE VERSION

What we are building. Mantle is a well-to-surface digital twin of a Baghewala heavy-oil well and its field. It keeps a live, uncertainty-aware estimate of the whole physical chain:



It forecasts where that chain is heading, searches thousands of **joint** CSS + SRP plans, rejects any plan that breaks a mechanical or thermal limit, and recommends the best remaining one. Every recommendation carries its reasoning, its uncertainty and its provenance, and gets a ledger entry that is later scored against what happened.

One-line pitch. A *flight simulator and autopilot for a heavy-oil well*. It shows the engineer the heat they can't see, the rod stress they can't feel, and the value they lose by optimising steam and pump separately.

Why it wins:

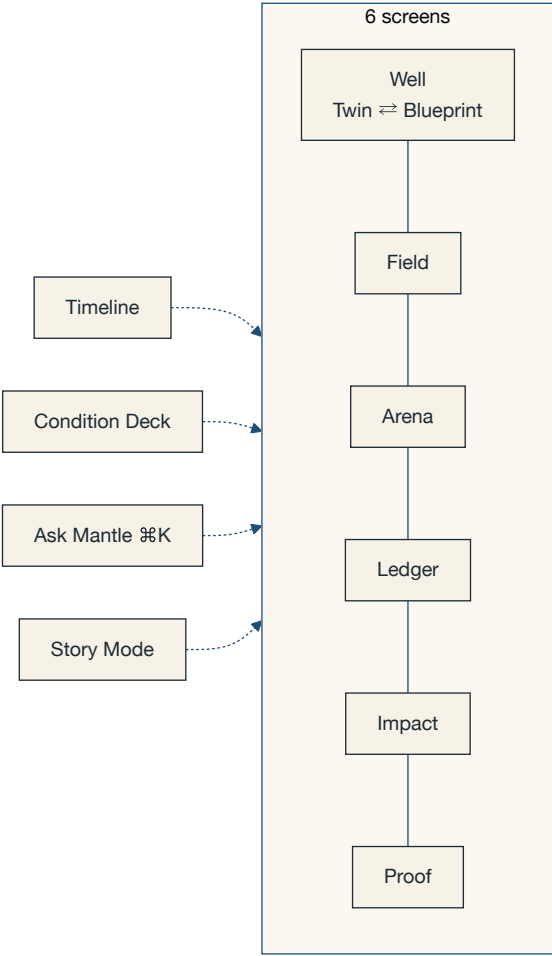
1. **You can see it.** A living 3D twin whose visuals *are* the physics: the thermal colour is the computed temperature field, particle speed is Darcy velocity, and rod colour is wave-equation stress. One key morphs it, in place, into a living engineering drawing.
2. **You can measure it.** A **Strategy Arena** races Mantle against static, heuristic, greedy, pump-off, sequential-optimal and reinforcement-learning strategies, plus an oracle upper bound, on identical simulated wells. The production-output graph and scoreboard are the evidence.

- 3. **It says what the PS says, as a number.** The **Coupling Dividend** is the value of optimising CSS and SRP *together* versus separately, decomposed into its parts.
- 4. **You can trust it.** An Ensemble Kalman Filter state estimator, P10–P90 everywhere, a deterministic Safety Gate, physics-vs-ML disagreement alarms, a hash-chained decision ledger and provenance tags on every number.
- 5. **It runs anywhere.** `docker compose up` brings up the whole stack, offline, on the demo laptop.

Stack. Next.js (App Router) + React 19 + TypeScript. React Three Fiber / Three.js for 3D, GSAP for choreography, D3/visx SVG charts. Python 3.12 managed with **uv** (a workspace of packages): FastAPI, NumPy/SciPy/Numba physics, LightGBM/PyTorch, pymoo, OR-Tools, Stable-Baselines3 for the RL contender, and our own EnKF. PostgreSQL + TimescaleDB, Redis, MinIO. Caddy as the single entry point. Everything in Docker Compose.

How we build. Every screen and component is first drafted as plain HTML/CSS, in several variants compared side by side, then promoted into the Next.js app. Tests exist for every layer, from physics golden values to pixel-level transition frames (§23).

Honesty stance. There is no public well-level Baghewala data. We use a physics-based synthetic field (*Baghewala-S*) calibrated to published anchors, keep the option to combine competitor or LLM-assisted data (§17), label every number by source, and never present simulated gains as field results.



2. THE PROBLEM, PROPERLY UNDERSTOOD

2.1 WHAT OIL INDIA ASKED FOR

See §0 for the verbatim statement and §0.1 for the line-by-line mapping.

2.2 THE PHYSICS CHAIN, IN PLAIN WORDS

Baghewala crude is **17–19° API** in the **Jodhpur Sandstone** at roughly **1,100–1,150 m**. Reservoir temperature is **46–48 °C**, reservoir pressure is low, asphaltene content is high, and viscosity is **roughly 10,000–13,000 cP at 50 °C** (OIL public figures). It barely moves on its own.

CSS injects steam for **about 14–21 days** (OIL), soaks the well shut-in, then produces with a **sucker rod pump**. Right after the soak the near-well zone is hot and the oil runs like thin syrup. Then the heat bleeds into the cap and base rock and leaves with the produced fluids. Viscosity climbs exponentially, and three things happen at once:

- 1. Inflow falls.** Mobility k/μ drops, so the well delivers less oil.
- 2. Downstroke drag rises.** The 1.1 km rod string has to fall through thickening oil. When viscous drag approaches the string's buoyant weight, the rods can't keep up with the carrier bar. That is **rod float**. Re-engagement produces **impact loading**, and that leads to fatigue and **rod failure**.
- 3. The pump runs out of balance.** Fillage drops (fluid pound), friction rises, and the insert pump can be pulled off its seat (**pump unsetting**).

The operator's levers are the **steam design** (how much, how hot and wet, how fast, how long to soak, when to stop producing and re-steam) and the **lift settings** (SPM, stroke length, VFD profile, plunger size). Today the two are tuned separately and reactively. **But the steam decision sets the viscosity path, and the viscosity path sets the safe pumping envelope.** They are one decision. That coupling is the whole problem, and Mantle is built around it.

2.3 THE ONE IDEA EVERYTHING HANGS ON

A heavy-oil CSS well is a thermal battery attached to a 1.1-km steel spring. Steam charges the battery. The pump spends it. The spring breaks if you spend it too fast while it's draining.

Every screen should make one part of that sentence visible.

3. EDITORIAL DECISIONS: WHAT WE KEPT, MERGED, CUT AND REFRAMED

The ChatGPT thread lists about 120 features, some overlapping and some weak. Our rule: **a feature survives only if it changes a decision, proves a claim, or makes the physics visible.** Decisions are final unless the team disagrees with a reason.

3.1 MERGED (SAME IDEA, SEVERAL NAMES)

THREAD ITEMS	MERGED INTO	WHY
Time Machine · Digital-Twin Replay · Telemetry Replay · Counterfactual Time Machine · 180-day horizon · "what the twin believed 23 days ago"	The Timeline (one global scrubber: past = replay of the recorded state, now = live, future = forecast fan; "Fork" creates a counterfactual branch at any point)	Five features were one control with three regions. One scrubber that works everywhere is stronger than five pages.
Why? button · SHAP · Root-Cause Explorer · Causal Graph · Decision provenance	Explain , three depths: <i>Why?</i> chip (top 3 drivers) → <i>Waterfall</i> (SHAP + physics sensitivities) → <i>Causal Trace</i> (the causal graph lit along the active path, with provenance)	Same question ("why this number?") at three zoom levels.
What-If Lab · Scenario Battle · A/B Strategy · Scenario Library · Scenario Cloning · Human-vs-AI · Training Mode	the Arena with modes: <i>Scenario builder</i> , <i>Battle</i> (N strategies race over 30/90/180 d), <i>Challenge</i> (a human plays the well against Mantle; doubles as the training simulator)	One engine, one results view, different framings.
Environmental Twin · Environmental Dashboard · Energy Intelligence · Water Intelligence · Three Impact Scorecards · Social Dashboard	Impact Ledger (§25)	One page, three columns, consistent method and provenance.
Daily Ops Brief · "Optimize My Shift" · "Which 5 wells today?" · Opportunity Map	Field Triage (ranked worklist + auto-generated morning brief)	Same output: a prioritized list with reasons.
Virtual sensors · Sensor confidence · Twin health · Data quality center · CSV auditor	Proof screen (twin health, data quality, provenance) with virtual sensors shown inline wherever the number appears	A number's trustworthiness belongs next to the number, plus one audit page.
Rod-float detector · Rod failure early warning · Pump unsettling predictor · Fluid pound · Casing integrity · Anomaly genome	Risk Stack : six <i>margins</i> on one physics basis, plus the dynocard classifier library	Unified presentation: every risk is "distance to a limit", not a bare probability.

3.2 REFRAMED (RIGHT IDEA, WRONG PHRASING)

AS WRITTEN	PROBLEM	REFRAMED AS
"Rod-float risk: 72%"	A probability with no physical basis is untestable and reads as made up.	Rod-float margin = 1 – (downstroke drag + dynamic term) / buoyant rod weight, shown as a bar to zero. The ML probability is secondary ("P(float within 7 d) = 0.31 ± 0.06").
"Maximize profit"	Invites the "dumb optimizer picks the dangerous plan" critique.	Maximize risk-adjusted net value : revenue – steam – power – water – expected failure cost – carbon cost (optional), under hard constraints.
"AI recommends SPM = 4.9"	A black-box point value.	A schedule (SPM, stroke and VFD profile over the next N days) with confidence, drivers and the binding constraint named.
"Autopilot mode" executes actions	Oil India will not let a student prototype touch field PLCs, and saying so costs credibility.	Modes are Observe · Advise · Autopilot (simulation only) . Autopilot exists in the Arena sandbox and in Story Mode to show closed-loop behavior. Field control is a stated future integration behind OIL's own control system.
"Estimated remaining recoverable oil"	Can't be estimated credibly without reservoir data.	Recovery trajectory over the simulated horizon ("the current strategy leaves ≈X bbl on the table over 180 d vs the plan").
"Reservoir–lift coupling index" (vague)	Undefined.	Coupling Dividend (defined in §15.6): V(joint optimum) – V(sequential optimum), in ₹ and bbl per cycle. It is measurable, and it's the PS thesis in one number.
"Social: fake empowerment metrics"	Correctly flagged in the thread.	Social = safety, heat-stress exposure, knowledge retention, skilling, energy security, water stewardship in the Thar. All grounded (§25.3).
Model metrics like "R² 0.999"	Competitors report near-perfect R² on their own simulator. Judges will poke at that.	Report metrics with the dataset stated ("vs Baghwala-S simulator, held-out wells, time-ordered split"), plus a calibration metric (coverage of P10–P90). Never claim field accuracy.
"Soil" as a condition	Surface soil barely affects the physics.	The Condition Deck separates Surface ground (visual, sand ingress, pad) from Formation (drives heat and flow). See §12.

3.3 CUT (OR PUSHED TO "VISION" SLIDES ONLY)

CUT	REASON
Separate Spring Boot / Java backend (PetroTwin)	No value; a second language and a second calculation path.
Login wall with demo passwords (several competitors)	Friction in front of judges. We use a persona switcher (Operator · Engineer · Manager · Viewer) without passwords, and document real SSO for deployment.
"Live SCADA" claims	We have none. Replaced by the Synthetic SCADA Engine with real protocols (MQTT/OPC-UA) (§18.8).

CUT	REASON
Full multiphase Beggs–Brill flowline hydraulics	Baghewala is dead heavy oil with little gas. We model single-phase liquid with water-cut emulsion viscosity and a gas/steam-flash flag. Beggs–Brill is a P3 plug-in.
Commercial reservoir-simulator coupling (CMG STARS)	Out of scope. We state it as a calibration path.
Twelve-plus top-level pages (Polaris, nishanth)	Cognitive overload. We use 6 screens + overlays (§4.2).
Voice assistant	Low value in a control room, and a risk in a noisy venue.

3.4 ADDED (NOT IN THE THREAD, AND WE THINK THEY MATTER)

ADDITION	WHY IT MATTERS
Stroke Shaper (VFD in-stroke speed profile)	A real lever for heavy-oil rod float that most teams ignore. It lifts the SPM ceiling without steam.
Solar-thermal steam as a steam-source condition	Rajasthan has some of India's best solar resource. Solar EOR runs commercially elsewhere (e.g. Oman). It changes CO ₂ /bbl and time-of-day injection scheduling, which gives us a genuine environmental optimization.
Asphaltene / wax deposition band along the tubing	The PS explicitly says "high asphaltene content" and no public repo models it. We use the wellbore temperature profile with an onset-temperature assumption (flagged ASSUMED).
Casing thermal-stress gauge ($\sigma = E \cdot \alpha \cdot \Delta T$)	Simple, correct and dramatic: a fully constrained casing heated 200 °C would see ~500 MPa. That explains why thermal connections and VIT matter.
Ensemble Kalman Filter state estimation	The honest way to "sync the twin to the real well". It gives P10–P90 and virtual sensors for free.
Field Steam Scheduler (CP-SAT/MILP)	One competitor planned it but never built it. Steam generators are a shared, limited resource, and deciding which well gets steam when is a field-scale economic decision.
Hash-chained decision ledger	Tamper-evident audit trail. Cheap, credible and industrial.
Provenance badge on every number (MEASURED · ESTIMATED · SIMULATED · ASSUMED · PUBLIC-SOURCE)	Our honesty policy made visible. Judges read it as maturity.
Safe Calibration Probe	Proposes the small, safe setpoint change that would teach the twin the most (active learning). It makes the loop self-improving on purpose.
Drawing export	The Blueprint sheet exports as a vector A3 PDF "well card". Engineers will pin it to a wall.
ISA-101 HMI and ISA-18.2 alarm practice	Grey at rest, color only for meaning; alarm priorities and flood suppression. Credibility with an operator audience.

3.5 REV B DECISIONS (29 SEP 2026)

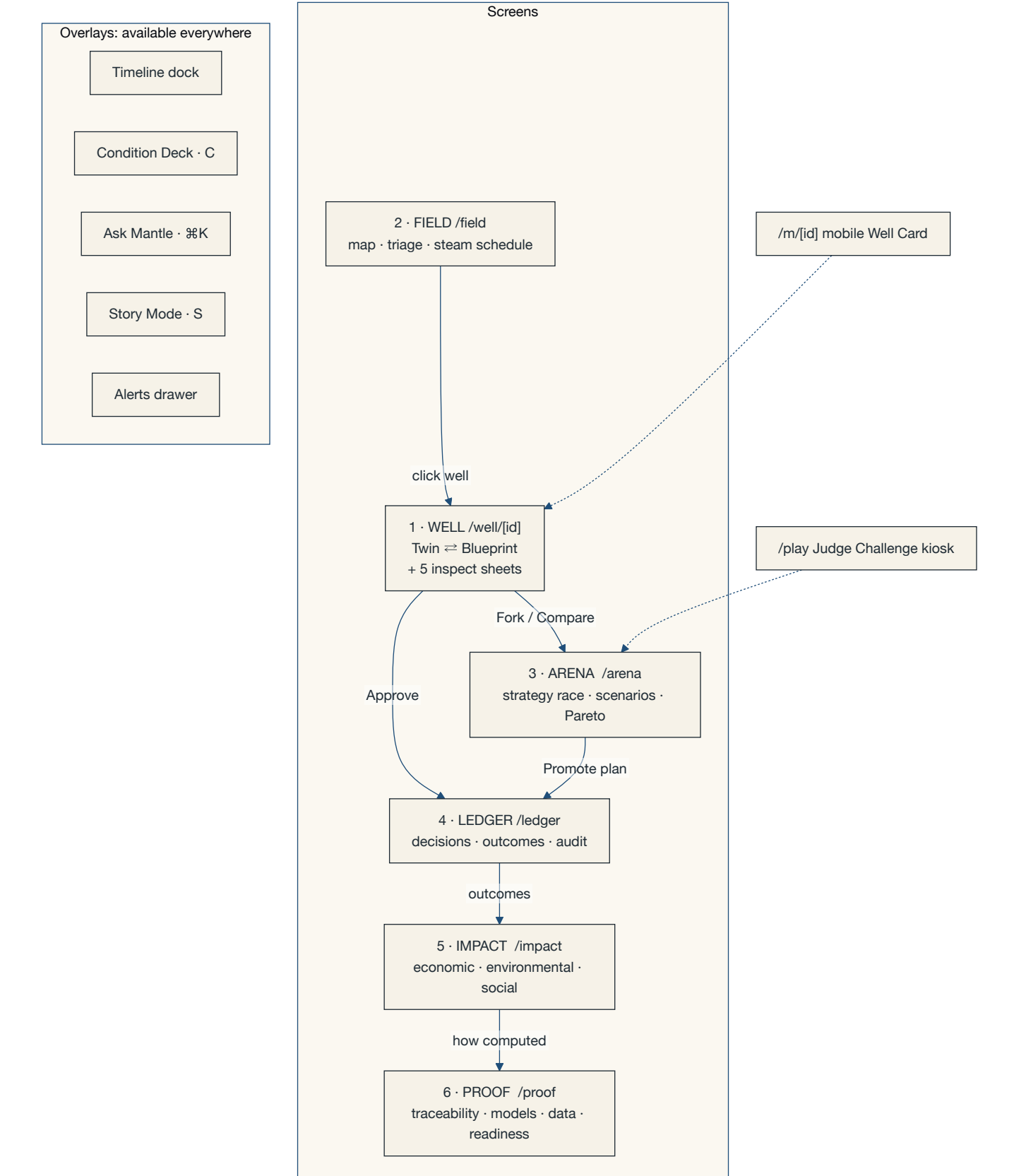
DECISION	CHOSEN	WHY	REJECTED ALTERNATIVE
Frontend framework	Next.js (App Router) + React 19 + R3F	Team choice. Nested layouts let one 3D canvas persist across routes, which the transitions need. Server components give fast first paint. output: "standalone" gives a small Docker image.	Vite SPA (Rev A)
Python tooling	uv everywhere (workspace, lockfile, Docker, CI)	Fast, reproducible, one tool for venv + deps + scripts	pip/poetry/conda
Run model	One command: docker compose up	Judges, mentors and new teammates get the same stack. Offline demo.	Manual multi-terminal setup (most competitors)
UI build process	HTML/CSS drafts first , several variants each, then promoted to React	Cheap exploration, side-by-side comparison, and a pixel reference for tests	Designing directly in React
Screen count	6 screens (Well, Field, Arena, Ledger, Impact, Proof) + overlays	Clean surfaces. Detail lives in hover cards, inspect sheets and dialogs.	7 destinations + Evidence (Rev A); 12+ pages (competitors)
Cycle & Lift pages	Folded into the Well screen as inspect sheets (click the reservoir → Reservoir sheet; click the pump → Lift sheet)	The object <i>is</i> the navigation	Separate pages
Evidence strategy	Strategy Arena : 9 strategies, identical wells, paired statistics, visual race	"What judges don't see doesn't exist"	Before/after numbers only
ML vs RL	Both exist . ML for estimation, surrogates, diagnosis and hazard. Shielded RL (PPO) is an Arena contender and an optional proposal generator. The primary controller is physics-based MPC.	RL is the obvious judge question. Showing it <i>competing</i> is stronger than claiming it or avoiding it.	Pure RL controller (unsafe, opaque); no RL (looks like we avoided it)
Data	Keep three options open : physics-synthetic (default), competitor-combined (licence-aware), LLM-assisted (text and scenarios, never raw physics)	Flexibility until the Phase-2 data decision	Committing now
On-screen text	No paragraphs on screens . Labels of three words or fewer, numbers with units; explanations only in hover or dialog	Visual cleanliness	Explanatory copy on the page

4. INFORMATION ARCHITECTURE

4.1 PERSONAS

PERSONA	QUESTION ON ARRIVAL	LANDS ON
Field operator	"Is anything going wrong now, and what do I do?"	Well (Twin), Alerts, mobile Well Card
Production engineer	"Why is this well behaving like this, and what's the best plan?"	Well (Blueprint + sheets), Arena
Asset manager	"Where are the money, the risk and the steam across the field?"	Field, Impact
Judge / mentor	"Is it real, is it novel, does it answer the PS?"	Story Mode, Arena, Proof

4.2 SIX SCREENS, FOUR OVERLAYS



#	SCREEN	ROUTE	SKIN	THE ONE QUESTION IT ANSWERS	HERO ELEMENT
1	Well	/well/[wellId] (? mode=twin blueprint)	Glass ⇌ Paper	What is this well doing, why, and what should we do?	The 3D twin / the living drawing
2	Field	/field	Paper	Which wells need attention, and where should steam go?	Field map + ranked triage
3	Arena	/arena	Paper	Does Mantle actually beat the alternatives?	Production-output race chart
4	Ledger	/ledger	Paper	What did we decide, and was it right?	Decision timeline + predicted-vs-realised

#	SCREEN	ROUTE	SKIN	THE ONE QUESTION IT ANSWERS	HERO ELEMENT
5	Impact	/impact	Paper	What does it do for money, environment, people?	Three columns of hero numbers
6	Proof	/proof	Paper	Why should anyone believe this?	Six evidence cards, each with a button

Also: /m/[wellId] (mobile Well Card), /play (Judge Challenge kiosk), /present (presenter remote). These are utility routes, not navigation.

4.3 PROGRESSIVE DISCLOSURE: THE RULES THAT KEEP SCREENS CLEAN

- 1. Default state shows only what answers the screen's one question. At most ~7 numbers per region, and no paragraphs anywhere on a screen.
- 2. Labels are three words or fewer. Units are small and dim. Numbers use Plex Mono with tabular figures.
- 3. Hover (300 ms) opens a Peek card: a sparkline, the top three drivers and one sentence. P pins it.
- 4. Click an object opens an Inspect sheet (right side, 420 px, tabbed). Click a decision opens a Focus dialog (modal).
- 5. Everything is reachable from %K, and every view has a deep link (§13.12).
- 6. Empty and loading states are drawings (blueprint lines drawing themselves), not "Loading..." text.
- 7. Colour carries meaning only (ISA-101): grey at rest; amber, orange or red only for alarms.

INTERACTION PATTERNS · QUIET BY DEFAULT, DETAIL ON DEMAND

1 HOVER CARD (PEEK) 300 ms

FLOAT MARGIN

18%

ROD-FLOAT MARGIN · 18%

hits 0 in ~6 d at current SPM

viscosity

SPM

stroke

+42%

+21%

+13%

click → Explain dialog · P = pin

2 INSPECT SHEET (CLICK A PART) RIGHT, 420 px

PUMP & RODS

Over Card Rod Hist

scene

PPRL 92 · MPRL 11 kN

fillage 78% · η 71%

Esc closes · O pins

3 FOCUS DIALOG (DECISIONS) MODAL

MODIFY RECOMMENDATION

SPM 4.8

kd 0.64

days 14

re-simulated: float margin 22% ✓

Approve Cancel

4 ASK MANTLE · %K

why is BGW-17 at risk?

Float margin is 18% and falling ~3%/d. 62% of it is viscosity (1,240 cP at 900 m); sandface T fell 11 °C in 14 d.

tools: get_state · explain_metric ✓ all 4 numbers verified

Show me Open plan

> fork at day 30

> inject: stuck T sensor

> go to Arena

5 CONDITION DECK (C) CARDS COMPOSE

FORMATION

Sandstone clean

Shaly

High-perm

Carbonate

CRUDE

API 19

API 17-18

API 14

WC 30%

CLIMATE

Summer noon

Oct dusk

Winter night

Sandstorm

STEAM

Gas OTSG

Solar hybrid

Recycle on

FAULT

Stuck T

Float event

Parted rod

Each card lists the parameters it sets, their provenance, and which outputs move.

Apply · 1.2 s

6 MOBILE WELL CARD (M/[ID]) OR AT WELLHEAD

BGW-17

18%

float margin

46 BOPD · 5.4 SPM

last decision #0184

Acknowledge

7 ALERTS DRAWER · ISA-18.2

P1 BGW-22 rod parted (card flatline) ack · shelf · open

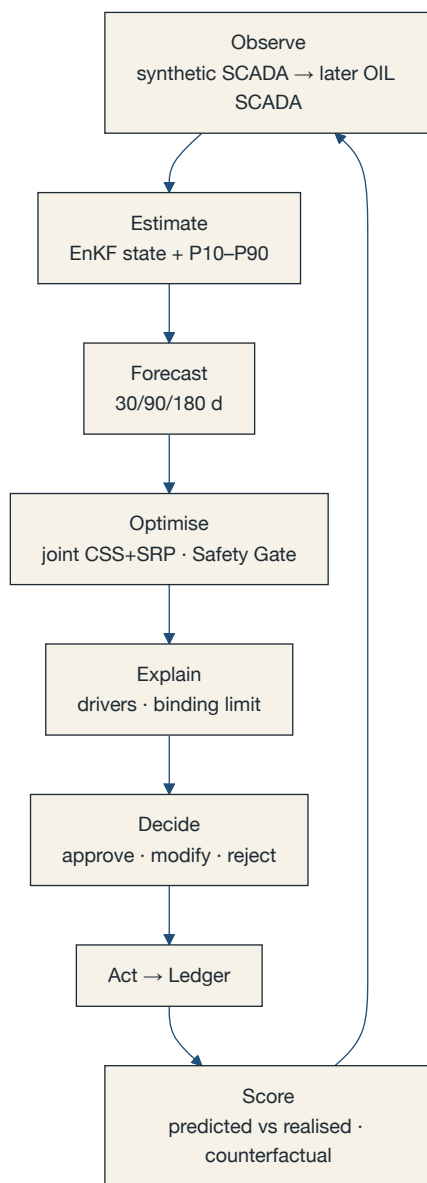
P2 BGW-17 float margin < 20% ack · shelf · open

P3 BGW-09 telemetry gap 14 h ack · shelf · open

P3 GEN-2 efficiency ~4% ack · shelf · open

flood suppression: 3 related alarms grouped

4.4 CORE LOOP



4.5 GLOBAL CONTEXT (SHARED BY EVERY SCREEN)

- **Well context:** BGW-17 · Cycle 6 · Day 41 · PRODUCTION. Clicking it opens the well switcher.
- **Time t :** one shared value. Scrubbing anywhere moves everything.
- **Scenario:** Baseline or a named branch. A coloured ribbon shows while you are off reality.
- **Conditions:** the active Condition Deck bundle, shown as a small chip.
- **Provenance mode:** a toggle that tints every number by source (MEAS/EST/SIM/ASSUM/PUB).
- **Persona:** Operator · Engineer · Manager · Viewer. This changes defaults (e.g. Operators land on Twin, Engineers on Blueprint), never permissions in the demo.

5. DESIGN LANGUAGE: GLASS AND PAPER

One design system with two material skins, and one token file (`packages/tokens/tokens.json`) feeding both the HTML drafts and the Next.js app. Components are identical in structure and behavior. Only the tokens change (`data-skin="glass" | "paper"`). The system should feel like **one instrument maker's product**: the Glass skin is the lit console, the Paper skin is the drawing on the engineer's desk.

5.1 SHARED FOUNDATIONS (BOTH SKINS)

- **Type family:** **IBM Plex**, open licence, with engineering heritage.
 - *Plex Sans* for UI text.
 - *Plex Sans Condensed*, UPPERCASE, +6% tracking, for labels, axis titles and drafting lettering.
 - *Plex Mono* with tabular figures for **every number**, so numbers never jitter while updating.
- **Type scale:** 11 / 12 / 14 / 16 / 20 / 28 / 40 px. Metric values are 20–28 px Mono. Units are 60% size and 60% opacity (`48.2 BOPD`).
- **Grid:** 8 px base. Paper pages also draw it on the background (§5.3).
- **Motion:** standard ease `cubic-bezier(.2,.7,.1,1)`, durations 120/240/420 ms. Numbers tween (count-up) at 240 ms. `prefers-reduced-motion` switches to cross-fades.
- **Semantic color is shared across both skins** (hue fixed, lightness tuned per skin):

TOKEN	MEANING	HUE
<code>thermal-*</code>	Temperature ramp	Perceptually uniform <i>inferno</i> -style ramp (black → purple → red → orange → pale yellow). Color-blind safe.
<code>steam</code>	Steam and injection	Cold cyan
<code>oil</code>	Crude and production	Deep amber / bitumen brown
<code>water</code>	Water and produced water	Blue
<code>ok · watch · high · critical</code>	ISA-18.2 priorities	Green · amber · orange · red. Color only carries meaning.
<code>twin</code>	Mantle plan / model	Cobalt
<code>baseline</code>	Historical practice	Warm grey
<code>forecast-band</code>	P10–P90	Twin hue at 18% alpha

- **Iconography:** 1.5 px stroke line icons (Lucide or Phosphor) and custom P&ID-style symbols for valves, pumps and gauges.
- **Provenance badges:** tiny caps tags after numbers, `MEAS · EST · SIM · ASSUM · PUB`. They are hidden by default, visible in Provenance mode, and always visible on the Proof screen.

5.2 GLASS SKIN (TWIN HOME ONLY)

The 3D scene is the backdrop, and the glass panels float above it like a head-up display.

- **Panel:** fill `rgba(18,22,30,0.38)`, `backdrop-filter: blur(22px) saturate(140%)`, 1 px border `rgba(255,255,255,0.14)`, top inner highlight `inset 0 1px 0 rgba(255,255,255,0.18)`, shadow `0 12px 40px rgba(0,0,0,0.35)`, radius 16 px. Heavier panels use 0.52 fill. Text must pass WCAG AA against the *worst* background frame, so the scene-color tint under panels is limited.
- **Text:** `#F4F1EA` primary, 72% secondary. Numbers are pure white.
- **Accent:** a thin 2 px underline glow in the semantic color. No colored fills on glass except alerts.
- **Frosted edge detail:** a subtle noise texture (2% opacity) in the panel stops banding in the blur.
- **Light from the scene:** panels take a faint tint from scene lighting via a CSS variable updated at 2 Hz from the renderer's average sky color. At dusk the glass warms, at night it cools. This is a small touch that makes the UI feel physically *in* the scene.
- **Performance rule:** at most 5 blurred surfaces on screen, and nothing blurred larger than 30% of the viewport (§24).

5.3 PAPER SKIN (BLUEPRINT AND EVERY OTHER PAGE)

A drafting sheet, not a web page with a beige background.

- **Paper:** `#F3EEE2` base with a very low-contrast fibre noise texture and a faint vignette.
- **Grid:** minor 8 px lines `rgba(31,78,121,0.07)`, major every 40 px `rgba(31,78,121,0.14)`, drawn as SVG pattern and pixel-snapped. It behaves like engineering graph paper.
- **Inks:** graphite `#1E2A33` (primary lines and text), drafting blue `#1F4E79` (construction lines, dimensions), red pencil `#B3362C` (limits, violations, revision clouds), amber `#C98A1B` (watch), green `#3F7D4E` (ok), steam `#2F8FA8`, oil `#6B4513`.
- **Line weights (ISO 128 spirit):** 0.35 mm (visible outlines) · 0.25 mm (details) · 0.18 mm (dimension and hatch lines) · dash-dot centre lines.

- **Panels are "plates"**: no shadows and no radius. 1 px graphite frame with a 4 px inner offset frame and a plate number in the corner ("PLATE 3 · DYNAMOMETER CARD"). A **title block** sits in the bottom-right of Blueprint and of every exported sheet.
- **Charts** use drafting conventions: tick marks inside the frame, condensed caps axis labels with units in brackets `LOAD [kN]` , hand-set annotations with leader lines, hatched regions for "forbidden" zones.
- **Revision clouds**: when a value changes because of a scenario or condition switch, the changed annotation gets a red scalloped cloud and a revision triangle `Δ2` for 3 s, then settles. This is how the paper shows change.

5.4 COMPONENTS THAT EXIST IN BOTH SKINS

COMPONENT	GLASS LOOK	PAPER LOOK
Metric chip	Frosted pill, big mono number, micro sparkline	Boxed value with unit, small hatch sparkline
Callout	Glass card with glowing leader line to a 3D anchor	Balloon (circled number) + leader + notes table
Tabs / segmented	Frosted pills	Index-card tabs with a folded-corner active state
Buttons	Glass, glow on focus	Outlined ink, "stamp" press effect
Timeline	Frosted dock, phase bands glowing	Ruled strip like a chart recorder, phase bands hatched
Alert	Glass with left color bar	Red-pencil margin note with ISA priority glyph
Recommendation card	Glass hero card	Engineering change notice (ECN) form layout

6. SCREEN SPECIFICATIONS

This is the build spec for every screen. Each screen lists **regions** (what's visible by default), **interactions** (hover, click, keys), **dialogs and sheets, data** (API and WebSocket bindings from §19), **states**, and **acceptance criteria**. Wireframes are not to scale. The HTML/CSS drafts (§21.3) turn them into real proposals.

6.1 THE SHELL (EVERY SCREEN)

REGION	CONTENTS	NOTES
Top bar	Logo · screen tabs 1–6 · well context · Twin/Blueprint toggle (Well only) · Conditions chip · ⌘K · Alerts bell · SIM badge · persona avatar	Glass on Well/Twin, index-card tabs on Paper
Timeline dock	Phase bands, now marker, forecast fan, event pins, play/speed, Fork	Present on Well and Field. Collapsed to a thin strip elsewhere.
Overlay layer	Peek cards, inspect sheets, dialogs, command palette, toasts	One overlay manager, one focus trap

Keyboard map

KEY	ACTION	KEY	ACTION
1–6	Go to screen	B	Twin ⇄ Blueprint
C	Condition Deck	⌘K / Ctrl K	Ask Mantle / command palette
Space	Play / pause time	← →	Step (stroke / hour / day, by zoom)
F	Fork at current t	L	Cycle lens (Twin)
X	X-ray (Twin)	P	Pin hovered Peek
S	Story Mode	?	Shortcut sheet
Esc	Close top overlay	.	Toggle provenance tint

6.2 SCREEN 1: WELL · TWIN MODE



Regions (default)

REGION	DEFAULT CONTENT
Scene	3D block-diagram well (§7, §8), current lens, ≤ 7 anchored callouts (§9)

REGION	DEFAULT CONTENT
Recommendation card	Verb-first action title ("Slow the downstroke"), the setting change, 3 deltas (oil, float risk, energy), confidence, Approve · Modify · Why?
Risk margins	6 bars: float, pound, unsettling, fatigue, casing thermal, deposition
Forecast	Oil rate now → +30 d, P10–P90 fan, red dashed "do nothing" ghost
Economy strip	₹/day · SOR · kWh/bbl · kg CO ₂ /bbl
Coupling Dividend + Thermal Battery	₹/cycle chip; battery icon = remaining useful reservoir heat (% and days)
Lens switcher	Physical · Thermal · Mech · Flow · Pressure · Risk

Interactions

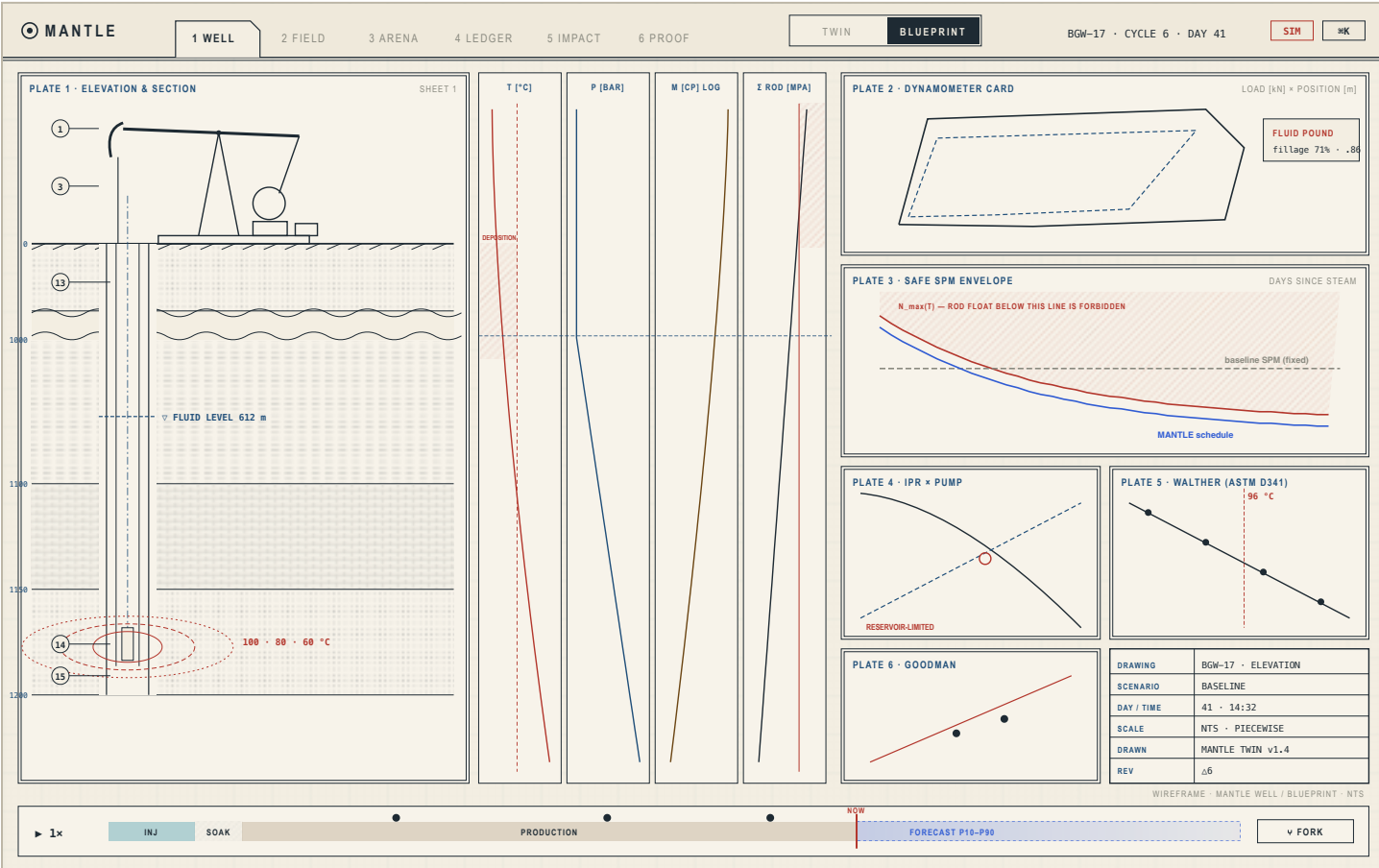
TARGET	HOVER (PEEK)	CLICK
Any 3D part	Outline glow + name + 2 numbers	Opens that part's Inspect sheet (§6.4)
Callout	Peek: 24 h sparkline + drivers	Opens the part's sheet on the relevant tab
Risk bar	Peek: margin trend, time-to-limit, top drivers	Explain dialog (Why? → waterfall → causal trace)
Forecast	Peek: horizon switch 7/30/90/180 d	Timeline jumps to forecast
Recommendation deltas	Peek: P10–P90 for each delta	—
Approve	—	Confirm toast; ledger entry; the card flips to "Applied (sim)"
Modify	—	Modify dialog : editable settings; live re-simulation of margins; approve/cancel
Why?	—	Explain dialog on the recommendation
Coupling Dividend	Peek: decomposition mini-waterfall	Goes to Arena, pre-filtered to this well
Thermal Battery	Peek: heat in, lost, used; days to cut-off	Reservoir sheet → Heat tab

Data: GET /wells/{id}/state?t=, GET /wells/{id}/forecast, GET /wells/{id}/thermal-field?t=, GET /recommendations?well_id=&status=open, WS /ws/wells/{id} (state @2 Hz, cards per stroke, alarms).

States: *loading*: the intro sequence (§13.12) plays until the GLB and state are ready. *stale* (no data > 30 s sim): a grey "STALE" badge on callouts. *sensor-excluded*: the affected callout shows **EST** and a dotted leader. *infeasible plan*: the Recommendation card shows "No safe plan: binding: rod float @ 900 m" in red pencil.

Acceptance: 60 fps on the demo laptop · ≤ 7 callouts · every number shows its provenance in Provenance mode · every interaction is reachable by keyboard.

6.3 SCREEN 1: WELL · BLUEPRINT MODE



Regions: Plate 1 elevation & section (balloons ①–⑮) · 4 depth tracks (T, P, μ , σ_{rod}) · Plate 2 dynamometer card · Plate 3 safe-SPM envelope · Plate 4 IPR $\times$ pump · Plate 5 Walther chart · Plate 6 Goodman · title block · (BOM and notes open in a sheet so the default view stays clean).

Interactions: hover a depth track $\rightarrow$ a crosshair across all tracks with a readout. Hover a balloon $\rightarrow$ part highlight + BOM row peek. Click a plate $\rightarrow$ it expands to full-sheet **Plate focus** (with export). Drag the timeline $\rightarrow$ revision clouds on changed annotations. **E** $\rightarrow$ export sheet (A3 vector PDF/SVG).

Data: GET /wells/{id}/profile?t= (depth tracks), GET /wells/{id}/cards?t=&n=, GET /wells/{id}/envelope, GET /wells/{id}/ipr?t=, GET /wells/{id}/bom.

Acceptance: vector-crisp at 200% zoom · the line-art pumping unit animates at SPM · plates match Twin numbers exactly at the same **t** (tested, §23).

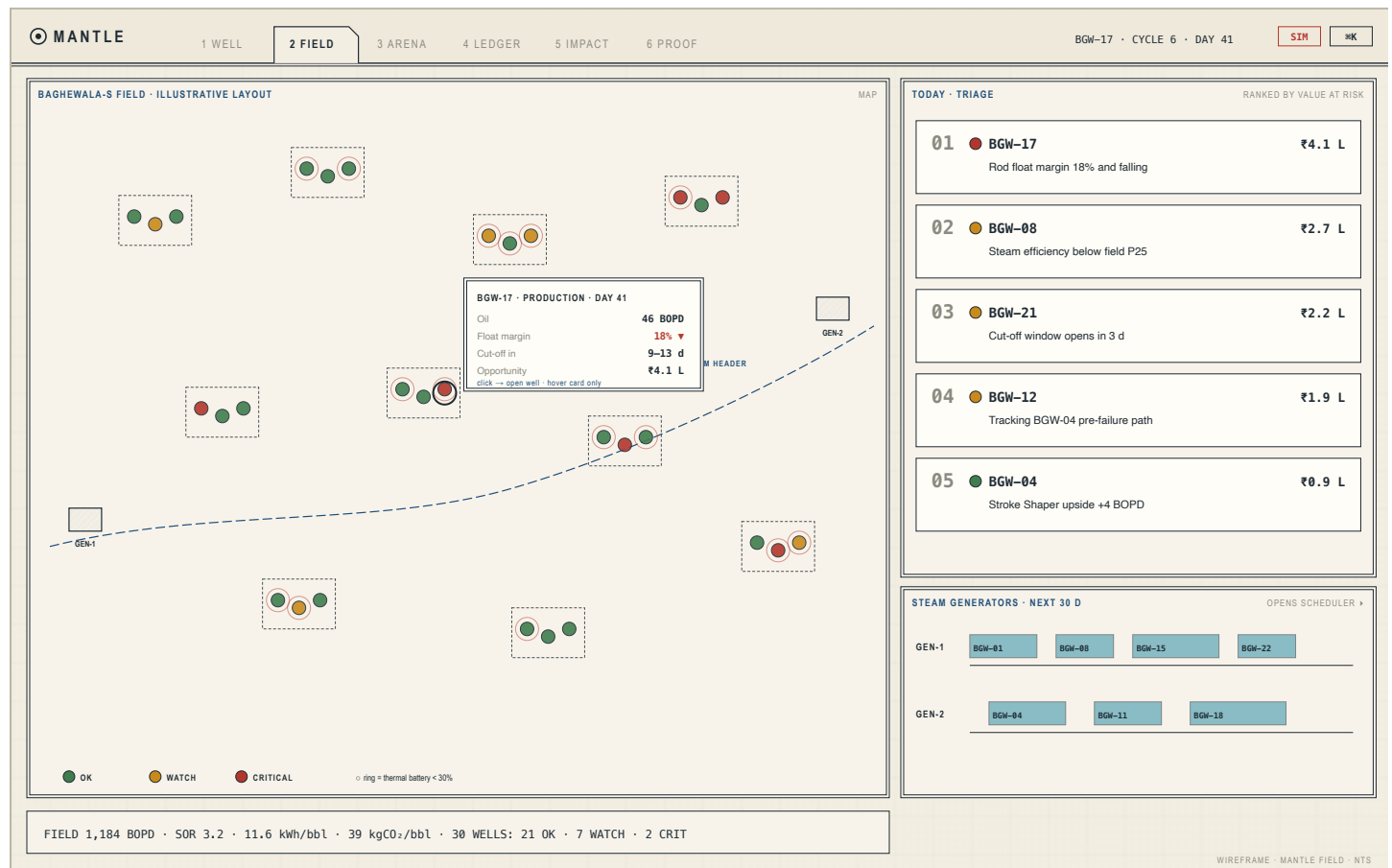
6.4 INSPECT SHEETS (INSIDE THE WELL SCREEN)

Clicking a part opens its sheet. **This is where the Rev A "Cycle" and "Lift" pages now live.**

SHEET (OPENED BY CLICKING...)	TABS	CONTENTS
Surface unit (beam, crank, motor, VFD)	Overview · Stroke Shaper · Energy	SPM, stroke, kd, torque % rating, polished-rod HP, motor kW; the Stroke Shaper polar speed diagram with an editable kd (what-if); energy per stroke
Wellhead & steam (wellhead, steam line)	Injection · Heat balance · Casing	Steam rate, pressure, quality (surface vs sandface); heat-balance Sankey; casing thermal-stress gauge; injection plan editor (volume $\leftrightarrow$ rate $\leftrightarrow$ duration locked)
Wellbore (tubing, casing, annulus)	Profiles · Deposition · Fluid level	Temperature/pressure/viscosity vs depth; asphaltene/wax band; fluid level and submergence history
Pump & rods (rods, pump) = Lift sheet	Card · Rod string · Risks · Schedule · Maintenance	Live surface/downhole cards + diagnosis; per-section stress/Goodman; the 6 margins with time-to-limit; the 14-day SPM/kd/stroke schedule; maintenance forecast + "delay it?" what-if
Reservoir (heated zone, rock) = Reservoir sheet	Heat · Cycle · Cut-off · History	Heated radius and mean T with bands; Thermal Battery; steam design editor; marginal steam curve (Δoil per +100 t); cut-off/re-steam window with cost-of-waiting; cycle history and degradation per cycle

Every sheet footer shows: Compare in Arena · Fork here · Ask about this.

6.5 SCREEN 2: FIELD



REGION	DEFAULT	HOVER	CLICK
Field map (cream cartographic)	Pads, wells as risk-coloured dots; thermal-battery ring; steam header and generators	Well Peek card (oil, float margin, cut-off in, opportunity)	Opens Well screen
Triage list	Top 5 wells ranked by value at risk: number, well, one-line reason, ₹	Peek: the evidence behind the reason	Opens Well at the relevant sheet
Generators strip	Next 30 d of injection slots per generator	Slot Peek: well, tonnes, window	Steam Scheduler dialog (Gantt, drag to override, re-solve, shadow price of an extra generator-day, what-if generator down)
KPI footer	Field BOPD · SOR · kWh/bbl · CO ₂ /bbl · health counts	—	—

More in dialogs: *Fleet table* (all 30 wells, sortable, T) · *Morning Brief* (generated 06:00 sim-time; print) · *Similarity* ("BGW-12 is tracking BGW-04's pre-failure path"; trajectory overlay) · *Field time-lapse* (the map "breathes" as wells heat and cool across a year).

Data: GET /field/summary, GET /field/triage, GET /field/map, GET /field/steam-schedule, POST /field/steam-schedule/solve, GET /wells/{id}/similar.

6.6 SCREEN 3: ARENA (THE EVIDENCE SCREEN)



6.7 SCREEN 4: LEDGER

MANTLE

1 WELL2 FIELD3 ARENA4 LEDGER5 IMPACT6 PROOF

BGW-17 · CYCLE 6 · DAY 41

SIM

DECISIONS

HASH-CHAINED

#0184

APPROVED-MOD

Slow downstroke 5.4→4.8

sha 9f3c_e21a ~ prev 71bd...

#0183

APPROVED

Advance CSS window →4 d

sha 9f3c_e21a ~ prev 71bd...

#0182

REJECTED

Hot-oil tubing 480-620 m

sha 9f3c_e21a ~ prev 71bd...

#0181

AUTO

Pause optimization (sensor)

sha 9f3c_e21a ~ prev 71bd...

#0180

PENDING

Plunger 2¼→2" at workover

sha 9f3c_e21a ~ prev 71bd...

#0179

APPROVED

Soak +12 h

sha 9f3c_e21a ~ prev 71bd...

ENGINEERING CHANGE NOTICE · #0184

ECN

WELL / TIME

BGW-17 · D41 14:32

ACTION

SPM 5.4 → 4.8, kd 0.58 → 0.64, 14 days

BINDING LIMIT

Rod-float margin ≥ 15% at 900 m

MODEL VERSIONS

physics 1.4.2 · enkf 0.9 · surrogate 2.1

APPROVED BY

Engineer (modified SPM 4.7 → 4.8)

REASON

Known well behaviour: delayed response

PREDICTED vs REALIZED · FLOAT MARGIN & OIL

PREDICTED ΔOIL

REALIZED ΔOIL (7 d)

FLOAT EVENTS AVOIDED

SCORE

-1.2 %

-0.8 %

3 (est.)

A · within band

Show provenance trace

Replay in Twin

CHAIN VERIFIED

CALIBRATION · PREDICTED VS REALIZED

P10-P90 coverage 83% (target 80%)

OVERRIDES · WHERE ENGINEERS DISAGREE

T < 60 °C, sandstone-B

70%

Water cut > 40%

45%

After hot-oiling

30%

Cycle ≥ 9

22%

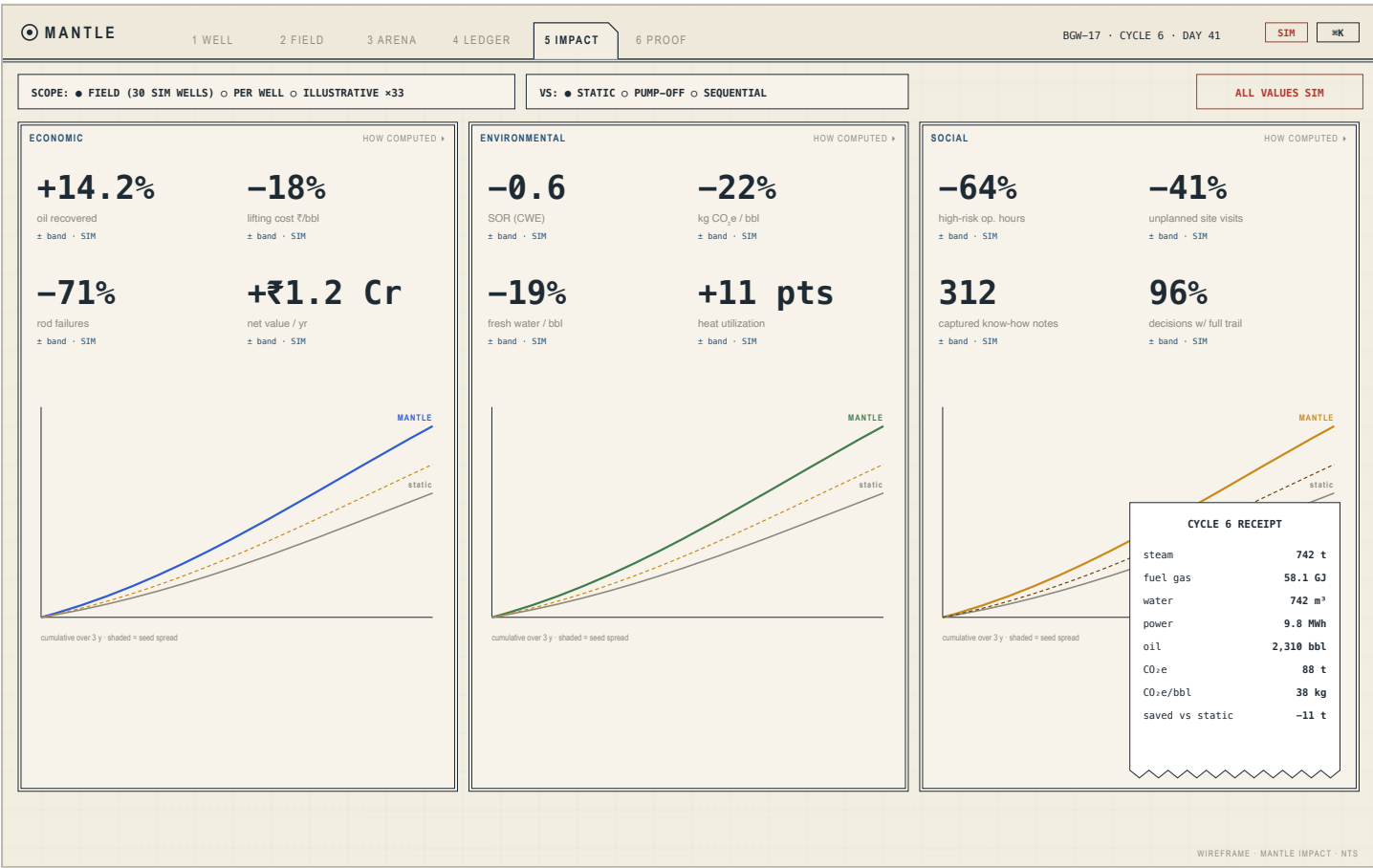
⚠ possible miscalibration: cold sandstone-B regime

WIREFRAME · MANTLE LEDGER · NTS

REGION	DEFAULT	CLICK
Decision timeline	Cards: id, action (≤ 6 words), status stamp, short hash	Selects the decision
ECN (engineering change notice)	Well/time, action, binding limit, model versions, approver, reason; predicted vs realised chart with counterfactual; score	Show provenance trace dialog; Replay in Twin
Calibration	Predicted vs realised scatter; P10–P90 coverage	Calibration dialog by model
Overrides	Regimes where engineers disagree with the model	Override analytics dialog
Chain seal	"CHAIN VERIFIED" stamp	Re-verify (GET /ledger/verify)

Data: GET /decisions, GET /decisions/{id}, GET /decisions/{id}/trace, GET /decisions/{id}/outcome, GET /ledger/verify, GET /analytics/overrides, GET /analytics/calibration.

6.8 SCREEN 5: IMPACT

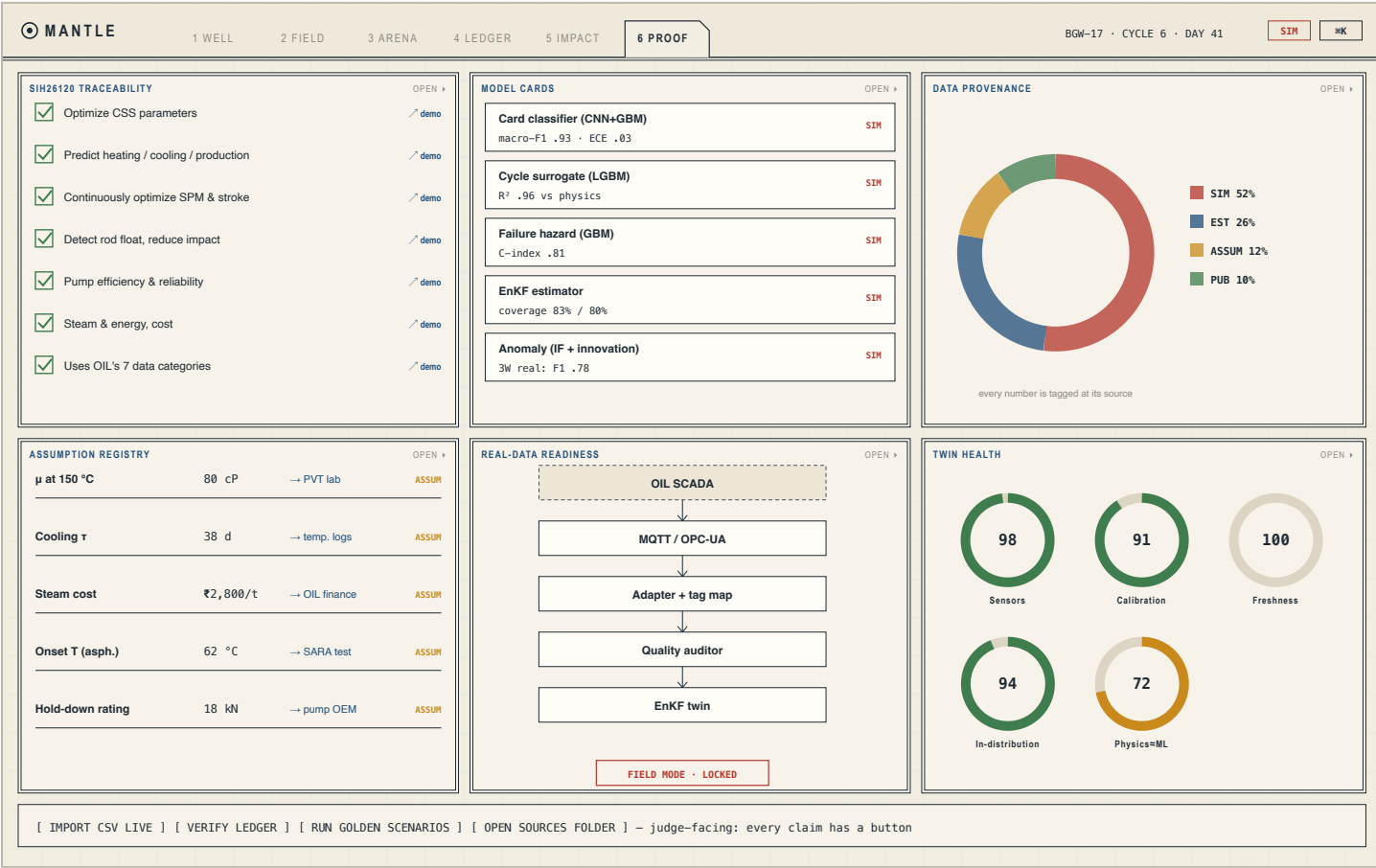


Three columns (Economic · Environmental · Social), four hero numbers each (value, ± band, SIM), and a cumulative chart vs baselines. The scope toggle is *Field (30 sim wells) · Per well · Illustrative × 33 wells*, and the baseline toggle is *Static · Pump-off · Sequential*.

Dialogs: *How computed* per number (formula, inputs, source links) · **Cycle receipt** (printable receipt: steam, fuel, water, power, oil, CO₂e, saved vs baseline) · *Social evidence* (heat-stress-weighted site visits avoided; high-risk hours).

Data: GET /impact/summary?scope=&baseline=, GET /impact/series, GET /impact/receipt?well_id=&cycle=.

6.9 SCREEN 6: PROOF



Six cards, each with one action button (judge-facing):

CARD	DEFAULT	BUTTON
SIH26120 traceability	Each PS line with a check	demo jumps to the proving view via deep link
Model cards	Model, headline metric, dataset tag	Opens the full model card
Data provenance	Donut: SIM / EST / ASSUM / PUB / MEAS share	Opens source registry
Assumption registry	Top assumptions with "replace with..."	Full registry
Real-data readiness	SCADA → MQTT/OPC-UA → adapter → auditor → twin; FIELD MODE locked	Import CSV live (auditor runs in front of judges)
Twin health	Rings: sensors, calibration, freshness, in-distribution, physics≈ML	Health detail

Footer actions: Import CSV live · Verify ledger · Run golden scenarios · Open sources.

Data: GET /evidence/traceability, GET /evidence/models, GET /evidence/provenance/summary, GET /evidence/assumptions, POST /data/imports, GET /data/imports/{id}/report, GET /twin/health.

6.10 OVERLAYS

OVERLAY	TRIGGER	CONTENTS
Timeline dock	Always (Well, Field)	Phase bands; past = replayed estimate; future = P50 with P10–P90 fan; pins (events, decisions); speed 0.25×–500×; horizons 24 h–180 d; Fork
Condition Deck	C	5 card columns (Formation, Crude, Climate & time, Completion & lift, Steam source) + Faults + Presets. Apply = 1.2 s scene cross-fade + revision clouds.
Ask Mantle	⌘K	Command palette + copilot answers (verified numbers, sources, "Show me")
Alerts drawer	Bell	ISA-18.2 priorities P1–P4; ack / shelf / open; flood grouping
Story Mode	S	Scripted demo beats, presenter notes on a second screen or phone (/present), deterministic seed
Settings	Avatar	Quality (auto/high/low), units (SI / field), sound, reduced motion, persona

6.11 UTILITY ROUTES

ROUTE	PURPOSE
/m/ [wellId]	Mobile Well Card: installable PWA; float-margin ring, rate, SPM, last decision, Acknowledge. Reached by QR at the (virtual) wellhead.
/present	Presenter remote: phone as clicker for Story Mode (next/prev beat, notes, timer), paired by QR, over WebSocket

ROUTE	PURPOSE
/play	Judge Challenge kiosk: 60-second game, operate BGW-17 for 60 sim-days with two sliders (SPM, re-steam day) vs Mantle; leaderboard

6.12 SCREEN STATE RULES (ALL SCREENS)

STATE	TREATMENT
Loading	Blueprint line-drawing skeleton (strokes draw on), never spinners
Empty	A small drawing plus a 3-word label ("No decisions yet")
Error	Red-pencil margin note with retry; details in a dialog
Offline / no LLM	LOCAL badge; copilot switches to templated answers
Stale	Grey STALE badge on affected numbers
Off-reality (scenario)	Coloured ribbon + scenario name in the top bar

7. TWIN BEHAVIOUR (3D SCENE)

7.1 LAYOUT

The Twin screen's layout, regions and interactions are specified in §6.2. This section covers how the 3D scene behaves.

7.2 CAMERA AND INTERACTION

- **Default pose:** three-quarter view from slightly above, looking at the wellhead, with the whole block diagram framed. There is a slow idle drift ($\pm 3^\circ$ azimuth over 40 s) so the scene always feels alive.
- **Orbit** with constraints (no going under the slab bottom, no clipping inside rock). **Zoom** to the wellhead, pumping unit or reservoir via **snap points**, with double-click on a part to fly to it.
- **Dive (scroll while hovering the wellbore):** the camera descends along the well axis. Callouts change with depth: surface → fluid level → pump → perforations → heated zone. A depth gauge on the left edge shows true MD/TVD.
- **Hover a part:** outline glow, tooltip with name and 2 key numbers. **Click:** the right stack switches to that part's panel (e.g. Rod String: taper table, stress at depth, fatigue).
- **X-ray toggle (X):** casing and tubing go translucent so the rods, pump and fluid become visible.
- **Split view (Scenario compare):** the canvas splits vertically into two synchronized twins (Baseline | Plan) with linked cameras and a divider you can drag. This is a signature demo moment.

7.3 LENSES (ONE MODEL, SIX ANALYTICAL TOOLS)

LENS	WHAT CHANGES IN THE SCENE	DATA SOURCE
Physical (default)	Photoreal PBR materials	none
Thermal	Cut faces and reservoir volume show the computed $T(r,z)$ field. Tubing is colored by the fluid temperature profile (Ramey). Steam lines glow. Everything else goes grey.	Thermal engine grid
Mechanical	Rod string colored by axial stress per section (tension warm, compression cold). Stress waves travel down the string (from the Gibbs solver) in slow-motion. Gearbox torque shown as a radial gauge ring on the crank.	Wave-equation solver
Flow	GPU particles for oil (amber) and water (blue) moving through the formation toward the perforations, speed $\propto$ computed Darcy velocity . When the reservoir cools, the particles visibly crawl. Fluid rises in the tubing in pulses at SPM.	Inflow + pump model
Pressure	Pressure field on the cut faces (reservoir → Pwf → pump intake) and the fluid level in the annulus as a bright meniscus plane.	Wellbore hydraulics
Risk	Scene desaturated. Risk "hot spots" pulse where margins are low: rod string sections near float, the pump (fillage/unsetting), the casing (thermal stress), the tubing band where asphaltene deposition is likely.	Risk Stack

7.4 PHASE-DEPENDENT BEHAVIOR (THE SCENE TELLS THE CSS STORY)

PHASE	SCENE BEHAVIOR
Injection	Pumping unit parked (horsehead at top, brake on). Rods are pulled or hung off, per the completion option. White steam particles stream down the VIT. A steam plume vents at the generator skid off-pad. The heated zone <i>grows</i> visibly (Marx–Langenheim radius). Callouts: steam rate, wellhead pressure, steam quality at surface vs sandface, casing thermal stress.
Soak	Everything still. Wellhead shut-in valve closed (red handle). The heated zone <i>diffuses</i> : the edge softens and the core cools slightly. A soak timer and the "optimal production start" countdown appear on the wellhead.
Production	Pump runs at SPM with the Stroke Shaper profile. Oil particles flow in. The fluid level oscillates. The heated zone slowly shrinks and cools; its color drifts from yellow to red to purple across the cycle.
Decline / cut-off	The same as production, but the Risk lens auto-suggests itself when margins shrink. When the marginal value theorem trigger fires, a subtle red "CUT-OFF WINDOW" ring appears on the timeline and a ghosted steam line lights up to foreshadow the next cycle.

7.5 SURFACE ENVIRONMENT (DRIVEN BY THE CONDITION DECK)

Sky, sun angle, ambient light, wind, dust and heat haze all respond to the selected climate and time of day (§12.4). The default demo condition is "Thar · late-October dusk", which gives the best glass contrast and the most beautiful light.

8. 3D ASSET BRIEF (HAND-OFF DOCUMENT)

For the 3D modeling and animation agent. This section is self-contained. Build a physically plausible, *schematic-realistic* Baghewala heavy-oil CSS well as a geological block diagram. Everything that moves will be driven procedurally by our physics engine at runtime. You build the rig, the pivots, the materials and the anchors, **not baked animations** (except where noted). Target: glTF 2.0 (.gltf), Three.js / React Three Fiber, WebGL2 with a WebGPU-ready path.

8.1 OVERALL CONCEPT

- A **floating geological block diagram**: a rectangular slab of earth about **60 m × 40 m in plan**, with the top face being desert terrain and the front and right faces cut to reveal the strata. The well descends through the cut corner so the full wellbore is visible on the cut faces (a quarter-section cutaway around the well axis).
- **Depth handling**. Real depth is ~1,150 m and the surface equipment is ~10 m, so use a **piecewise depth scale** with drafting-style **break bands**:
 - **Zone A, surface to 30 m**: true scale 1:1.
 - **Zone B, 30 m to 1,050 m**: compressed about 25:1. Two **"depth break" ribbons** (a thin gap with a zig-zag/sine edge, like the break symbol on an engineering drawing) cross the slab. Strata continue across them.
 - **Zone C, 1,050 m to 1,200 m (cap rock, reservoir, base)**: about 1:3. The reservoir zone is the visual climax at the bottom.
 - The final slab's visual height is ~55 m. Provide the **mapping function** (true depth → model Y) as a JSON table in the export so the frontend places depth-driven elements correctly.
- **Style target**: *photoreal materials, illustrative composition*. Think of a high-end museum cutaway or a Nat Geo infographic. Clean bevels, believable wear, no noise for its own sake. Rock faces read as real sandstone, carbonate and shale, but with crisp, slightly idealized bedding lines, like a textbook plate.

8.2 SURFACE: PAD AND ENVIRONMENT

- **Terrain top**: Thar desert sand with low aeolian ripples (displacement plus normal map), sparse dry scrub (khejri-like thorny shrubs, 3–5 instances, low poly with alpha cards), a few pebbles. Sand color warm ochre #C9A46A to #E0C28E.
- **Well pad**: a compacted gravel/caliche rectangle, ~20 m × 16 m, slightly raised, with a darker oil-stained patch near the wellhead. There is a concrete plinth under the pumping unit.
- **Perimeter**: a low chain-link fence on 2 sides only (so it doesn't block the camera), a gate, and a small hazard signboard *with generic text* (no real company logos or branding).
- **Instrumentation pole**: an RTU cabinet on a pole with a small solar panel, an antenna and a status LED (emissive, driven by alarm state).
- **Steam line**: an insulated steam line (aluminium-clad lagging with visible band clamps) enters from the slab edge on pipe sleepers, through an isolation valve (handwheel), to the thermal wellhead. The **off-slab steam generator** is implied: the line exits the slab edge with a clean cap.
- **Production flowline**: an insulated flowline leaves the wellhead to the opposite slab edge. Include a check valve and a pressure gauge.
- **VFD cabinet**: a weatherproof enclosure near the motor with cable tray, a small HMI window (emissive screen that we texture at runtime) and a sunshade canopy.
- **Lighting fixtures**: 1 pole light (emissive at night).

8.3 PUMPING UNIT: CONVENTIONAL BEAM UNIT (PRIMARY ASSET)

Build an API-style **conventional crank-balanced beam pumping unit**, roughly a 320-size unit (overall ~9–10 m tall, ~12 m long). **Every moving part is a separate node with its pivot at the true joint.**

- **Base / skid**: a steel I-beam skid bolted to the plinth.
- **Samson post**: an A-frame tripod (3 legs, cross bracing, ladder with safety cage on one leg, a small platform at the saddle bearing).
- **Walking beam**: a box/I-beam with a stiffener pattern. The pivot is the **saddle bearing** on top of the Samson post.
- **Horsehead**: a curved arc front with a wire-rope groove. It should be a separate node so it can swing for workover (optional).
- **Bridle (wireline hanger)**: two steel wire ropes from the horsehead arc to the **carrier bar**. Rig as a **2-bone chain or spline** so it can go **slack** (rod float state). Provide a blend shape `bridle_slack` with 0–1 sag.
- **Carrier bar, polished rod clamp, polished rod**: a chrome polished rod (high metalness, low roughness) passing down into the **stuffing box** on the wellhead.
- **Equalizer and tail bearing**: at the rear of the beam.
- **Pitman arms (×2)**: connect the equalizer to the crank pins.

- **Cranks (x2) with counterweights:** crank arms with 2–4 bolted counterweight blocks each. The crank pivot is the gearbox output shaft.
- **Gear reducer:** a cast gearbox housing with an oil sight glass and a breather.
- **Prime mover:** an electric induction motor on sliding rails, a **V-belt drive** with a perforated belt guard (the pulleys inside are visible through the perforations), and a brake lever and drum.
- **Paint:** industrial safety yellow with a dark grey structure (or oxide red, as a variant). The **counterweights** have black and yellow hazard stripes. Edges get subtle wear (edge-mask metal showing through), dust accumulation on upper surfaces, and a hint of oil streaking down the wellhead.
- **Kinematics requirement:** we solve the **four-bar linkage** (crank → pitman → beam → horsehead) at runtime. Provide exact pivot locations as **named empties** (`PIV_crank`, `PIV_crankpin_L/R`, `PIV_equalizer`, `PIV_saddle`, `PIV_horsehead_arc_center`) and link lengths in metadata. Do **not** bake the motion.

8.4 PUMPING UNIT: HYDRAULIC LONG-STROKE VARIANT (SECONDARY ASSET)

OIL operates both conventional and hydraulic SRP units, so build a **hydraulic long-stroke unit** as a swap-in variant:

- A vertical tower (~11 m) with a hydraulic cylinder, a cable-over-sheave or direct-acting rod, a hydraulic power pack skid (tank, pump, motor, accumulator) and hoses.
- It uses the same wellhead interface and a named anchor for the polished rod.
- Moving nodes: the cylinder rod and the sheave, with a linear pivot.

8.5 WELLHEAD (THERMAL)

- A **thermal wellhead** (rated for steam, "320 °C class" in look): tubing head, casing head, master valve, wing valves (one to the steam line, one to the flowline), a tee with the **stuffing box** on top, a pressure gauge (dial face as a runtime texture), a temperature transmitter, and a sample point.
- Valve handwheels are separate nodes with rotation pivots (open/close animation during phase changes).
- Insulation jackets on the upper wellhead body (quilted blanket look).

8.6 SUBSURFACE: THE CUT BLOCK

- **Strata (top to bottom, schematic and indicative for the Bikaner–Nagaur basin; label as schematic):**
 1. Aeolian sand / alluvium (loose, warm ochre)
 2. Tertiary sediments (variegated clays/siltstones)
 3. Nagaur Group, red-brown sandstone/siltstone
 4. Bilara Group carbonates/evaporites (grey dolomite with white anhydrite/halite streaks). This is the **cap rock** and it gets a subtle crystalline sparkle in the normal map.
 5. Upper carbonate interval (thin, darker, oil-stained patches). This is the variant heavy-oil zone for the Condition Deck.
 6. **Jodhpur Sandstone (reservoir):** buff-to-pinkish cross-bedded sandstone. Visible cross-bedding lines. **Oil-stained** darker brown saturation pattern in the pay interval.
 7. Basement below (dark, crystalline).
- Each stratum is a **separate mesh** (`STRAT_01_aeolian` ... `STRAT_07_basement`) with **world-space triplanar PBR materials** (no UV stretching on cut faces) and a **thin bright edge line** along the bedding contacts (the "textbook" feel).
- **Cut faces** have a very slight emissive lift so the section reads even at night.
- Depth tick marks are **embossed on the front cut edge** every 100 m (true depth) and are also exported as anchors (`DEPTH_0100` ... `DEPTH_1200`).

8.7 WELLBORE AND COMPLETION (SECTION VIEW)

Build the wellbore as **concentric tubulars in quarter-section**, so the cut plane shows the layers:

- **Conductor** (short, large, cemented), **surface casing** to ~Zone B top, **production casing** to the reservoir base with a **thermal-grade look** (premium connection collars every ~12 m true, shown as slightly thicker rings, fewer in the compressed zone).
- **Cement sheath** between casing and formation: light grey, with a granular normal map.
- **Vacuum-insulated tubing (VIT):** inner and outer walls with a visible gap (the vacuum annulus) at the cut. Joints every ~12 m. A **bare-tubing variant** is needed as a Condition Deck swap.
- **Tubing anchor / catcher** above the pump.
- **Rod string:** sucker rods of **3 taper sections** (e.g. 1", 7/8", 3/4") plus **sinker bars** at the bottom. Rod couplings are visible as bulges. **Rod guides / centralizers** are molded polymer, every few rods, off-white. The string is **segmented into 24 nodes** (`ROD_SEG_01` ... `ROD_SEG_24`, each with a pivot at its top) so we can drive per-segment displacement (stretch and wave) and per-segment stress color.

- **Downhole insert pump (x-ray-ready):** a **barrel** (transparent-able), **plunger** (chrome) with the **traveling valve** (ball and seat) at its bottom, the **standing valve** (ball and seat) at the barrel bottom, and the **seating nipple / hold-down** (a mechanical cup-type look).
 - Valve balls are separate nodes (they lift ~5 mm, exaggerated ×3, when open).
 - A **fluid-fill mesh** inside the barrel with a morph target `fill_level` 0–1 (we drive fillage).
 - A **gas/steam pocket** mesh above the fluid (`barrel_gas`) for interference states.
- **Perforations:** a helical shot pattern in the casing across the pay. Each perforation tunnel is a small cone into the formation with a darker "flow mouth" decal.
- **Annulus fluid:** a mesh from the pump intake up to the **dynamic fluid level**, with a separate top-surface node `FLUID_LEVEL_SURFACE` (a flat ring with a meniscus shader) that we move in Y.
- **Tubing fluid:** inner column mesh with a vertex-color gradient we update (temperature/viscosity along depth).

8.8 RESERVOIR VISUALS (THE HERO)

- **Heated zone:** an **axisymmetric volume** around the perforations. Deliver it as:
 1. a set of **8 nested shells** (`HEAT_SHELL_1..8`, radial iso-surfaces at normalized radii 0.125 ... 1.0, smooth, low poly, with edge-fade alpha), and
 2. a **disc slice** on each cut face (`HEAT_SLICE_FRONT`, `HEAT_SLICE_SIDE`) that we shade with a data texture. At runtime we scale the shells and write $T(r,z)$ into a 256×64 float texture. The shader maps it to the thermal ramp on the cut faces and gives the shells a soft volumetric glow.
- **Steam chamber (injection phase):** a translucent, slightly turbulent volume near the perforations (a raymarched noise volume or layered soft particles) that grows with injected heat.
- **Oil flow particles:** no mesh needed. Provide **8 spline "streamline" guides** converging radially on the perforations along the pay (`FLOW_GUIDE_01..08`). We spawn GPU particles along them.
- **Cap-rock and base heat loss:** thin glowing gradients at the reservoir top and bottom contacts (so heat loss to the over- and underburden is visible), driven by a uniform.

8.9 ENVIRONMENT, SKY AND ATMOSPHERE

- **Sky:** a physically based sky (Preetham/Hosek or an HDRI set) with **4 lighting presets as HDRIs** (CC0 sources such as Poly Haven are fine): *Desert noon*, *Golden hour / dusk (default)*, *Night (clear, stars)*, *Overcast / dust haze*. Sun direction is exposed as a runtime parameter.
- **Ground fog / dust layer:** a height-fog volume, parameterized (for sandstorm).
- **Heat haze:** a screen-space distortion mask over hot pipes (the steam line and wellhead during injection). Provide a `HAZE_EMITTERS` group of simple proxy meshes.
- **Steam vents:** 2 emitter locations (steam generator edge, wellhead vent) as empties `FX_STEAM_*`.
- **Sandstorm:** emitters along the windward slab edge (`FX_SAND_*`).
- **Night:** emissive windows on the VFD HMI, the pole light, the RTU LED and the beacon on top of the Samson post (red aviation-style blink).

8.10 STATES AND ANIMATION (WHAT MOVES AND HOW WE DRIVE IT)

All motion is **procedural and driven by runtime parameters**. You provide rigs, pivots, morphs and material hooks. Bake only the idle micro-motions listed.

ELEMENT	DRIVEN BY	REQUIRED FROM ASSET
Crank, pitman, beam, horsehead	Crank angle $\theta(t)$ from SPM + Stroke Shaper profile; four-bar solver	Pivots, link lengths
Polished rod, carrier bar	Horsehead arc kinematics	Named nodes
Bridle slack	Rod-float state (0–1)	<code>bridle_slack</code> morph / 2-bone rig
Rod segments	Per-segment displacement $u(z,t)$ and stress $\sigma(z,t)$ from the wave equation	24 segment nodes, a vertex attribute for stress color
Plunger, valve balls	Pump kinematics, valve open/close logic	Nodes with local Y translation
Barrel fluid	Fillage per stroke	<code>fill_level</code> morph
Gas pocket	Interference state	<code>barrel_gas</code> scale
Annulus fluid level	Fluid level (m) → model Y	<code>FLUID_LEVEL_SURFACE</code>
Heated-zone shells and slices	$T(r,z)$ field, $r_h(t)$	Shell meshes, slice meshes with UVs in (r,z)
Valve handwheels	Phase changes	Rotation pivots
Beacon, LEDs, HMI screens	Alarm state, time of day	Emissive materials with named slots
Baked idle only	Scrub sway ($\pm 2^\circ$, 6 s loop), fence flag flutter, RTU antenna micro-sway	Small baked clips

Special states we must be able to show (verify the rig supports them):

- 1. **Normal pumping** at 2–9 SPM.
- 2. **Rod float:** the carrier bar separates from the polished rod clamp by a visible gap, the bridle sags, then a snap-back. We add a white impact flash and a shockwave ring on the rod string.
- 3. **Fluid pound:** the plunger hits the fluid surface mid-downstroke. We add a ripple and a load spike.
- 4. **Gas/steam interference:** a gas pocket in the barrel compresses on the downstroke.
- 5. **Pump unsetting:** the whole pump assembly lifts ~0.3 m (exaggerated) off the seating nipple. Needs a pump-assembly parent node.
- 6. **Parted rod:** a separation between two segments. The lower part drops.
- 7. **Injection:** the unit is parked and the steam flow is on.
- 8. **Soak:** everything is still and the wellhead is shut-in.
- 9. **Workover (optional, vision):** the horsehead is swung back.

8.11 ANCHORS FOR LABELS AND CALLOUTS

Provide **empties** (zero-size nodes) at the exact point where a leader line should touch. Name them exactly as follows, since the frontend binds metrics by these names:

ANCHOR_horsehead	ANCHOR_polished_rod	ANCHOR_crank
ANCHOR_gearbox	ANCHOR_motor	ANCHOR_vfd
ANCHOR_wellhead	ANCHOR_steam_inlet	ANCHOR_flowline
ANCHOR_casing_upper	ANCHOR_fluid_level (moves with FLUID_LEVEL_SURFACE)	
ANCHOR_rod_mid	ANCHOR_rod_bottom	ANCHOR_tubing_mid
ANCHOR_pump	ANCHOR_standing_valve	ANCHOR_perforations
ANCHOR_heated_zone	ANCHOR_caprock	ANCHOR_reservoir_far

8.12 LINE-ART LAYER (REQUIRED FOR THE BLUEPRINT TRANSITION)

The 2D Blueprint is drawn from **the same model**, so the asset must also contain **clean drafting geometry**:

- A collection **LINE_***: **hand-curated polylines** (not auto-generated from every triangle edge) that describe each part's **side-elevation outline** as seen from the +X axis (the Blueprint camera looks down -X). Include silhouettes, key internal edges (beam web, crank circle, counterweight outlines), **centre lines** (flagged `centre=true`) and **hidden lines** (flagged `hidden=true`).
- **Section hatching regions** (**HATCH_***) as closed planar polygons on the cut plane for each stratum, cement and steel section, with a `hatch` custom property naming the ISO-style pattern (`sandstone_dots`, `limestone_brick`, `shale_dashes`, `steel_45`, `cement_stipple`, `evaporite_cross`).
- The line collection is parented to the same moving nodes, so the line-art pumping unit **animates identically**.
- **Custom properties** on line nodes: `weight` (thick/medium/thin), `layer` (outline/detail/centre/hidden/dimension).

8.13 MATERIALS AND TEXTURES

- **PBR metallic-roughness.** Textures at 2K max (1K for small parts), **KTX2 / Basis-compressed**. Use texture atlases where sensible. Rock uses **triplanar** shaders (we can do this in code; supply tileable albedo, normal and roughness sets per stratum).
- **Material list (named, so we can override per lens):** `M_paint_yellow`, `M_paint_grey`, `M_steel_bare`, `M_chrome`, `M_wire_rope`, `M_rubber_belt`, `M_concrete`, `M_gravel`, `M_sand`, `M_insulation_alu`, `M_insulation_blanket`, `M_cement`, `M_tubular_steel`, `M_VIT_outer`, `M_rod_steel`, `M_rod_guide_poly`, `M_pump_barrel` (supports transmission), `M_fluid_oil`, `M_fluid_water`, `M_strat_01..07`, `M_glass_hmi`, `M_emissive_led`.
- **Oil:** near-black brown with a strong specular and thin-film edge sheen. When heated, the shader lightens toward amber (we drive `u_temperature`).

8.14 BUDGETS AND DELIVERY

- **Triangles:** ≤ 350k total in the hero LOD. Pumping unit ≤ 90k, wellhead ≤ 25k, rock block ≤ 40k (detail comes from textures), wellbore ≤ 60k. **LOD1** at ~40% and **LOD2** at ~15% for the fleet thumbnails and split view.
- **Draw calls:** ≤ 120 in the hero view. Merge static meshes by material.
- **File:** a single `mantle_well.glb` ≤ 25 MB with meshopt compression, plus `mantle_well_hydraulic.glb` (the variant unit only), plus `depth_map.json`, `kinematics.json` (pivot positions, link lengths, stroke length options: 100/120/144 in), and `anchors.json`.
- **Units:** metres, Y-up, the well axis at the origin (X=0, Z=0), and the ground surface at Y=0.
- **Naming:** `PREFIX_part_detail`, no spaces. Every animated node has its origin at the pivot.
- **Acceptance test:** loads in `three/examples` glTF viewer without errors. All pivots rotate cleanly (no wobble). The line-art layer overlays the orthographic side view within 1 px at 1080p. Renders at 60 fps at 1440p on an M1/M2 MacBook with a baseline post-processing chain.

8.15 LOOK REFERENCES (DESCRIBE, DON'T COPY)

- Museum cutaway illustrations of oil wells (clean section, labelled strata).
- Textbook geological block diagrams (floating slab, bedding lines, depth ticks).
- Industrial product renders of pumping units (clean studio lighting, believable wear).
- *Mood*: golden hour over the Thar, warm sand against cold steel, and a glowing ember of heat deep underground.

9. METRICS AND LABELLING SYSTEM

9.1 DECISION: ATTACH THE FEW, DOCK THE MANY

We **do** attach metrics to the object like a scientific diagram, but **only the few that are physically located**, and never more than **seven at once**. Everything else lives in the glass side panels.

The reasons: attached labels make causality legible ("this number is *this* part"), and they are what makes the scene look like a scientific plate rather than a game. Past seven labels, the scene turns into a sticker book and callouts start occluding the physics.

Rules

1. A metric earns a callout only if it is a **property of a place** (load *at* the polished rod, fluid level *in* the annulus, temperature *in* the heated zone). Field-level or economic numbers never get callouts.
2. **Max 7 visible**. The set changes with the **phase** and the **lens** (tables below).
3. **Leader lines**: 1 px, with a small circle at the anchor end. Horizontal-first elbow routing with 12 px min spacing. Labels lay out in screen space with a light force-directed solver: they repel each other, are attracted to their anchor's side of the screen, and never overlap the right-hand glass stack.
4. **Occlusion aware**: when an anchor is hidden behind geometry, the leader goes dashed and the card dims to 50%.
5. **Level of detail**: when zoomed far out, cards collapse into a dot with just the number. When zoomed in, cards expand to a value, a trend arrow, a 24 h micro-sparkline and a provenance tag.
6. **Changing values** tween. **Alarmed values** gain a colored left bar and a gentle pulse (ISA priority 1–2 only).
7. **Click a callout** to open that part's panel in the right stack. **Shift-click** to pin it (pinned callouts survive phase changes).

9.2 CALLOUT SETS (TWIN VIEW)

Production phase (default):

#	ANCHOR	METRIC	EXAMPLE
1	Polished rod	PPRL / MPRL, rod-float margin	92 / 11 kN · margin 18%
2	Crank	SPM (+ downstroke share) and gearbox torque % of rating	5.4 SPM · kd 0.58 · 71% torque
3	Motor / VFD	Power, frequency	18.6 kW · 42 Hz
4	Flowline	Oil rate, water cut, wellhead temperature	46 BOPD · WC 22% · 58 °C
5	Fluid level	Level and submergence	612 m · subm. 380 m
6	Pump	Fillage, volumetric efficiency, hold-down margin	78% · η 71% · HD 2.3×
7	Heated zone	Sandface temperature, viscosity, heated radius	96 °C · 210 cP · r_h 11 m

Injection phase: (1) steam inlet: rate t/d, pressure, quality · (2) wellhead: cumulative steam t / target · (3) casing upper: thermal stress % of yield · (4) tubing mid: steam quality at depth · (5) perforations: sandface quality and temperature · (6) heated zone: heated radius growth · (7) caprock: heat loss rate.

Soak phase: (1) wellhead: shut-in time / optimal soak · (2) heated zone: average T and diffusion · (3) caprock: loss rate · (4) perforations: pressure build-up.

Lens overrides: in the Mechanical lens the rod callouts show stress at depth (@742 m: 84 kN · σ 118 MPa · Goodman 0.64). In the Flow lens they show inflow vs pump capacity (the bottleneck).

9.3 WHAT LIVES IN THE SIDE PANELS (TWIN)

- **Recommendation card:** action, schedule, expected Δ (oil, SOR, energy, risk), confidence, binding constraint, Approve / Modify / Reject.
- **Risk Stack (margins):** rod float · fluid pound · pump unsetting · rod fatigue (Goodman) · casing thermal · deposition. Each shows the margin bar, trend and "time to limit" (e.g. float margin hits 0 in ~6 d at current SPM).
- **Forecast:** oil rate P10/P50/P90 for 30 d, with the "if we do nothing" ghost line.
- **Economy strip:** ₹ net/day, cycle SOR (CWE), kWh/bbl, kg CO₂e/bbl, m³ water/bbl.
- **Coupling Dividend:** +₹4.1 L / cycle · +6.8% oil · SOR -0.4 (vs sequential).
- **Twin health:** a small ring (sensor integrity, calibration, data freshness).

9.4 DIFFERENT METRICS IN THE BLUEPRINT

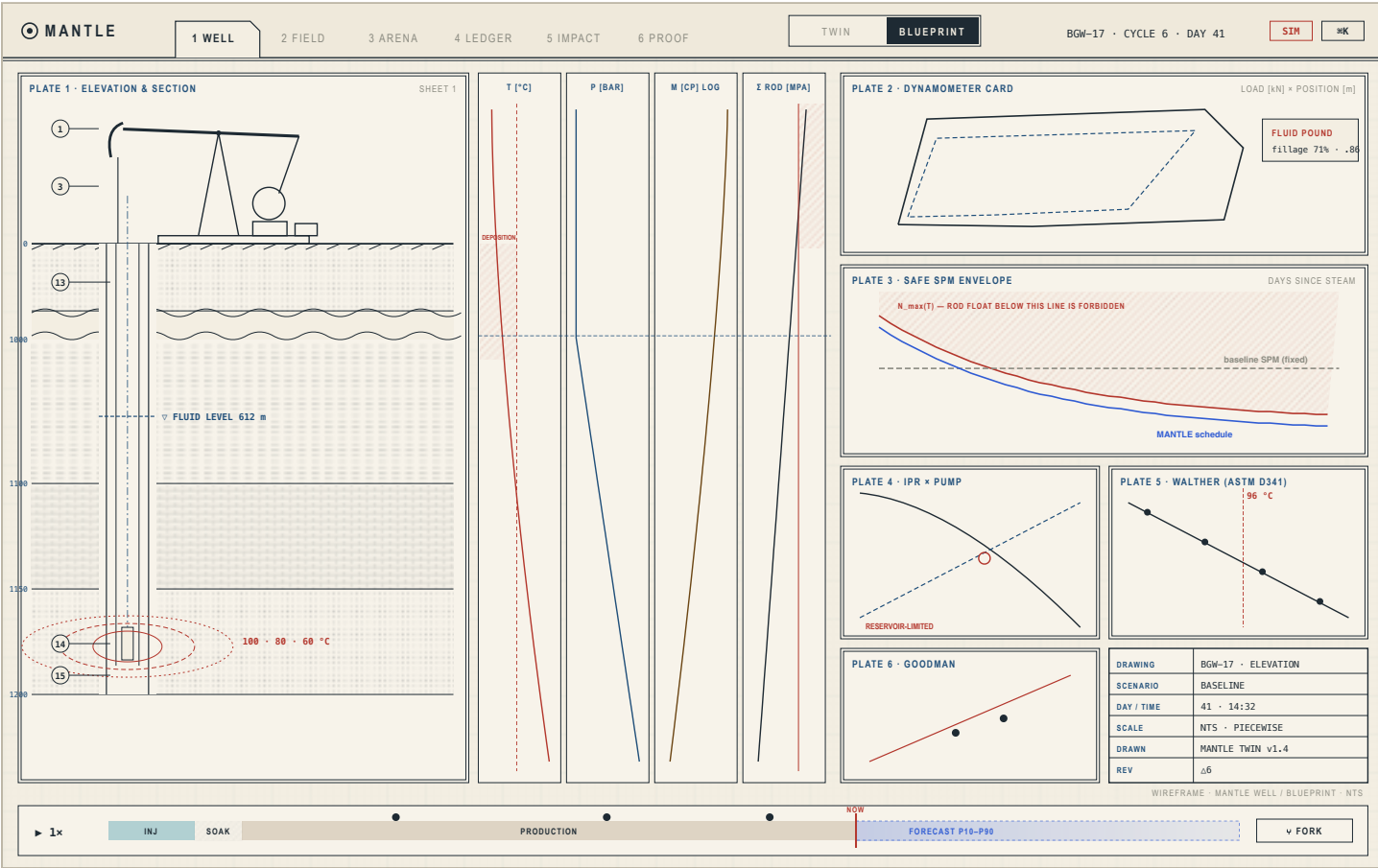
The Twin answers "*how is it doing?*". The Blueprint answers "*why, and how close to the limits?*" The Blueprint's metrics are engineering quantities, drawn as profiles, envelopes and design tables (full list in §10.3). The **same underlying state** feeds both, and the transition retypesets the shared numbers so the viewer sees continuity.

10. THE BLUEPRINT VIEW (2D TECHNICAL SHEET)

10.1 CONCEPT

A **living engineering drawing**: an A-series landscape sheet on cream graph paper, drawn in graphite and drafting-blue ink. It shows the same well at the same instant as the Twin, **still animated**. The line-art pumping unit keeps nodding at SPM, the rod-string stress bands pulse, the fluid-level line bobs, and the dynamometer card traces itself stroke by stroke. It should feel like a drawing that is quietly alive.

10.2 SHEET LAYOUT



- **Plate 1, Elevation and section (left, ~45% width):** the line-art well (from the **LINE_*** layer) with depth breaks, section hatching per stratum, **balloon callouts** ①–⑮ referencing the **Bill of Materials**, dimension lines (stroke length, pump setting depth, perforation interval, fluid level), and heated-zone **isotherms** drawn as contour lines (100 °C, 80 °C, 60 °C) with inline labels, like a topo map.
- **Depth tracks (log-style strips aligned to the same depth axis):**
 1. **Temperature profile.** Tubing fluid, annulus and formation T vs depth (Ramey), with the asphaltene/wax **onset temperature** as a red dashed vertical line and the **deposition-risk interval** hatched red where the profile crosses it.
 2. **Pressure profile.** Tubing and annulus pressure vs depth, P_{wf}, pump intake pressure, and the fluid level marker.
 3. **Viscosity profile.** $\mu(z)$ on a log axis, following the temperature.
 4. **Rod stress envelope.** $\sigma_{\max}$ and $\sigma_{\min}$ per depth over the last stroke, with the **Goodman allowable** as a red limit line and sections within 10% hatched.
- **Plate 2, Dynamometer card:** the surface card (graphite) and the downhole card (drafting blue, dashed) from the wave equation. A ghost "healthy" card in faint hatch. The **diagnosis label** is set in a drafting box ("FLUID POUND · fillage 71% · conf. 0.86"). The card traces live.
- **Plate 3, SPM envelope:** days since steam on X. On Y, the **safe SPM ceiling** $N_{\max}(t)$ (from viscosity, rod weight and the downstroke share), the actual SPM (baseline) and the Mantle schedule. The forbidden region above $N_{\max}$ is hatched red. This is the single most explanatory chart for the PS: **"the safe speed shrinks as the well cools."**
- **Plate 4, IPR x pump capacity:** the inflow performance curve at the current temperature (and ghost curves at +15 d and +30 d), the pump displacement line, and the operating point marked $\oplus$. Annotated **"RESERVOIR-LIMITED"** or **"PUMP-LIMITED"**.
- **Plate 5, Walther chart (ASTM D341 paper):** log-log(v) vs log T, with the lab points (or assumed anchors, tagged ASSUM), the fitted line, and today's operating temperature marked. It looks exactly like the chart engineers use.
- **Plate 6, Goodman diagram:** alternating vs mean stress for the top rod of each taper, the operating points, and the allowable line with service factor.

- **Optional plates (tabbed, same slot):** heat-balance Sankey drawn as a drafting flow diagram (steam energy → surface loss → wellbore loss → reservoir → caprock loss → produced heat), the **casing thermal-stress** mini-diagram, the **Stroke Shaper** polar speed profile, and the **pump-off / fillage** history.
- **Bill of Materials and rod taper table:** parts with sizes, grades and ratings (API rod grade, lengths, weights). Values that are assumptions are *italic with ASSUM tag*.
- **Notes block:** numbered drafting notes generated from the state ("3. Rod float margin < 20% below 900 m: consider SPM ≤ 4.8 or kd ≥ 0.6"). These are the recommendation rewritten as an engineer would write it.
- **Title block:** well, cycle, day, time, scenario/branch name, revision number (increments on every condition or plan change), "DRAWN: Mantle twin v1.4 · CHECKED: [engineer on approval]", scale "NTS (depth piecewise, see breaks)", provenance legend.

10.3 BLUEPRINT METRICS (ENGINEERING SET)

Pump displacement PD · effective plunger stroke (after rod stretch) · rod stretch · PPRL · MPRL · load range · card area (work per stroke) · polished-rod HP · hydraulic HP · system efficiency · peak gearbox torque vs rating · counterbalance effect · Goodman ratio per taper · N_max (safe SPM) · float margin per depth · hold-down margin · Pwf · pump intake pressure · submergence · fluid gradient · heat-loss breakdown (surface / wellbore / caprock) · steam quality at sandface · heated radius · average heated-zone T · μ at sandface and at the pump · mobility ratio μ_hot/μ_cold · Boberg–Lantz J_hot/J_cold · casing thermal stress % of yield · deposition interval (m) · IPR slope (productivity index).

10.4 INTERACTIONS ON PAPER

- Hover a balloon to highlight the part and its BOM row. Hover a track to show a depth crosshair across all tracks, with a readout (@ 742 m · 71 °C · 34 bar · 1,120 cP · σ 118 MPa).
- Drag the **timeline**: the whole sheet updates. Changed annotations get **revision clouds**.
- **Export**: vector PDF (A3) and SVG, with the title block filled. A "Print well card" button.
- **Compare**: overlay a scenario in drafting blue on top of the baseline in graphite. This is the classic "revision overlay".

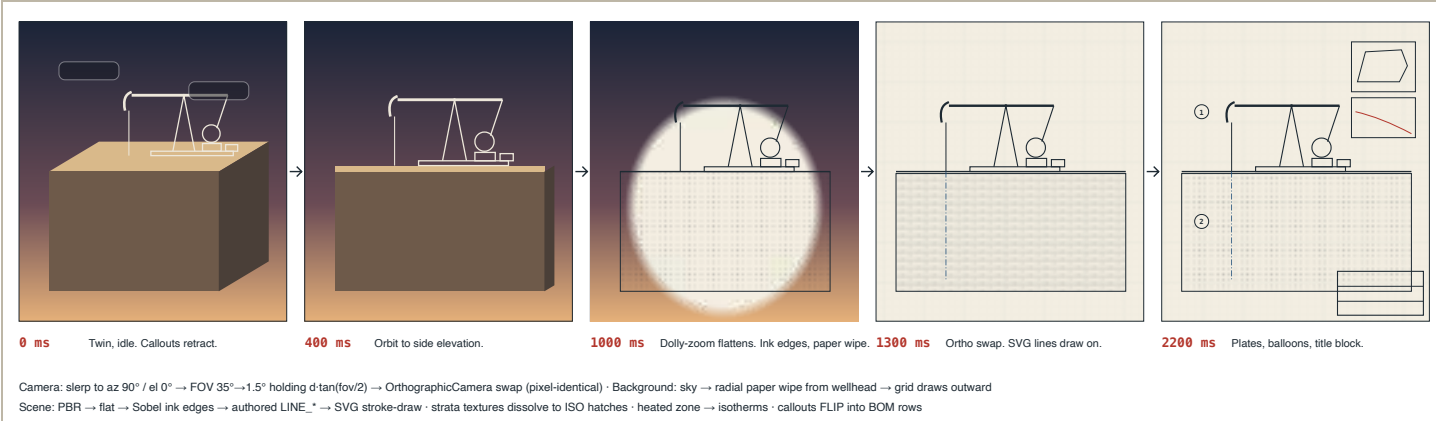
11. THE TWIN ⇄ BLUEPRINT TRANSITION (SHOT-BY-SHOT)

11.1 INTENT

The Twin should **become** the Blueprint in place. The well never leaves its spot on screen. The camera swings to a pure side elevation, perspective flattens into orthographic, light and material drain into ink, and the world around the well (sky, desert, glass panels) turns into paper, grid and plates. The viewer's eye stays locked on the well, so they feel they are *looking at the same object in a different language*. Target duration **2.2 s** (adjustable 1.6–3.0 s), fully interruptible and reversible.

Trigger: the **TWIN** | **BLUEPRINT** toggle, the **B** key, or a "Show the engineering" beat in Story Mode.

11.2 CHOREOGRAPHY



Timeline in milliseconds. Easing is `power3.inOut` unless noted.

T (MS)	CAMERA	SCENE / 3D	UI / 2D	BACKGROUND
0–150	Idle drift stops. Controls lock.	Callouts retract: cards shrink into their anchor dots (80 ms stagger).	Right-hand glass stack slides 24 px right and fades to 0 (blur ramps down for performance).	—
150–700	Orbit from the current pose to azimuth 90° (pure +X side), elevation → 0°, roll 0. The path is a slerp with slight overshoot damping. The well axis stays pinned to the screen point it will occupy in the Blueprint (the Plate 1 centre line).	Lenses fade to Physical. Particle systems ease out (steam and oil particles slow and fade).	Top bar glass de-saturates.	Sky begins to desaturate. Sun direction swings to side light to flatten shadows.
700–1250	Dolly-zoom to orthographic: move the camera back along −X while narrowing FOV so the well's on-screen height <i>h</i> stays constant: $d(t) \cdot \tan(fov(t)/2) = const$. FOV goes 35° → 1.5°. Perspective visibly <i>flattens</i> : the Samson post legs straighten and the block's depth faces collapse to a sliver. This is the "wow" beat.	Material drain: PBR shading cross-fades to a flat albedo (light-grey fill), then to white. A screen-space edge pass (Sobel on depth + normals) fades in in graphite, and the authored LINE_* layer fades in on top. The terrain's top face and the non-cut faces fade out. The cut face remains. Strata textures dissolve into hatch patterns (the hatch shader cross-fades from texture to procedural hatch using a noise-threshold dissolve).	The glass left rail morphs: rounded pill icons become index-card tabs (FLIP transform), and the fill goes from glass to paper.	Paper wipe: a cream "ink-bleed" mask expands radially from the wellhead. Its edge has a subtle fibrous noise so it looks like paper soaking in, and the sky is replaced behind it.
1250–1300	Camera swap: at FOV 1.5° we swap to an OrthographicCamera whose frustum height equals the perspective frustum height at the focal plane. The swap is pixel-identical.	—	—	—
1300–1800	Ortho camera holds. A subtle 1.02 → 1.0 "settle" scale.	WebGL lines hand off to SVG : the same LINE_* polylines, projected with the ortho matrix into the SVG viewBox, draw on with stroke-dashoffset (outline first, then details, then centre lines), in perfect registration. WebGL line opacity goes to 0 as SVG completes. Depth break symbols snap in.	Plates slide in from the right like drafting sheets placed on the desk (40 ms stagger, 8 px shadow that fades). Plates draw their frames first, then their axes, then the data. Depth tracks extend downward from the ground line.	Grid draws in: minor lines fade in, then major lines draw outward from the well axis (scaleX from centre).

T (MS)	CAMERA	SCENE / 3D	UI / 2D	BACKGROUND
1800– 2200	—	Heated zone becomes isotherm contour lines (marching squares on $T(r,z)$) with inline labels. Balloons ①–⑮ pop in with leader lines, retypesetting the Twin callout values into BOM rows (a FLIP animation of the numbers from their old screen positions into the table cells).	Title block stamps in (scale 1.1 → 1.0 with a tiny rotation settle, like a rubber stamp). Notes type in (typewriter, 8 ms/char, capped at 400 ms).	Paper vignette settles.
2200+	Idle.	The line-art pumping unit continues nodding : SVG paths are updated every frame from the same kinematic solver.	Dyno card begins tracing live.	—

Reverse (Blueprint → Twin): play the same timeline in reverse with two changes. (1) The paper wipe *collapses* into the wellhead, like ink draining. (2) The dolly-zoom reversal is 20% faster so that depth "pops" back, which feels good. Particles restart at the end.

Interruptibility: any input during the transition reverses from the current progress (GSAP timeline `reverse()`), with no jumps.

Reduced motion: a 250 ms cross-fade between two pre-rendered states, no camera move.

Sound (optional, off by default, the demo toggles it on): a soft mechanical "latch" at the camera swap, a paper slide per plate, and a stamp thud on the title block. It should stay very quiet.

11.3 HOW IT WORKS TECHNICALLY

- **One source of truth for geometry:** the GLB provides meshes, `LINE_*` polylines and `HATCH_*` polygons. The same nodes are animated by the kinematic solver in both views.
- **Perspective → ortho continuity:** keep the target's focal distance d_o and frustum height $h = 2 \cdot d_o \cdot \tan(fov/2)$. Each frame, set `fov(t)` and `d(t) = h / (2 \cdot \tan(fov(t)/2))`. At the end, create the ortho camera with `top = h/2`, `bottom = -h/2`, `left/right = ±h \cdot aspect/2`, at the same position and orientation. The projection is identical to within a pixel.
- **3D → SVG registration:** the projection `P \cdot V` of the ortho camera maps world points to NDC, then to the SVG viewBox (the same pixel size as the canvas). Polylines are projected once at the swap and reprojected per frame only for moving nodes (the pumping-unit parts and the fluid level).
- **Edge pass:** a post-processing effect (postprocessing library) with Sobel on normal + depth buffers into an ink-colored line mask. It is used only during the 700–1300 ms window to bridge the materials. Final lines are SVG.
- **Hatch dissolve:** a custom `ShaderMaterial` per stratum: `mix(textureSample, hatchPattern(worldPos, patternId), dissolve)`, where `dissolve` uses a noise threshold so the change looks like the pattern "growing" through the texture.
- **Paper wipe:** a full-screen CSS layer (`mask-image` with an animated radial gradient and a noise texture) above the canvas and below the UI. The canvas background is set to transparent after the wipe completes, to save GPU.
- **Orchestration:** a single **GSAP timeline** owns all tracks (camera, uniforms, CSS variables, SVG draw), so the whole thing can be seeked, reversed and time-scaled. Theatre.js is used in development to *author* the camera curve and easing visually, then exported as keyframes the GSAP timeline plays.
- **State continuity:** the physics state and time `t` don't change during the transition. Only presentation changes. Numbers that appear in both views are FLIP-animated between their DOM positions.

11.4 OTHER TRANSITIONS IN THE PRODUCT

- **Twin → any Paper page:** the same paper-wipe mask, but with no camera move, 600 ms.
- **Paper page → Paper page:** plates reshuffle (FLIP), grid stays, 240 ms.
- **Condition switch in the Twin:** a 1.2 s scene cross-fade (sky, strata materials, particles) with the relevant callouts showing a before/after tick.
- **Condition switch in the Blueprint:** changed plates get revision clouds. The title-block revision number increments.
- **Scenario fork:** the Twin splits into two views with a vertical "tear" line (the two halves slide apart 12 px, then settle).

11.5 NEXT.JS IMPLEMENTATION NOTES

- The R3F `<Canvas>` lives in the **root layout** (`SceneHost`, client-only via `next/dynamic` with `ssr: false`), so route changes never remount WebGL. The Twin ⇌ Blueprint toggle is a `?mode=` search-param change on the same route, with no navigation.
- The Blueprint SVG layer is a sibling client component positioned over the canvas. It subscribes to the same camera store, so the ortho projection matrix is shared.
- The paper-wipe between the Twin and other screens uses the View Transitions API where available (Next.js `unstable_ViewTransition` / `document.startViewTransition`), falling back to the CSS mask layer.
- Server components render the paper shell and the first data. The 3D and animated layers hydrate after. The intro sequence (§13.12) covers the gap.

12. THE CONDITION DECK: SWITCHING GROUND, CRUDE, CLIMATE, EQUIPMENT AND FAULTS

12.1 PRINCIPLE

A condition is a **named bundle of parameters that changes the physics and the picture together**. If a switch only changes the look, it doesn't belong here. It belongs to the visual settings. The Deck opens as a glass sheet in the Twin and as a paper index-card drawer elsewhere. Each card shows the physical parameters it sets, their provenance, and which outputs they affect.

Conditions compose: *Formation* × *Crude* × *Completion* × *Lift* × *Steam source* × *Climate* × *Fault*. **Presets** bundle interesting combinations for the demo.

12.2 GROUND: SURFACE VS FORMATION

Most of what "soil" suggests has no effect on the physics, so we split it into two layers.

Surface ground (visual and minor physics):

OPTION	VISUAL	PHYSICS EFFECT
Aeolian sand (default)	Ochre ripples	Sand ingress into the stuffing box raises polished-rod friction slightly (ASSUM). Surface flowline heat loss follows ambient temperature.
Gravel pad / caliche	Compacted, lighter	None beyond visuals
Wet (monsoon)	Darkened sand, puddles	Higher convective loss on bare surface lines. Pad access delay adds to maintenance time.

Formation (drives heat and flow):

OPTION	KEY PARAMETERS (INDICATIVE, ALL ASSUM UNLESS PUB)	WHAT IT CHANGES
Jodhpur Sandstone, clean (default)	$\varphi \approx 0.22$, $k \approx 800$ mD, $h_{\text{net}} \approx 18$ m, $pc \approx 2.4$ MJ/m ³ K	Baseline
Jodhpur Sandstone, shaly / tight	$k \approx 150$ mD, higher clay	Lower injectivity (longer injection for the same steam), lower J, steeper decline. The optimizer shifts toward smaller slugs and earlier cut-off.
Jodhpur Sandstone, high-perm channel	$k \approx 2,000$ mD	High injectivity, faster heat convection away, risk of steam breakthrough to offset wells (fleet effect)
Upper carbonate (heavy oil variant)	Lower φ , fractured, crude $\sim 38,000$ cP (a public OIL EOI figure, PUB)	Dramatic viscosity. Rod float binds early. Shows why CSS alone isn't enough.
Unconsolidated / sand-prone	Sand production index $\uparrow$	Plunger/barrel wear rate $\uparrow$, pump efficiency decay, sand-fill events

Each option re-textures the reservoir stratum (grain size, cross-bedding, fractures for the carbonate) and changes the hatch pattern on the Blueprint.

12.3 CRUDE AND FLUID

OPTION	PARAMETERS	EFFECT
API 19° (lighter end)	$\mu_{50} \approx 8,000$ cP	More forgiving envelope
API 17–18° (default)	$\mu_{50} \approx 10,000\text{--}13,000$ cP (OIL PUB)	Baseline
API 14° (heaviest wells)	$\mu_{50} \approx 15,000$ cP+	Tight envelope, float-prone
Water cut 10% / 30% / 60%	Emulsion multiplier (Richardson form, ASSUM)	Viscosity $\uparrow$ at moderate water cut, liquid load $\uparrow$, heat carried away $\uparrow$
High asphaltene	Onset T_{ao} $\uparrow$ (ASSUM)	Deposition band grows up the tubing. Hot-oiling recommendation appears.

Visual: crude darkness and sheen in the tubing, the particle color, and the barrel fluid.

12.4 CLIMATE AND TIME (THAR DESERT)

OPTION	SCENE	PHYSICS EFFECT
Summer noon (48 °C ambient)	Harsh overhead sun, strong heat haze, bleached sky	Lower surface heat loss. Motor derating at high ambient (efficiency $\downarrow$). Worker heat-stress index (social metric) $\uparrow$, so the value of each avoided site visit $\uparrow$.
Late-October dusk (default)	Golden light, long shadows	Nominal
Winter night (4 °C)	Stars, lit RTU, cold blue light	Surface flowline cooling $\uparrow$, so wellhead viscosity $\uparrow$ and flowline back-pressure $\uparrow$ (insulation value visible)
Sandstorm (andhi)	Dust fog, low visibility, particles	Telemetry packet loss (data-quality demo), sand ingress, deferred site visits

OPTION	SCENE	PHYSICS EFFECT
Monsoon shower	Wet surfaces, overcast	Access delays, higher surface losses

Time-of-day also drives the **solar steam availability** below.

12.5 COMPLETION AND LIFT EQUIPMENT

OPTION	EFFECT
VIT (default) vs bare tubing	Wellbore heat loss (Ramey U-value) and steam quality at the sandface. You can literally see the thermal lens change along the tubing.
Conventional beam unit (default) vs hydraulic long-stroke unit	Stroke length range (long-stroke reduces SPM for the same PD, which reduces float risk), speed-profile flexibility, efficiency
Rod string design A / B / C (taper, grade)	Buoyant weight, drag, Goodman margins
Plunger size 1¾" / 2¾" / 2¾"	PD, fluid load
VFD on / off	Stroke Shaper available or not

12.6 STEAM SOURCE (THE ENVIRONMENTAL LEVER)

OPTION	EFFECT
Gas-fired once-through steam generator (default)	~η 85% (ASSUM), CO ₂ from fuel combustion
Diesel/crude-fired	Higher CO ₂ /t steam, higher cost
Solar-thermal + gas hybrid	Daytime steam from solar and night top-up from gas. The optimizer prefers daytime injection and shifts cycle timing. CO ₂ /bbl drops sharply. Visual: a ghosted mirror field at the slab edge catches the sun.
Produced-water recycling on/off	Fresh-water use per tonne of steam

12.7 FAULTS (THE ANOMALY INJECTOR)

Inject one or more faults, then watch detection, diagnosis and response.

FAULT	WHAT THE SIMULATOR DOES	WHAT MANTLE SHOULD DO
Temperature sensor stuck / drifting	Frozen or ramping reading	Flag inconsistency with the EnKF innovation. Exclude the sensor. Pause optimization for that well. Explain.
Sudden cooling (early water breakthrough)	Faster T decline	Re-forecast. Tighten the SPM envelope. Advance the re-steam window.
Rod float event	Wave solver shows float	Alarm P1. Card classifier shows "rod float". Recommend kd ↑ / SPM ↓ .
Fluid pound	Fillage < 70%	Card diagnosis. Recommend SPM ↓ or timer (pump-off control).
Steam/gas interference	Gas in the barrel	Card diagnosis, recommend gas-anchor / pump-intake depth change
Pump unsetting	Hold-down margin < 1	Alarm. Workover recommendation.
Parted rod	Load collapses	Card flatline. P1 alarm. Production stops.
VFD fault	Frequency oscillation	Diagnose electrical vs mechanical
Steam pressure loss	Injection pressure drop	Heat-delivery estimate ↓ . Extend injection or accept a smaller slug.
Telemetry dropout (sandstorm)	Missing packets	Data-quality degrade. Virtual sensors take over (labelled EST).

12.8 PRESETS (FOR THE DEMO AND TRAINING)

- "Golden-hour baseline": default well, Cycle 6, Day 41.
- "Summer squeeze": 48 °C noon, API 16°, water cut 30%, day 55. Float margins collapse.
- "The carbonate problem": upper carbonate, 38,000 cP. Why CSS alone isn't enough.
- "Sun-powered steam": solar hybrid. Shows the CO₂/bbl and schedule shift.
- "Night of the sandstorm": telemetry loss plus a stuck temperature sensor. Shows safety behavior.
- "Bare tubing vs VIT": a side-by-side split twin.

13. FEATURE CATALOGUE BY MODULE

Priority tags: **P0** = the product doesn't work or answer the PS without it · **P1** = what makes it clearly better than competitors · **P2** = the "how did students build this" tier · **P3** = vision / post-hackathon. We intend to build P0–P2.

13.1 WELL SCREEN: TWIN AND BLUEPRINT

FEATURE	TIER	NOTES
3D block-diagram well, procedurally animated at the real SPM	P0	§7, §8
Attached callouts (≤ 7 , phase-aware) and glass side stack	P0	§9
Lenses: Physical / Thermal / Mechanical / Flow / Pressure / Risk	P0 (Thermal, Mechanical) · P1 (others)	§7.3
Phase-dependent scene (injection, soak, production, decline)	P0	§7.4
Blueprint sheet with depth tracks, 6 plates, BOM, notes, title block	P0	§10
Twin $\rightleftharpoons$ Blueprint transition	P0	§11. This is our signature, so treat it as core.
Dive camera along the wellbore with depth gauge	P1	
X-ray mode (translucent casing/tubing)	P1	
Split twin (Baseline vs Plan), synchronized	P1	
Stress waves visible on the rod string (from the wave equation)	P1	Slow-motion factor shown on screen
Drawing export (vector A3 PDF/SVG)	P1	
Scene light-tinted glass UI	P2	
Hydraulic long-stroke unit variant	P2	

13.2 FIELD

FEATURE	TIER	NOTES
Field map (cream cartographic style, pads, wells as glyphs showing phase and risk)	P0	Synthetic layout, labelled "illustrative". The location context is Bikaner district, Rajasthan.
Fleet table (sortable: opportunity, risk, SOR, energy/bbl, CO ₂ /bbl, phase, days to cut-off)	P0	
Field Triage + Morning Brief (ranked worklist with reasons; brief generated at 06:00 sim-time, exportable)	P1	Brief text is templated from tool results (with an optional LLM polish that passes through the verifier)
Field Steam Scheduler (Gantt across generators; CP-SAT allocation of steam slots to wells under capacity, crew and cut-off windows)	P1	Our biggest field-level economic feature
Cross-well similarity ("BGW-12 is tracking BGW-04's pre-failure trajectory"; nearest neighbours in trajectory-embedding space)	P2	DTW distance or a learned embedding
Well fingerprint (radar: thermal response, injectivity, cooling rate, float sensitivity, degradation per cycle)	P1	Derived from EnKF-calibrated parameters, which makes it real, not decorative
Opportunity budget ("where should the next ₹10 L of intervention go?"; a knapsack over candidate interventions)	P2	
Well Card (mobile page via QR at the wellhead: state, alarms, last decision)	P2	Social impact: field-crew access

13.3 RESERVOIR SHEET (CSS)

FEATURE	TIER	NOTES
Cycle timeline and history (all cycles: steam t, days, soak, peak rate, 60-d oil, SOR, heat efficiency)	P0	
Steam design panel: volume, rate, pressure, quality, soak. Coupled fields (duration = volume / rate, locked relations shown).	P0	Enforces physical coupling
Heat balance (steam energy → surface loss → wellbore loss → reservoir → caprock loss → produced heat), as a Sankey	P0	"Heat delivery efficiency", "Heat utilization"
Heated-zone growth and cooling ($r_h(t)$, $\bar{T}(t)$) with P10–P90	P0	
Cut-off and re-steam timing (marginal-value trigger, window with uncertainty, cost of waiting vs cost of early start)	P0	Explicit PS item
Marginal steam curve (Δ oil per extra 100 t, diminishing returns, economic stop point)	P1	"Steam ROI"
Cycle degradation model (response vs cycle number, per well; remaining economic cycles over the horizon)	P1	
Casing thermal-integrity gauge ($\sigma = E \cdot \alpha \cdot \Delta T$ vs yield; ramp-rate advice)	P1	
Solar-aware injection schedule (if the steam source is a hybrid: day/night steam mix, CO ₂ impact)	P2	

13.4 LIFT SHEET (SRP)

FEATURE	TIER	NOTES
Live surface and downhole dynamometer cards (Gibbs wave equation)	P0	
Card diagnosis library (normal, fluid pound, gas/steam interference, rod float, parted rod, TV leak, SV leak, pump tagging (hitting down), pump unsetting, over-travel)	P0 (5 classes) · P1 (all)	CNN on card image plus Fourier-descriptor GBM, ensembled
SPM safe envelope N_max(t) with actual vs plan	P0	The chart that explains the PS
Risk Stack (float, pound, unsetting, Goodman fatigue, casing thermal, deposition) as margins plus time-to-limit	P0	
Receding-horizon SPM/stroke/plunger schedule (daily re-solve)	P0	
Stroke Shaper (VFD in-stroke speed profile; downstroke share kd; polar diagram)	P1	Our lever. The optimizer can use it.
Bottleneck analysis (inflow vs pump capacity; reservoir-limited vs pump-limited)	P0	
Rod-string designer (taper sizes, grades; Goodman per section; buoyant weight vs drag)	P1	Hardware options enter the optimizer
Rod string depth map (click any depth: tension, compression, drag, temperature, fatigue index)	P1	
Maintenance forecast calendar (risk-ranked inspections; maintenance what-if: delay cost vs risk)	P1	
Pump-off control emulation (a baseline everyone in industry uses; we compare against it honestly)	P1	
Energy per stroke, motor loading, power factor	P1	

13.5 ARENA: SCENARIOS AND STRATEGIES

FEATURE	TIER	NOTES
Strategy Arena (9 strategies race on identical wells; \$16)	P0	The evidence backbone
Sandbox (change any lever or condition; simulate 30/90/180 d; result deltas)	P0	Coupled inputs enforced; infeasible settings are rejected with the reason
Battle (up to 6 strategies: Baseline, Max-oil, Min-energy, Min-SOR, Min-risk, Mantle; racing trajectories and cumulative curves)	P0	
Pareto explorer (3D/2D fronts over oil, SOR, energy, risk; brush to pick a region; the optimizer returns the plan)	P1	
Forks from the Timeline (branch at any past day; counterfactual replay)	P1	The Replay + counterfactual feature
Scenario library (save, clone, tag, share link, compare any two)	P1	
Challenge mode (the user operates the well for 60 sim-days vs Mantle; scored on oil, SOR, damage, CO ₂ ; leaderboard)	P2	Training simulator and a great stall/demo toy
Autopilot (simulation only): closed-loop Mantle runs the well in fast-forward	P1	Shows stability

13.6 LEDGER

FEATURE	TIER	NOTES
Recommendation queue (per well, priority, expiry)	P0	
Approve / Modify / Reject (modify re-simulates; reject requires a reason code + note)	P0	
Decision Ledger (hash-chained, append-only: inputs snapshot, model versions, plan, approvals, outcomes)	P0	Tamper-evident
Outcome scoring (predicted vs realized at T+1/7/30 d; calibration over time)	P1	
Counterfactual estimate per decision ("vs not acting": estimated benefit ± band from the EnKF ensemble)	P1	
Knowledge capture (structured override reasons become features and rules; "known well behaviours" library)	P1	
Override analytics ("the model is overridden 70% of the time when T < 60 °C on sandstone-B: possible miscalibration")	P2	
Recommendation provenance trace (plan → models → inputs → assumptions → evidence)	P1	

13.7 TIMELINE (GLOBAL OVERLAY)

FEATURE	TIER
Phase-banded strip for the current and previous cycles; events and decisions as pins	P0
Scrub past (replay the estimated state), now (live), future (P50 path with P10–P90 fan)	P0
Playback 0.25×–500× (sim-time), step by stroke / hour / day	P0
"If we do nothing" ghost trajectory vs plan	P0
Fork (creates a scenario branch at t)	P1
Horizons 24 h / 7 d / 30 d / 90 d / 180 d	P0

13.8 ASK MANTLE (COPILOT AND COMMAND PALETTE)

FEATURE	TIER	NOTES
Command palette (jump anywhere, run actions: "fork at day 30", "inject stuck sensor")	P0	
Tool-calling copilot grounded in twin tools (§18.7)	P1	
Numeric verifier (every number in an answer must trace to a tool result; otherwise the answer is blocked or rewritten)	P1	
Answer cards with "Show me" (highlights in the Twin, opens the plate)	P1	
Offline fallback (templated answers from the same tools when there is no network)	P1	Venue-proof

13.9 IMPACT

See §25. Economic, environmental and social ledgers vs baseline policies, per well and per field. P0 for economic and environmental, P1 for social.

13.10 PROOF

FEATURE	TIER
Provenance mode (tints every number by source)	P0
Assumption Registry (each constant: value, basis, "replace with...", status)	P0
Model cards (purpose, data, split, metrics, calibration, limits)	P0
Data import (CSV/Parquet/ZIP) + Data Quality Auditor (gaps, duplicates, impossible values, unit mismatch, drift)	P1
Twin Health (sensor integrity, calibration, freshness, OOD, physics–ML disagreement)	P1
SIH Traceability page (Appendix A, live links to the features)	P0
Real-data readiness diagram and adapter status (SIM mode ⇌ FIELD mode switch, disabled)	P1

13.11 STORY MODE

FEATURE	TIER
Deterministic scripted demo (seeded), keyboard stepping, presenter notes on a second screen	P0
Auto-pilot demo loop for the exhibition stall	P1
"Judge's questions" quick links (each opens the evidence for a common question)	P1

13.12 NEW IN REV B: CREATIVE FEATURES

FEATURE	WHAT IT IS	WHY IT EARNS ITS PLACE	TIER
Intro: "the drawing comes alive"	On first load, the Blueprint line drawing draws itself (masking GLB streaming), then runs the reverse transition into the 3D Twin	Turns loading time into the first wow. It also teaches the two views in 3 seconds.	P1
Thermal Battery	A battery gauge for remaining useful reservoir heat (% and days to cut-off), computed from the heat balance	The core metaphor as UI. Instantly understood.	P0
Ghost Well	In compare mode, a translucent cyan "ghost" pumping unit runs the alternative plan next to the real one and visibly drifts out of phase	Counterfactuals you can see	P1
Strategy Race	Animated cumulative-oil bar race and synchronised production curves in the Arena	The evidence becomes a spectacle	P0
Judge Challenge (/play)	60-second game: a judge operates the well against Mantle, with a leaderboard on the stall screen	Engagement, and judges feel the problem is hard	P1
Presenter remote (/present)	Phone as clicker plus notes for Story Mode	Smooth demo	P1
Cycle receipt	A printable, receipt-styled summary of each cycle's steam, fuel, water, power, oil and CO ₂ e vs baseline	Makes the environmental impact tangible	P1
Deep-link snapshots	The URL encodes well,  , scenario, lens, camera and open sheet, so any exact view can be shared	Proof buttons, Story Mode and bug reports all use it	P0
"Since you last looked"	On return, a small digest of changes (alarms, decisions, forecast shifts)	Operators' real need	P2
Shift handover note	Auto-generated one-page PDF of the shift: state, changes, open decisions	Operational realism, and a social-impact talking point	P2
Black-box recorder	The 10 minutes around any P1/P2 event are auto-saved as a replayable clip (state + cards)	Incident forensics	P1
Explain-by-contrast	"Why this plan instead of X?" shows the delta in drivers and constraints between two plans	Better than generic explanations	P2
Cycle time-lapse export	Renders the Twin through a whole cycle to WebM (for reports and the pitch video)	Makes the pitch video free	P2

FEATURE	WHAT IT IS	WHY IT EARNS ITS PLACE	TIER
Units toggle	SI ↔ oilfield units (bbl/m³, bar/psi, °C/°F)	Engineers will ask	P1
Field "breathing" time-lapse	The field map animates a year: wells glow as they're steamed and fade as they cool	One of the prettiest views of CSS	P2
Blueprint revision flip	Flip through the drawing's revision history like a stack of sheets	Shows change over time in the Paper idiom	P2
Sonification (off by default)	Soft rhythmic pump sound; a distinct click on impact loading	"Hear the rod float" moment in the demo	P3

14. PHYSICS ENGINE

One Python package, `mantle.physics`, is the **only** calculation path. Forecasts, the optimizer, the copilot and exports all call it. The frontend never re-implements physics. The only exception is pure geometry (pumping-unit kinematics for rendering). Each relation below lists its source standard or paper. Everything tagged ASSUM goes into the Assumption Registry.

14.1 STEAM AND SURFACE LINES

- **Steam properties:** IAPWS-IF97 (the `iapws` or `CoolProp` library). Wet-steam enthalpy $h = h_f + x \cdot h_{fg}$. Heat injected $Q = m \cdot (h - h_{fw})$.
- **Surface line loss:** insulated-pipe conduction plus external convection to ambient, where ambient comes from the Condition Deck. Steam quality drops along the line.
- **Steam-generator fuel and CO₂:** $\text{fuel} = Q_{\text{steam}} / \eta_{\text{gen}}$. CO₂ = fuel energy × emission factor (natural gas ≈ 56 kg CO₂/GJ, IPCC default; ASSUM until OIL data). For the solar hybrid, the solar fraction comes from a time-of-day DNI profile (ASSUM, typical western Rajasthan).

14.2 WELLBORE HEAT TRANSFER (INJECTION AND PRODUCTION)

- **Ramey (1962)** transient wellbore heat loss with the **Willhite (1967)** overall heat-transfer coefficient U. **VIT** is modelled via an apparent thermal conductivity (ASSUM range 0.006–0.06 W/m·K).
- Outputs: steam quality and temperature at the sandface (injection), and the fluid temperature profile T(z) during production (which feeds viscosity along the tubing, rod drag per depth and the deposition band).
- **Casing thermal stress:** fully constrained $\sigma = E \cdot \alpha \cdot \Delta T$ ($E \approx 207$ GPa, $\alpha \approx 12.4 \times 10^{-6}$ /°C, so ~2.6 MPa per °C of ΔT), compared with casing grade yield. This drives the ramp-rate advice.

14.3 RESERVOIR THERMAL MODEL

- **Injection: Marx–Langenheim (1959)** heated area $A(t) = (\dot{Q} \cdot M \cdot h) / (4 \cdot k_{ob} \cdot M_{ob} \cdot \Delta T) \cdot G(t_D)$, $G(t_D) = e^{\{t_D\}} \cdot \text{erfc}(\sqrt{t_D}) + 2\sqrt{t_D/\pi} - 1$. Heated radius $r_h = \sqrt{A/\pi}$.
- **Soak and production:** an **axisymmetric r–z finite-volume conduction–convection model** (a 256 × 64 grid). It includes conduction to the over- and underburden, convective heat removal with produced fluids and the initial condition from the injection stage. It is solved implicitly (SciPy sparse) and runs in < 50 ms per simulated day. **This grid is what the 3D Thermal lens and the Blueprint isotherms render.**
- A fast **lumped fallback** (Boberg–Lantz average-temperature form) is used inside the optimizer's inner loop, with the grid model used to calibrate it.

14.4 FLUID PROPERTIES

- **Viscosity–temperature:** **Walther / ASTM D341**, $\log_{10} \log_{10}(v + 0.7) = A - B \cdot \log_{10} T$, fitted per well to anchor points (PUB: $\mu \approx 10,000$ – $13,000$ cP at 50 °C; ASSUM high-temperature anchor). **Andrade** ($\ln \mu$ linear in $1/T$) is the cross-check.
- **Density:** API gravity → SG = 141.5 / (API + 131.5), with a thermal-expansion correction.
- **Emulsion:** effective viscosity $\mu_e = \mu_o \cdot \exp(k \cdot \phi_w)$ (Richardson form, k ASSUM) up to the inversion point.
- **Asphaltene / wax deposition proxy:** deposition risk where $T(z) < T_{\text{onset}}$ (ASSUM, per well) weighted by flow velocity (low shear → more deposition). Output: a risk band (m) and a hot-oiling or VIT recommendation.

14.5 INFLOW

- **Boberg–Lantz (1966)** stimulated productivity: $J_{\text{hot}}/J_{\text{cold}} = \ln(r_e/r_w) / [(\mu_h/\mu_c) \cdot \ln(r_h/r_w) + \ln(r_e/r_h)]$.
- **Composite radial Darcy** with skin and relative permeability. **Vogel (1968)** below the bubble point is optional (low gas).
- **Depletion:** material balance on drained pore volume, which feeds cycle-over-cycle degradation.
- **Production = min(inflow at the pump-intake drawdown, pump capacity × fillage).** The bottleneck analysis comes from which term binds.

14.6 WELLBORE HYDRAULICS

- The fluid gradient (oil + water, temperature-dependent density) sets the **pump intake pressure, Pwf, dynamic fluid level** and **submergence**.
- Tubing friction uses viscosity along depth (laminar in heavy oil, Hagen–Poiseuille in the annulus). It contributes to the upstroke load.
- Fluid level evolves as a **material balance in the annulus**: $dV/dt = q_{\text{inflow}} - q_{\text{pump}}$. This is what makes pump-off and fluid pound emerge naturally.

14.7 SRP MECHANICS

- **Pumping-unit kinematics:** exact **four-bar linkage** for conventional units (API Spec 11E geometry). The polished-rod position, velocity and acceleration come from crank angle $\theta(t)$. **VFD speed profile** $\dot{\theta}(\theta)$ implements the **Stroke Shaper** (a downstroke share k_d from 0.5 to 0.7, smooth profile).
- **Rod string dynamics:** the **Gibbs (1963) damped wave equation** $\partial^2 u / \partial t^2 = a^2 \cdot \partial^2 u / \partial z^2 - c \cdot \partial u / \partial t$ with $a = \sqrt{E/\rho} \approx 5,000$ m/s. Solved by finite differences (Numba JIT, CFL-stable) per taper section, with **viscous drag per section** (Couette annulus, below) and the pump boundary condition (valve logic, fillage, gas compression). Outputs: surface and downhole cards, stress $\sigma(z,t)$, displacement $u(z,t)$ (drives the 3D rod segments) and impact spikes.
- **Downhole card from a surface card** (for measured data): Gibbs / **Everitt–Jennings** inversion. This is used when real cards arrive.
- **Fast analytical layer** (used in the optimizer inner loop): **API RP 11L** plus **Mills acceleration** $\alpha = S \cdot N^2 / 70,500$:
 - $PPRL = W_{rf} + F_o + W_r \cdot \alpha + F_{drag}$, $MPRL = W_{rf} - W_r \cdot \alpha - F_{drag}$
 - $W_{rf} = W_r \cdot (1 - 0.128 \cdot G)$, $F_o = 0.340 \cdot G \cdot D^2 \cdot H$, $PD = 0.1166 \cdot S \cdot N \cdot D^2$ (bbl/d)
- **Viscous rod drag (Couette):** $F = 2\pi \cdot \mu(z) \cdot v \cdot L / \ln(D_t/D_r)$, integrated over depth with the $\mu(z)$ profile, where v is the rod velocity.
- **Rod-float criterion:** the rods follow only if buoyant weight > drag + friction. **Float margin** $m_f = 1 - (F_{drag} + F_{fric}) / W_b$ (at peak downstroke velocity). **Safe downstroke velocity** $v_{max} = (1 - margin_{req}) \cdot W_b / C_{drag}$, and the **SPM ceiling** $N_{max} = 120 \cdot k_d \cdot v_{max} / (\pi \cdot S)$, corrected by a factor fitted to the wave solver. *This is the envelope on Plate 3.*
- **Fatigue: Modified Goodman** per taper: $S_{a,allow} = (T/4 + 0.5625 \cdot S_{min}) \cdot SF$, and the ratio $S_{range} / S_{a,allow}$ is the Goodman loading.
- **Gearbox torque:** net torque from the torque factor $\times$ (polished-rod load – counterbalance effect). The peak goes against the API 11E rating.
- **Pump unsetting:** hold-down margin = hold-down rating / (upstroke plunger friction + fluid-load transient). Plunger friction rises with μ at the pump (ASSUM correlation).
- **Motor and energy:** polished-rod HP from card area $\times$ SPM. Motor input = PRHP / ($\eta_{surface} \cdot \eta_{motor}(load)$). Energy per barrel follows.

14.8 CARD DIAGNOSIS PHYSICS (TRAINING DATA FOR THE CLASSIFIER)

The wave solver generates cards for each fault class by changing boundary conditions: incomplete fillage (pound), gas compression (interference), float (lagged downstroke), a parted rod (load collapse), valve leaks (leak term), tagging (bottom contact), unsetting (hold-down release). Noise, sensor offset and card shape augmentations make the classifier robust.

14.9 CYCLE ECONOMICS AND THE CUT-OFF RULE

- **Daily margin:** $m(t) = \text{oil} \cdot \text{price} - \text{steam cost (amortized)} - \text{power} - \text{water} - \text{maintenance hazard cost} - \text{carbon cost (optional)}$.
- **Cut-off by the marginal value theorem:** end production when $m(t)$ falls below the **best achievable average margin per day of a fresh cycle, including injection and soak downtime**. Take the argmax over the next cycle's design. This makes the steam plan and the pump plan one decision.
- **SOR** in cold-water-equivalent barrels: $SOR = (\text{steam } t \times 6.2898) / \text{oil bbl}$, reported per cycle and cumulatively. Instantaneous SOR (iSOR) is also available.

14.10 NUMERICAL HYGIENE

SI units internally, with `pint` at the API boundary. Every function carries a docstring citation and a unit test against a published worked example (API RP 11L examples, Marx–Langenheim tables, IAPWS check values). Property-based tests (Hypothesis) check the invariants: T never rises during soak without a source, viscosity is monotone in T, PD is non-negative, and energy balance closes within 1%.

15. LEARNING, ESTIMATION AND OPTIMIZATION LAYER

Physics first, ML where it earns its place: **calibrating**, **correcting residuals**, **being fast** and **recognizing patterns**. One engine per job.

15.0 DO WE HAVE AN ML MODEL? AN RL MODEL? YES, AND HERE IS THE EXACT INVENTORY

Mantle is **physics-first and ML-everywhere-it-helps**. The primary controller is model-predictive control (MPC) on the physics twin. ML makes it calibrated, fast, perceptive and honest about uncertainty. RL is present as a serious contender and an optional proposal generator, never as an unshielded controller.

#	MODEL	KIND	INPUTS → OUTPUTS	TRAINED ON	USED IN	VISIBLE WHERE
M1	EnKF state estimator	Bayesian data assimilation (64–128 member ensemble)	telemetry → hidden state, parameters, P10–P90	online, per well	everything downstream	Twin bands, virtual sensors, Proof → Twin health
M2	Cycle surrogate	LightGBM + MLP ensemble, conformal wrapper	well params + plan + conditions → oil, SOR, energy, risk, value	~200k simulated cycles	optimiser pre-screen	Arena (Pareto), Proof → Model cards
M3	Residual learner	LightGBM on physics residuals	state → correction	residuals (synthetic model-form error now, real later)	forecasts	Proof → "physics vs ML"
M4	Dynocard classifier	1D-CNN + Fourier-descriptor GBM ensemble, calibrated	card → 10 classes	~500k wave-equation cards + augmentations; Polaris MIT cards as external test	alarms, diagnosis	Plate 2 label; Lift sheet; Alerts
M5	Failure hazard	Discrete-time survival GBM	loading history → P(rod failure / unsettling in 7/30 d)	hazard-driven synthetic failures	risk margins, maintenance	Risk Peek; Maintenance tab
M6	Anomaly detector	EnKF innovation tests + Isolation Forest + physics–ML disagreement	windows → anomaly score + type	synthetic + Petrobras 3W real data (validation)	Safety pause	Alerts; Proof
M7	Production forecaster (P2)	Temporal Fusion Transformer	daily history + known steam schedule → quantiles	synthetic history	comparison vs EnKF forecast	Proof → Model cards
M8	Shielded RL policy	PPO (Stable-Baselines3) in a Gymnasium env; actions filtered by the Safety Gate	observation → Δ SPM, kd, re-steam trigger	20–50 M simulated steps, domain-randomised	Arena contender ; optional proposal generator	Arena (purple line)
M9	Copilot LLM	Tool-calling LLM + numeric verifier	question → grounded answer	pre-trained (API)	Ask Mantle	⌘K

The RL environment (`packages/r1`, Gymnasium API):

- **Observation (daily):** EnKF posterior summary (sandface T, μ at the pump, r_h , fluid level, fillage, float margin, Goodman ratio), cycle day, phase, days since steam, recent oil, SPM, kd, economics prices, and condition flags.
- **Action:** Δ SPM $\in [-0.5, +0.5]$, kd $\in [0.5, 0.7]$, `resteam` $\in \{0, 1\}$ (only valid in production). A discrete-continuous hybrid is handled via a parameterised action head or a discretised grid.
- **Reward:** daily risk-adjusted margin (oil value – power – steam amortisation – expected failure cost) – $\lambda \cdot$ (constraint-violation magnitude).
- **Shield:** actions pass through the same **Safety Gate** as Mantle. Unsafe actions are projected to the nearest safe action and counted (reported in the Arena).
- **Training:** domain randomisation over wells and conditions; curriculum from easy (clean sandstone) to hard (carbonate, summer); 5 training seeds; evaluation on held-out wells.

Why MPC is primary and RL is a contender. MPC is explainable (it names its binding constraint), sample-efficient (it needs no millions of well-days) and safe by construction. It also adapts instantly to new conditions via the EnKF. RL shows what a black-box learner can do. If it ever beats MPC in the Arena, we can promote it as a *proposal generator* whose suggestions still pass through physics verification and the Safety Gate. **Either way, the Arena decides, in public.**

15.1 STATE ESTIMATION: THE ENSEMBLE KALMAN FILTER (THE HEART OF "DIGITAL TWIN")

- **Ensemble:** 64–128 members of the coupled physics model, each with slightly different uncertain parameters.
- **State and parameters (augmented):** heated-zone temperature field (reduced to 6 modes), r_h , a viscosity-curve multiplier, J (productivity) multiplier, skin, fluid level, rod-friction coefficient, pump-leak coefficient, and the heat-loss multiplier.
- **Observations:** wellhead temperature and pressure, oil and water rate (test-separator or tank-dip cadence), motor power, SPM, card features (PPRL, MPRL, area, fillage), echometer fluid level (when available).
- **Update cadence:** hourly for fast states and daily for parameters. Inflation and localization are applied.
- **Outputs used everywhere:**

1. P10/P50/P90 for any forecast (run the ensemble forward),
2. **virtual sensors** (downhole viscosity, fillage, heated radius, rod stress at depth) tagged EST with their spread,
3. **innovation checks** for sensor-fault detection (a reading inconsistent with the ensemble is suspicious),
4. **well fingerprints** (posterior parameters),
5. **counterfactuals** (re-run the posterior ensemble under the alternative action).

15.2 SURROGATES (SPEED FOR THE OPTIMIZER)

- **Cycle surrogate:** a gradient-boosted (LightGBM) and small-MLP ensemble that maps (well parameters, steam design, SRP schedule, conditions) → cycle outcomes (oil, SOR, energy, max risk, net value). It is trained on ~200k simulated cycles from Baghewala-S. It is used to pre-screen thousands of candidates, and **the top-k are re-verified with full physics** before any recommendation. The surrogate never has the last word.
- **Residual learner:** LightGBM on physics residuals $y_{\text{obs}} - y_{\text{phys}}$ (per well, pooled with well-ID effects). It absorbs systematic model error once real data exists. On synthetic data it is trained with deliberate "model-form" perturbations so the machinery is demonstrable.

15.3 PATTERN MODELS

- **Card classifier:** a 1D-CNN on the resampled (position, load) loop plus **Fourier shape descriptors** into LightGBM, ensembled, with calibrated probabilities (temperature scaling). The classes are the §13.4 library.
- **Failure hazard:** discrete-time survival model (gradient-boosted hazard) for rod failure and pump unsetting within 7/30 d. Inputs: cumulative Goodman-loading cycles, float events, impact counts, sand index, card-class history.
- **Anomaly detection:** (a) EnKF innovation tests, (b) Isolation Forest on multivariate telemetry windows, (c) physics-ML disagreement. Alerts are classified by type (sensor / thermal / mechanical / hydraulic / electrical).
- **Forecasting (P2):** a Temporal Fusion Transformer on daily production with the steam schedule as a known future input. It is compared with the EnKF forecast and reported honestly.

15.4 JOINT OPTIMIZER

- **Decision variables:**
 - **CSS:** steam mass, injection rate (→ duration), injection pressure, quality (bounded by the generator), soak duration, production cut-off, next-cycle start.
 - **SRP:** SPM(t) schedule (piecewise per 3–5 days), stroke length, downstroke share $k_d(t)$, plunger size (discrete), rod design (discrete, optional), pump-off timer.
- **Objectives (multi-objective):** maximize risk-adjusted net value · minimize SOR · minimize energy/bbl · minimize $\text{CO}_2\text{e/bbl}$ · minimize peak risk. Or scalarize with operator-set weights (the "objective sliders").
- **Hard constraints (the Safety Gate, deterministic and never learned):** float margin $\geq m_{\text{req}}$ at every step · Goodman ≤ 1.0 (× SF) · gearbox torque $\leq$ rating · PPRL $\leq$ structure rating · casing thermal stress $\leq$ yield/SF · injection pressure $\leq$ fracture limit (ASSUM) · SPM, stroke and frequency within equipment range · steam availability (from the Field Scheduler).
- **Method:**
 1. **NSGA-II/III (pymoo)** over the surrogate for the Pareto front (~5k evaluations in seconds),
 2. **full-physics verification** of the Pareto set,
 3. the **Safety Gate** filters, and each rejection is recorded with its reason,
 4. the chosen point is refined with **CMA-ES / Nelder–Mead** on full physics. If nothing is feasible, the output is **"INFEASIBLE"** with the binding constraints named. We never substitute a fallback answer (a flaw found in a competitor).
- **Receding horizon (MPC-style):** each sim-day, re-solve the SPM/ k_d schedule for the next 14 days with the updated EnKF state. Only the first step is "recommended". This is continuous optimization in the literal sense.

15.5 STROKE SHAPER OPTIMIZATION

For each day, choose k_d and the smooth speed profile that maximize production (upstroke-speed dependent) subject to the downstroke float margin and gearbox torque (speed changes add inertial torque). The result is shown as a polar speed diagram. The gain appears as the "Stroke Shaper contribution" in the Coupling Dividend breakdown.

15.6 THE COUPLING DIVIDEND (HEADLINE METRIC, DEFINED)

Let $V(c, s)$ be the risk-adjusted net value over the horizon for CSS design c and SRP policy s .

- **Current practice (sequential):** $c_{\text{seq}} = \text{argmax}_c V(c, s_{\text{hist}})$, then $s_{\text{seq}} = \text{argmax}_s V(c_{\text{seq}}, s)$ (SRP tuned afterwards, reactively).
- **Joint:** $(c^*, s^*) = \text{argmax}_{\{c,s\}} V(c, s)$.

- **Coupling Dividend** $CD = V(c^*, s^*) - V(c_{seq}, s_{seq})$, reported in ₹/cycle, bbl/cycle and ΔSOR, with P10–P90 from the ensemble.
- **Decomposition (Shapley over the two decision groups):** $CD = \text{CSS-side gain} + \text{SRP-side gain} + \text{interaction gain}$. The interaction share is the literal answer to "why must CSS and SRP be optimized together".

It is shown on the Twin (chip), in the Arena (Battle row) and in Impact (field total).

15.7 FIELD STEAM SCHEDULER

- **Problem:** G generators with capacity (t/day) and N wells, each with a candidate cycle design and a feasible start window (from the cut-off rule), plus crew and rig constraints and an optional solar-availability profile.
- **Solver:** OR-Tools **CP-SAT** (interval variables per injection, cumulative capacity constraints). The objective is field risk-adjusted value over 180 d. It re-solves when any well's window moves.
- **Output:** a Gantt chart, per-well start dates, the value of an extra generator-day (shadow price), and the "what if generator 2 is down for 10 days".

15.8 UNCERTAINTY, CALIBRATION AND TRUST

- **Conformal prediction** (MAPIE) wraps the surrogates and ML models with coverage-guaranteed intervals (on the synthetic distribution). **Coverage is reported** on the Proof screen.
- **OOD detection:** Mahalanobis distance in standardized feature space plus the ensemble spread. Beyond the threshold, the recommendation is downgraded to "advisory: outside validated range".
- **Physics–ML disagreement:** $|ML - physics| / physics > 15\%$ raises a flag, shows both values, and the recommendation requires an engineer's reason to approve.
- **SHAP** for the ML models. **Local physics sensitivities** (finite differences on the full model) for physics outputs. Both feed **Explain**.
- **Safe Calibration Probe (P2):** candidate small setpoint changes (e.g. ± 0.3 SPM for 24 h) scored by **expected information gain** on the most uncertain parameters (ensemble variance reduction) minus the value lost. The best one is proposed as an optional "learning action".

16. STRATEGY ARENA: CONCRETE EVIDENCE THAT MANTLE WINS

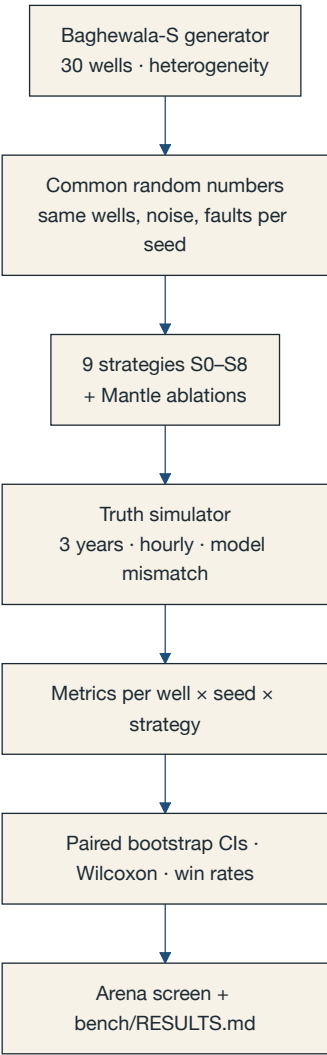
Judges believe what they can see. The Arena puts every reasonable way of running these wells on the same simulated field, under the same conditions, and **plots their production output side by side**.

16.1 THE CONTENDERS

ID	STRATEGY	DEFINITION	WHAT IT REPRESENTS	WHY INCLUDE IT
S0	Static (historical practice)	Fixed steam slug (e.g. 800 t), fixed soak (5 d), fixed SPM for the whole cycle, fixed cycle length (120 d). Reactive SPM cut only after a failure.	Today's practice described in the PS	The baseline
S1	Tuned static	The same fixed rules, but constants grid-searched per well on training seeds	"Just tune the constants"	Shows the gains aren't mere tuning
S2	Heuristic playbook	Operator rules: SPM −0.5 if fillage < 70% or a float alarm; +0.3 if fillage > 95%; re-steam when rate < 15 BOPD	Experienced-operator reactive control	Realistic human baseline
S3	Pump-off controller	Industry-standard POC: idle when fillage drops, fixed steam plan	Current industry automation	The honest baseline for energy
S4	Greedy (myopic)	Each day, choose the SPM that maximises <i>today's</i> production within hard limits; re-steam when today's margin < 0	Short-term maximiser	Shows why myopia burns heat and rods
S5	Sequential optimum (silo)	Optimise CSS assuming nominal SRP, then optimise SRP given that CSS	"Optimised separately", exactly the PS's complaint	Baseline for the Coupling Dividend
S6	Shielded RL (PPO)	M8 above	Learned black-box control	The inevitable judge question
S7	Mantle (joint MPC)	EnKF + joint optimiser + receding horizon + Stroke Shaper + Safety Gate	Our system	—
S8	Oracle	Hindsight optimum with true parameters and perfect forecasts (long-horizon optimisation on the true simulator)	The ceiling	Report Mantle as % of oracle

Ablations of Mantle: no EnKF (open-loop), no Stroke Shaper, sequential instead of joint, no Safety Gate (disqualified if any violation), surrogate-only (no physics verification).

16.2 PROTOCOL (FAIR BY CONSTRUCTION)



- **Same information:** every strategy except the oracle sees only the noisy telemetry. Mantle and RL go through the same estimator interface. Static and heuristic strategies see raw telemetry, as today.
- **Same constraints:** hard limits are enforced by the truth simulator for everyone (failures happen). The Safety Gate is part of a strategy, not the environment.
- **Same budget:** each strategy's tuning uses only training seeds. Evaluation uses held-out seeds **and held-out wells**.
- **Scale:** 30 wells × 20 seeds × 3 years (≈ 18 CSS cycles per well per seed). Stress sets use the Condition Deck presets (§12.8).
- **Model mismatch:** the truth simulator uses parameters the strategies don't know, and includes model-form perturbations (e.g. a different viscosity-curve shape), so Mantle's physics is not a copy of the truth.

16.3 METRICS

GROUP	METRICS
Production	cumulative oil; oil rate over time; recovery over horizon
Economics	net risk-adjusted value; lifting cost ₹/bbl
Efficiency	SOR (CWE); kWh/bbl; heat utilisation
Environment	kg CO ₂ e/bbl; fresh water m ³ /bbl
Reliability	rod-failure count; pump-unsetting count; float hours; impact events; Goodman damage; downtime
Safety	constraint violations (must be 0 for gated strategies)
Relative	% of oracle; Coupling Dividend (S7 – S5)

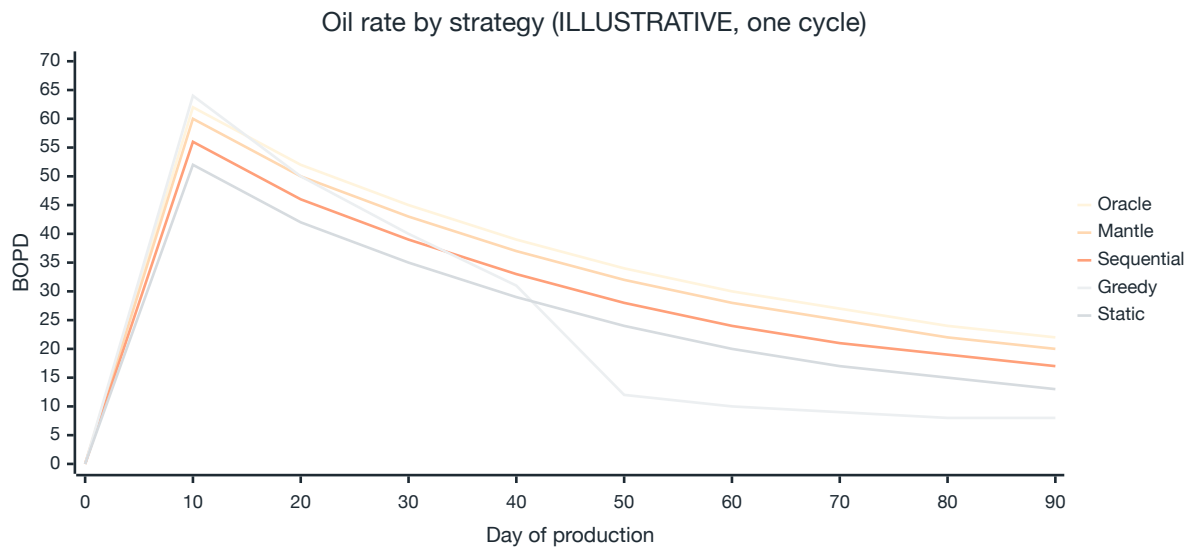
Statistics: per-well paired differences vs S0 with 95% paired-bootstrap CIs; Wilcoxon signed-rank; win rate (share of well × seed pairs where Mantle > contender); effect sizes. Everything is reported with N.

16.4 VISUALS (WHAT JUDGES SEE)

1. **Production output graph (hero):** oil rate vs day over 3 cycles, one line per strategy, seed bands on hover, failure markers. This is the graph asked for, and it's the first thing on the Arena screen.

- 2. **Cumulative oil race:** an animated bar race.
- 3. **Scoreboard:** deltas with CIs and % of oracle.
- 4. **Value vs risk scatter:** each dot = well × seed × strategy.
- 5. **Coupling Dividend decomposition** and **ablation bars**.
- 6. **Battle replay:** split-screen twins of any two strategies, with the ghost well.

Illustrative shape of the hero chart (placeholder values; the real chart is generated from `bench/` runs):



16.5 WHAT WE MUST SHOW TO CLAIM "MANTLE WINS"

CLAIM	PASS CRITERION
Mantle beats today's practice	Net value vs S0: CI lower bound > 0; win rate ≥ 80% of well × seed pairs
Mantle beats the best simple baselines	Net value > max(S1–S4) with CI lower bound > 0
Joint beats silo	Coupling Dividend (S7 – S5) > 0, CI excludes 0
Safer, not just richer	Failures and float hours ≤ S3; zero constraint violations
Close to the ceiling	% of oracle reported (target ≥ 85%)
Robust	Holds on ≥ 4 of 6 condition presets; failures explained honestly where it doesn't

If a criterion fails, the Arena says so. That honesty is itself a strong argument in front of technical judges.

16.6 LIVE AND PRE-COMPUTED RUNS

- **Official run:** pre-computed in `bench/`, versioned (seed manifest, code hash), and loaded instantly by the Arena.
- **Live mini-run:** 3 wells × 3 seeds × 1 cycle, run in front of judges (< 30 s), streaming progress ("S4 greedy: rod failure on BGW-07 day 58").
- **CI regression:** a small arena runs in CI. Mantle's margin over S0 must stay within a tolerance band, or the build fails (\$23).

17. DATA STRATEGY AND DATASETS

17.1 THE SITUATION

The PS says OIL has "sufficient historical and operational data", but **no dataset is attached to the public listing**, and **no public well-level Baghewala telemetry exists**. Every public competitor uses synthetic data. We will too, but we will do it better and say so more clearly:

Physics-based synthetic field (Baghewala-S) → calibrated to public anchors → real-data-ready ingestion → EnKF calibration the day real data arrives.

If OIL shares data during the hackathon (ask the nodal centre/mentor on day 1), the canonical schema (§17.3) takes it with no code change.

17.2 PUBLIC ANCHORS (FOR CALIBRATION AND CREDIBILITY)

ANCHOR	VALUE	SOURCE (VERIFY BEFORE CITING ON SLIDES)
API gravity	17–19°	SIH26120 statement
Reservoir temperature	46–48 °C	SIH26120 statement
Reservoir	Jodhpur Sandstone, ~1,150 m	Oil India, Rajasthan fields page
Dead-oil viscosity	~10,000–13,000 cP at 50 °C	Oil India, Rajasthan fields page
Steam injection phase	~14–21 days, injectivity-dependent	Oil India, Rajasthan fields page
Lift	Conventional + hydraulic SRPs, thermal wellheads, VIT	Oil India, Rajasthan fields page
Pump setting	~1,200–1,300 m production depths	OIL procurement documents (cited in the ChatGPT thread)
Upper carbonate crude	~38,000 cP (one reported figure)	OIL EOI 011/2022 (per a competitor's source index)
Thermal connections	ISO/PAS 12835 qualification, 320 °C steam class	OIL Drilling EOI 2025-26 (per a competitor's source index)
Field output milestone	~1,202 BOPD across ~33 wells (Apr 2026)	OIL announcement (per competitor READMEs; verify)
Geology	Baghewala characterization (stratigraphy, rheology)	ACT SHARP consortium report D4.1 (2022) (per a competitor's source index)
Technical papers	SPE papers on Baghewala CSS (e.g. SPE APOG 2023)	Referenced by competitors; obtain and read

Action: one team member owns "source truth". They obtain each primary document, confirm the numbers, and file them in `evidence/sources/` with page references. The app's PUB tags link to these.

17.3 CANONICAL SCHEMA (THE CONTRACT WITH REAL DATA)

Designed to map one-to-one onto OIL's seven data categories in the PS. Tables are TimescaleDB hypertables where time-series.

TABLE	GRAIN	KEY COLUMNS	OIL CATEGORY
wells	well	id, pad, lat/lon (illustrative), spud date, TD, perf top/bottom, formation variant, API, μ anchors	Well completion & reservoir
completions	well × version	casing program, tubing type (VIT/bare), anchor depth, pump depth, pump type/size, unit type/size	Well completion
rod_strings	well × version × section	size, grade, length, weight, guides	Well completion
pvt_lab	sample	T, μ , ρ , water cut, asphaltene onset (if any)	Fluid properties
css_cycles	well × cycle	start/end of injection, soak, production; steam t; avg pressure, quality; cut-off reason	CSS cycle records
steam_readings	well × time	rate, pressure, temperature, quality (est.)	Steam injection parameters
production_readings	well × time	oil, water, gas, wellhead P/T, test flag	Production history
srp_readings	well × time	SPM, stroke, VFD Hz, motor kW/A, PPRL/MPRL, fillage, runtime	VFD and SRP operating data
dyno_cards	well × time	surface card points (JSON/array), downhole card (derived), class, confidence	VFD and SRP data
events	well × time	failure type (rod part, unsetting, float, pound), workover, downtime h, cost	Rod failure & pump unsetting history
pressure_surveys	well × time	echometer fluid level, BHP surveys	Pressure data
conditions	well × time	active Condition Deck bundle (for synthetic runs)	—

TABLE	GRAIN	KEY COLUMNS	OIL CATEGORY
twin_state	well × time	EnKF posterior summary (means, spreads)	—
forecasts	well × issue time × horizon	P10/P50/P90 per variable	—
recommendations, decisions	id	plan JSON, gate results, approvals, hash, prev_hash	—
provenance	value ref	source type (MEAS/EST/SIM/ASSUM/PUB), source id, model version	—

The full DDL is in §20. The schema is published as JSON Schema + Parquet specs, with a data dictionary (units, ranges, required/optional).

17.4 BAGHEWALA-S: THE SYNTHETIC FIELD GENERATOR

- **Field:** 30 wells on 8 pads, heterogeneity sampled per well (formation variant, k, h, API/μ curve, completion, unit size, rod design, cycle count 1–12).
- **Span:** 3 years, hourly SRP/production, per-stroke cards sampled every 30 min, daily tests, CSS cycles with realistic scheduling.
- **Three policies simulated on identical wells and seeds:**
 1. **Historical practice** (fixed steam slug, fixed SPM, reactive changes after alarms),
 2. **Historical + pump-off controller** (the honest industry baseline for energy),
 3. **Mantle** (closed loop). All impact claims are **policy-vs-policy on the same simulated wells, mean ± sd over seeds**.
- **Realism layers:** sensor noise (per instrument class), drift, stuck sensors, packet loss (sandstorm days), unit conversions gone wrong (deliberately, for the auditor), operator overrides, maintenance crews with limited availability, steam-generator downtime.
- **Failures:** hazard-based (not scripted) from cumulative Goodman cycles, float/impact counts and sand, so failure *learning* is possible.
- **Outputs:** the canonical schema (Parquet + DB), a data card, and a seed manifest. Everything is reproducible with one command.

17.5 COMPETITOR DATASETS (WHAT'S PUBLIC, AND HOW WE MAY USE IT)

Reviewed 29 Sep 2026. **Licence matters:** a repo with no licence is "all rights reserved" by default. We may **read and compare** but must not copy code or data into our product without the author's permission. We use these datasets as **plausibility benchmarks** for Baghewala-S (distributions, ranges) and cite them. For MIT-licensed repos we may reuse with attribution.

REPO (LICENCE)	PUBLIC DATA	CONTENTS	OUR USE
Dhayananth1511/Polaris-SIH26120 (MIT)	7 CSVs	8 wells; telemetry (3,200 rows); 32 CSS cycles; production (2,560); SRP ops (1,472); dyno cards (38,400 points: normal, float, pound, gas) ; 65 failure events	Reusable with attribution: card shapes as a secondary classifier test set, schema cross-check
dasher06/SIH2026_SRP_Performance_Model (MIT)	synthetic_srp_cards.csv (32.7 MB) + metrics	Physics SRP cards; also trained on a real 7-well "NK" SRP SCADA dataset (not in repo)	Reuse cards as an external test. Ask the author where the NK data came from. A real SCADA source would be gold.
XploY04/Aavartan (none)	data_schema.json (schema 3.1), parameter_dictionary.csv (87 KB), formula_catalog.csv, demo + flagship datasets	The best-documented schema and formula catalogue among competitors	Read-only reference: compare our schema coverage and formula list
Apaar-Gupta/SIH-2026 (none)	training_data.csv (3,000 cycles), hourly_training_data.csv (24,000 rows)	Steam-driven hourly forecast data	Benchmark distributions
nishanth24vv (none)	15 wells × 365 d time series, SQLite	Wellbore hydraulics fields	Benchmark
Bhuvanesh0097/wellwise-sih (none)	baghewala_synthetic_v1_corrected.csv (7.5 MB)	Temperature/rate/float risk	Benchmark
sandhya-parekar (none)	Sensor readings (1.5 MB), well data (1.6 MB), SRP, CSS, alerts	5 wells, 90 days	Benchmark
n230837 (none)	wells_dataset.csv (12.6 MB), SRP features/timeseries, SRP operating envelope	Large synthetic set	Benchmark; envelope comparison
naveed1756/SIH120PROT (none)	Dyno-card JSON library, 9 classes (normal, fluid pound, rod float, rod parted, steam/gas interference, SV leak, tagging, ...)	Class taxonomy	Confirms our class list; read-only

REPO (LICENCE)	PUBLIC DATA	CONTENTS	OUR USE
ZotacMaster (none)	Synthetic CSS/production/reservoir/SRP CSVs + TimescaleDB schema		Reference
varshini2419 (none)	<code>well_static.csv</code> , <code>constraints_registry.csv</code> , RAG index over OIL public documents	Source list is valuable	Use the <i>source list</i> to find primary documents (the facts are public)
Rohit-girish-Belagali (none)	Generated at runtime: 24 wells × 1,095 d × 3 policies, CSV export	Policy comparison approach	Method reference

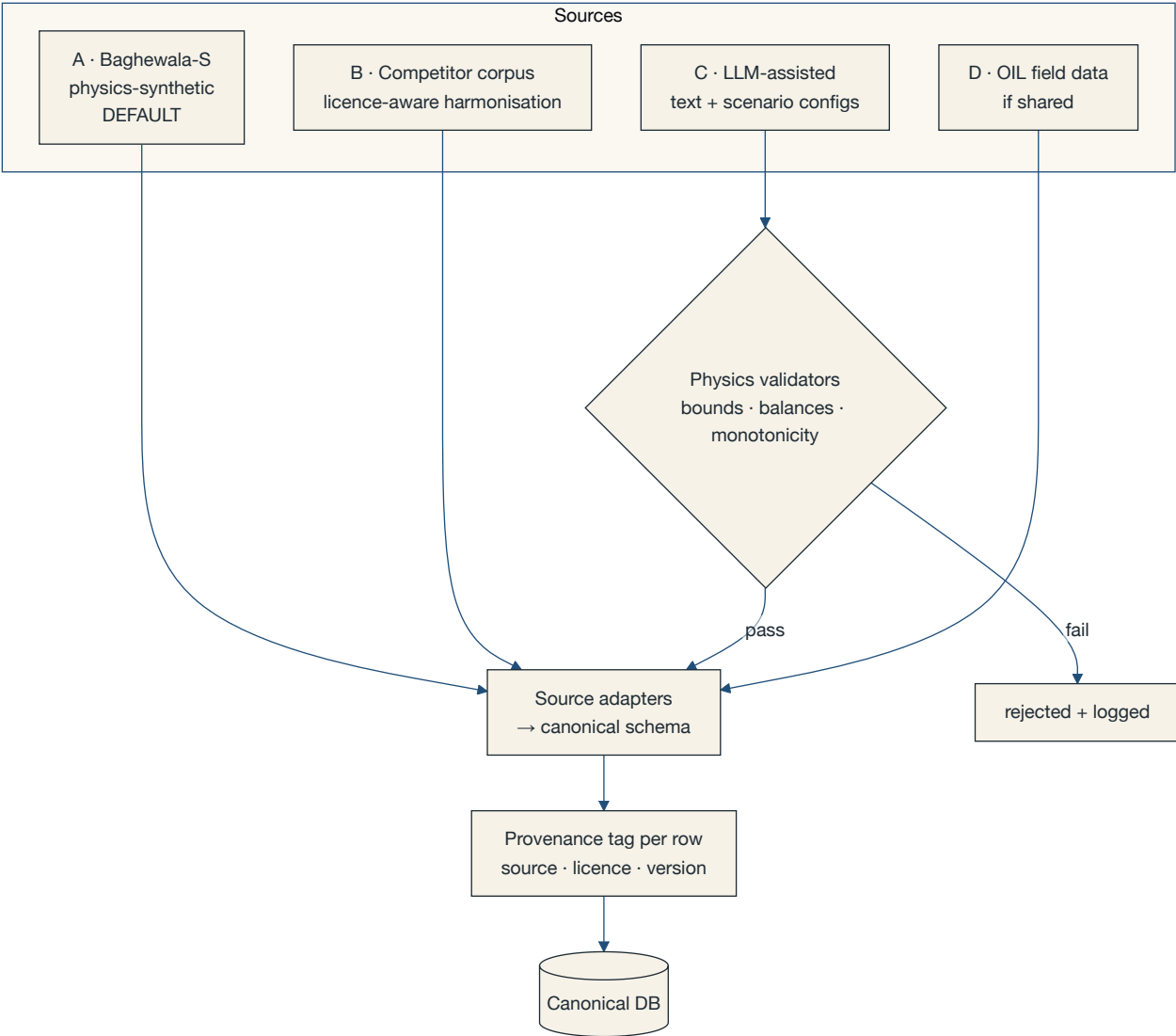
Real, licensed public data we should use:

DATASET	LICENCE	WHY
Petrobras 3W (github.com/petrobras/3W)	Apache-2.0	Real, expert-labelled well-event time series (offshore wells, not SRP). Use it to validate our anomaly-detection pipeline on real sensor pathologies (drift, frozen, spikes) and to show method credibility on real data.
BYU-PRISM USTAR-Artificial-Lift	none (research code)	Model-predictive rod-pump control reference (Hansen et al. 2018). Read for method only.
Public dynamometer-card datasets (Kaggle/papers)	varies	Search, verify licence and provenance, then use as an external test for the card classifier. Only use items with clear licences.

17.6 REAL-DATA READINESS

- **Adapters:** CSV/Parquet/ZIP upload · **MQTT** (Sparkplug B payloads) · **OPC-UA** client (asyncua) · historian export (PI/Aspen, as CSV).
- **Unit and tag mapping UI:** map a customer tag (e.g. BGW17.SRP.PPRL_KN) to a canonical field with a unit.
- **Mode switch:** `SIMULATION` ⇌ `FIELD`. Field mode is shown as available but locked ("requires OIL data-sharing agreement").

17.7 DATA SOURCE OPTIONS (KEPT OPEN UNTIL THE PHASE 2 DECISION)



OPTION	WHAT IT GIVES	RULES	BEST USE
A. Physics-synthetic (Baghewala-S)	Unlimited, internally consistent, labelled data incl. failures and faults	Calibrated to public anchors; seeds versioned	Training, Arena, demo (default)
B. Competitor corpus	Independent simulators' views of the same field	MIT repos (Polaris, dasher06): reuse with attribution. Unlicensed repos: only with written permission from authors; harmonise via adapters; keep the source column	Cross-simulator validation: train on ours, test on theirs. If our models hold up on four independent teams' simulators, that's a strong robustness argument.
C. LLM-assisted	Operator notes, maintenance logs, shift reports, failure narratives, alarm texts; scenario configurations (parameter sets) that the physics then simulates	Never LLM-generated time series as truth (they break energy and mass balance). Every numeric row passes physics validators; tagged SIM-LLM.	Realistic text for the copilot, knowledge capture and Ledger demos; scenario diversity
D. OIL data	Real calibration	Data-sharing agreement; stays on-prem; never committed	EnKF calibration, residual learning

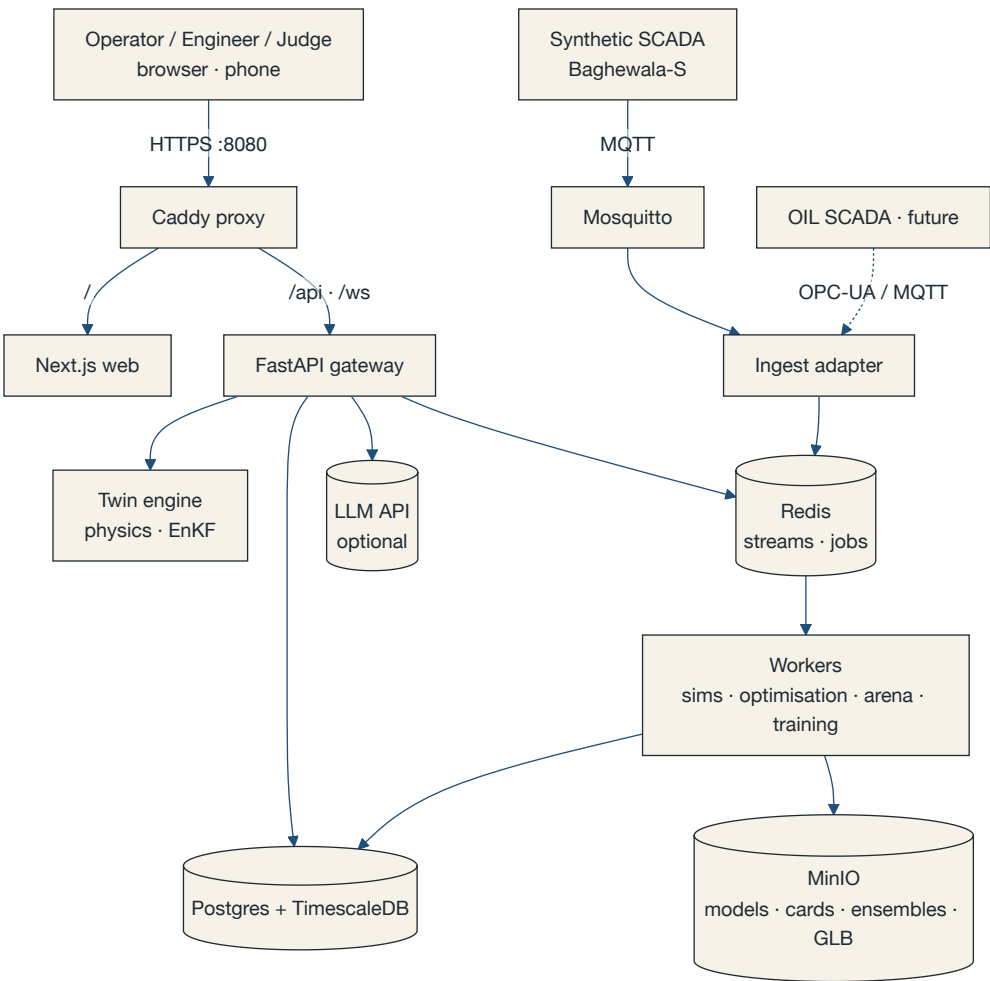
Decision point: end of Phase 2 (§27), recorded as ADR-0007. The adapters and provenance tags mean any mix works without code changes.

Permission request template for unlicensed competitor data (send via GitHub issue or email):

Hi, we're Team Mantle (SIH26120). We'd like to use <file> from your repo only as an external validation set, with credit in our README and app. Would you be OK with that? Happy to share our results with you.

18. TECHNICAL ARCHITECTURE

18.1 SYSTEM CONTEXT

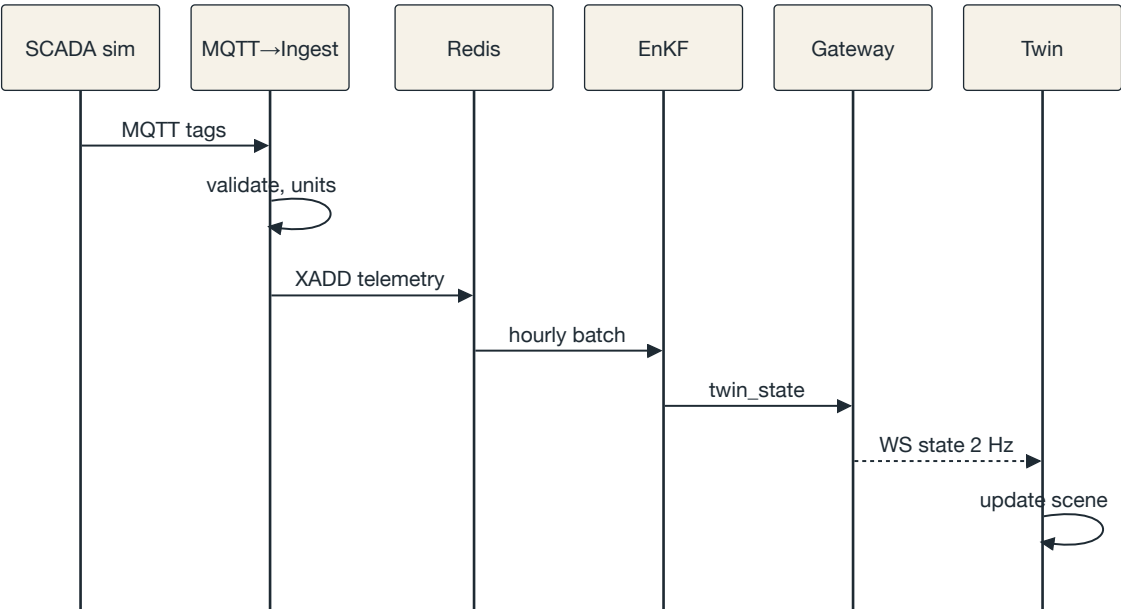


18.2 CONTAINERS AND RESPONSIBILITIES

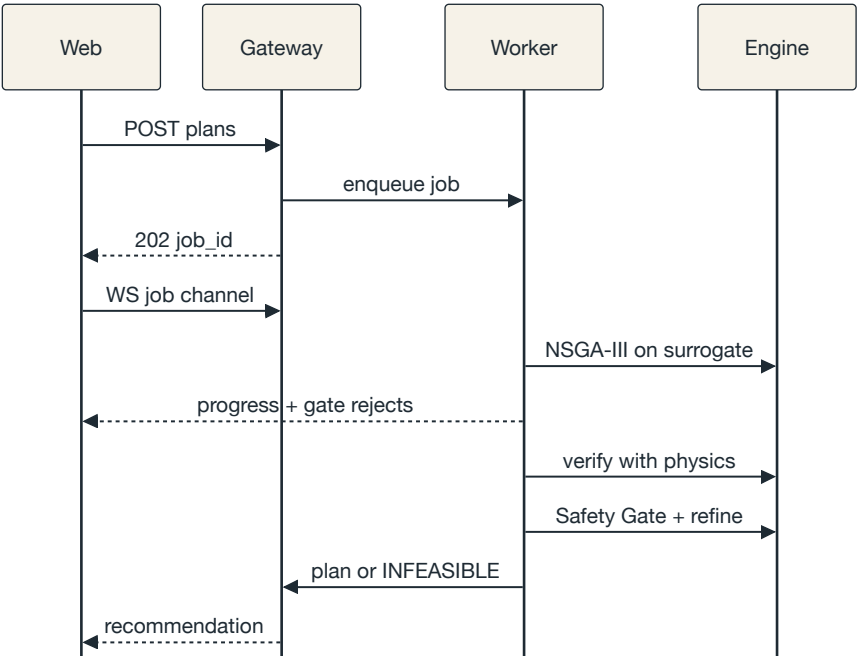
SERVICE	TECH	RESPONSIBILITY	SCALES BY
proxy	Caddy 2	Single entry (:8080), routes / → web, /api + /ws → gateway; gzip/zstd; local TLS optional	—
web	Next.js (App Router, standalone), React 19, R3F	All UI; server components for first paint; client components for 3D, charts and live data	replicas
gateway	FastAPI (uvicorn), Pydantic v2	REST + WebSocket; auth (persona now, OIDC later); validation; orchestrates engine calls; enqueues jobs	replicas
worker	Python (Arq on Redis)	Long jobs: scenario sims, optimisation, Arena runs, training, exports	replicas / queues
scada	Python	Baghewala-S real-time publisher (MQTT), replay, fault injection	per field
ingest	Python	MQTT/OPC-UA/CSV adapters → canonical rows → Redis Streams + DB; data-quality auditor	per source
estimator	Python (inside worker or its own)	EnKF updates per well (hourly fast states, daily parameters)	per well shard
mosquitto	Eclipse Mosquitto	MQTT broker (Sparkplug B topics)	—
db	PostgreSQL 16 + TimescaleDB	Canonical data, ledger, scenarios, results	vertical
redis	Redis 7	Streams (telemetry bus), pub/sub (WS fan-out), job queue, hot cache	—
minio	MinIO	Object store: models, ensembles, card archives, GLB, exports	—
mlflow (profile ml)	MLflow	Experiment tracking + model registry	—
drafts (profile drafts)	static server	HTML/CSS draft gallery	—

18.3 KEY RUNTIME FLOWS

Telemetry → twin → screen



Plan request (joint optimisation)



18.4 FRONTEND ARCHITECTURE (NEXT.JS)

- **App Router layout tree.** One persistent 3D canvas lives in the root app layout, so moving between screens never tears down WebGL. The paper-wipe transition (§11.4) plays between routes.

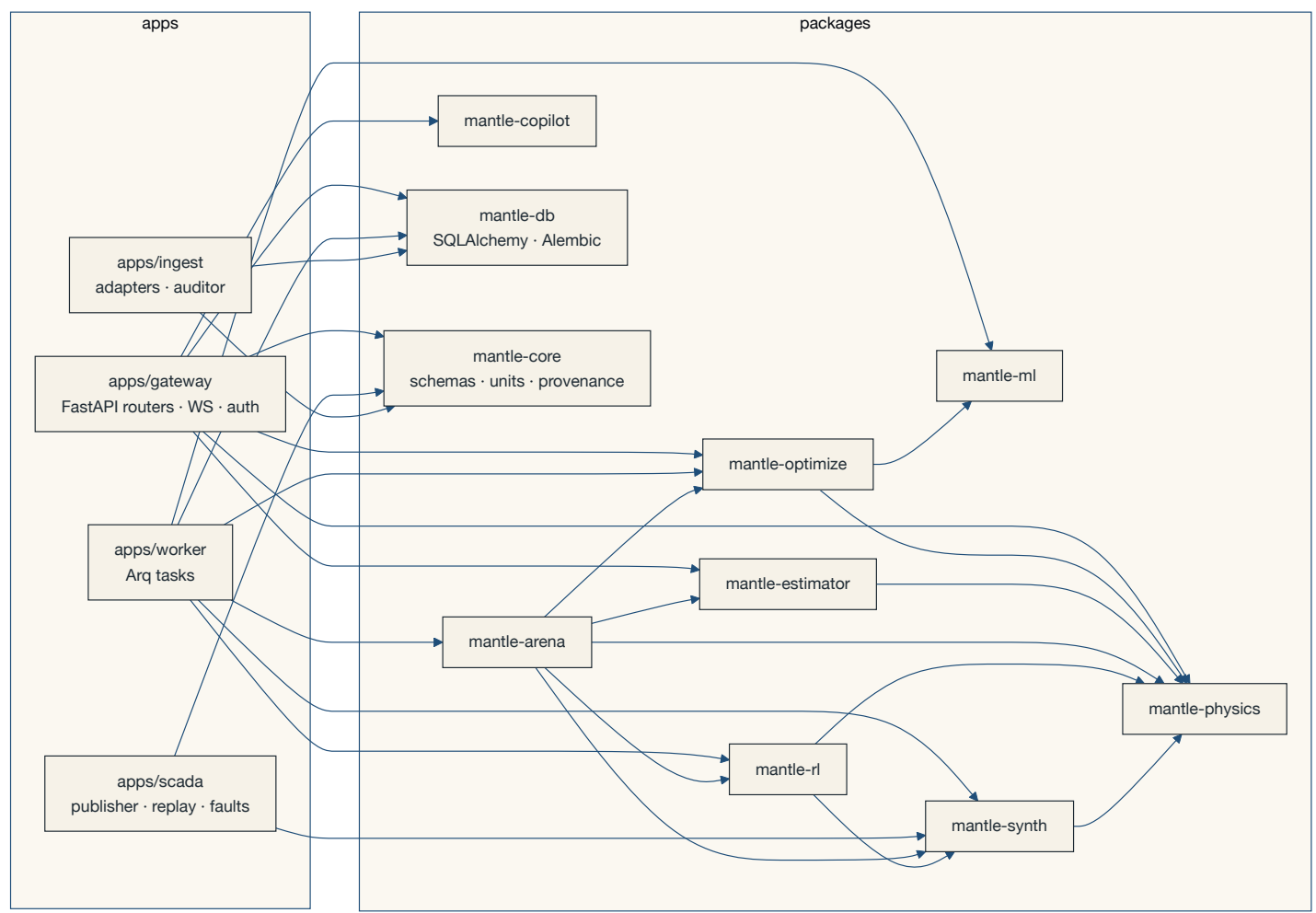
```
app/
  layout.tsx           ← fonts, tokens, providers (Query, stores), <SceneHost/> (client, persistent)
  (screens)/
    layout.tsx         ← shell: top bar, tabs, overlays, timeline
    well/[wellId]/page.tsx ← Twin/Blueprint (mode via searchParams)
    field/page.tsx
    arena/page.tsx
    ledger/page.tsx
    impact/page.tsx
    proof/page.tsx
    m/[wellId]/page.tsx ← mobile well card (PWA)
    present/page.tsx · play/page.tsx
    api/health/route.ts ← web liveness only (backend is FastAPI)
```

- **Server vs client components.** Page shells and first-paint data (well list, summary) are server components that fetch from the gateway over the internal Docker network. The 3D scene, charts, timeline and anything live are client components ("use client").

The R3F canvas is loaded with `next/dynamic({ ssr: false })`.

- **State:** Zustand stores (`time`, `well`, `scenario`, `conditions`, `ui`, `camera`). TanStack Query for REST (cache keys include `t` buckets). A single WebSocket manager multiplexes channels.
- **API client:** TypeScript types generated from the gateway's OpenAPI (`openapi-typescript`) + a thin fetch wrapper. Zod validation at the boundary in development.
- **3D:** R3F + drei + postprocessing. Custom GLSL (thermal field, hatch dissolve, stress bands, meniscus, heat haze). Kinematics in a Web Worker. WebGL2 baseline with an optional WebGPU renderer.
- **Charts:** D3 + visx SVG (Paper), regl-scatterplot for dense clouds.
- **Styling:** CSS Modules + design tokens as CSS variables, generated from `packages/tokens/tokens.json` (the same file feeds the HTML drafts).
- **Choreography:** GSAP (+ Flip). Theatre.js is used only in development to author camera paths.
- **Maps:** MapLibre GL with the custom cream style; offline tiles bundled.
- **PWA:** the mobile Well Card is installable (service worker scoped to `/m`).

18.5 BACKEND ARCHITECTURE (PYTHON, UV WORKSPACE)



- **uv workspace:** a root `pyproject.toml` declares `[tool.uv.workspace] members = ["apps/*", "packages/*"]`. One `uv.lock`. Every package has its own `pyproject.toml` and tests. Commands: `uv sync`, `uv run pytest`, `uv run --package mantle-gateway uvicorn ...`.
- **Python 3.12** (the widest wheel support across Numba, PyTorch, OR-Tools and pymoo at kickoff; revisit 3.13 in Phase 0).
- **Libraries:** FastAPI, Pydantic v2, SQLAlchemy 2 + Alembic, asyncpg, redis-py, arq, NumPy, SciPy, Numba, pint, iapws, pandas/polars, LightGBM, PyTorch, scikit-learn, MAPIE, SHAP, pymoo, OR-Tools, Stable-Baselines3 + Gymnasium, asyncua, paho-mqtt/aiomqtt, structlog, OpenTelemetry.
- **One calculation path:** only `mantle-physics` computes physics. The gateway, worker, Arena, RL environment and copilot all import it.

18.6 CROSS-CUTTING CONCERNS

CONCERN	APPROACH
Config	12-factor env vars (Appendix E), <code>pydantic-settings</code> ; <code>.env.example</code> committed
Auth	Demo: persona header, no passwords. Deployment: OIDC (Keycloak/Entra) bearer tokens; RBAC per role.

CONCERN	APPROACH
Observability	structlog JSON logs; OpenTelemetry traces (gateway → worker → engine); Prometheus metrics; optional Grafana (profile <code>obs</code>)
Security	No secrets in repo; LLM key only server-side; dependency audits (pip-audit, pnpm audit); container scan (Trivy); CORS locked to proxy origin
Determinism	Seeds everywhere; <code>demo</code> profile pins the seed manifest; golden scenarios are regression-tested
Units	SI inside physics; <code>pint</code> at the API boundary; the API documents units per field
Time	Two clocks: wall time (UTC ISO 8601) and simulation time <code>t_sim</code> (seconds since field epoch). Every record carries both where relevant.

18.7 COPILOT ARCHITECTURE ("ASK MANTLE")

```
user question → router (intent, well/time context)
  → LLM with tools (Anthropic Messages API, tool use; e.g. Claude Sonnet 5.5,
    with Haiku 4.5 for routing and short answers)
  → tools: get_state · get_forecast · explain_metric · get_card_diagnosis ·
           run_scenario · optimize · compare_wells · get_decisions ·
           get_assumptions · field_triage · steam_schedule
  → draft answer
  → NUMERIC VERIFIER: extract every number+unit; each must match a tool result
    (tolerance-aware) or a registered constant; otherwise regenerate once, then
    fall back to a templated answer
  → answer card with sources + "Show me" actions (highlight in Twin, open plate)
```

- The copilot **cannot** approve or apply anything. It can *propose* a scenario or open a recommendation.
- **Offline mode:** the same tools with templated natural language (no LLM), so the demo never depends on venue Wi-Fi. An optional local small model (via Ollama) is a P3 nice-to-have.

18.8 SYNTHETIC SCADA ENGINE (TELEMETRY, HONESTLY)

- Runs Baghewala-S forward in (accelerated) real time and **publishes telemetry through the same path real data would use**: MQTT topics (`mantle/BGW-17/srp/ppr1`) into the ingest adapter, then Redis Streams, the EnKF and the WebSocket.
- **Replay mode:** plays historical synthetic records with transport controls (0.25×–500×).
- **Fault injection API** (the Anomaly Injector) perturbs the generator, never the twin, so detection is genuine.

19. API REFERENCE (V1)

The gateway's OpenAPI document (`/api/v1/openapi.json`) is the machine-readable source of truth. This section is the design contract it must satisfy. Contract tests enforce it (§23).

19.1 CONVENTIONS

TOPIC	RULE
Base	<code>/api/v1</code> behind the proxy; WebSockets at <code>/ws</code>
Format	JSON, <code>snake_case</code> keys; UTF-8; gzip/zstd
Time	<code>t</code> = simulation seconds since the field epoch (integer). Wall times are ISO 8601 UTC (<code>*_at</code>). Query params accept <code>t</code> or <code>at</code> .
Units	Domain units, fixed per field and documented (°C, bar, cP, kN, m, BOPD, t, kWh, kg CO ₂ e, ₹). Every <code>Quantity</code> carries <code>unit</code> .
IDs	<code>well_id</code> like <code>BGW-17</code> ; other resources use ULIDs (<code>dec_01J...</code>)
Pagination	Cursor: <code>?limit=50&cursor=...</code> → <code>{items, next_cursor}</code>
Async jobs	<code>202 Accepted</code> + <code>{job_id}</code> . Poll GET <code>/jobs/{id}</code> or stream WS <code>/ws/jobs/{id}</code> . <code>Idempotency-Key</code> header required on job-creating POSTs.
Errors	RFC 9457 <code>application/problem+json</code> (<code>type</code> , <code>title</code> , <code>status</code> , <code>detail</code> , <code>instance</code> , <code>errors[]</code>)
Auth	Demo: <code>X-Persona: operator engineer manager viewer</code> . Deployment: <code>Authorization: Bearer <OIDC JWT></code> .
Versioning	URL major version. Additive changes only within v1.
Provenance	Any measured, estimated or simulated number is returned as a <code>Quantity</code> (below) unless it's inside a bulk series
Rate limits	Copilot 20/min/persona; jobs 10 concurrent per client

19.2 COMMON TYPES

```
type Source = "MEAS" | "EST" | "SIM" | "SIM-LLM" | "ASSUM" | "PUB";
interface Band { p10: number; p50: number; p90: number }
interface Quantity { value: number; unit: string; source: Source; band?: Band;
  model_version?: string; stale?: boolean }
interface Series { t: number[]; values: Record<string, number[]>; // columnar, compact
  units: Record<string, string>; source: Source;
  bands?: Record<string, { p10: number[]; p90: number[] }> }
interface Problem { type: string; title: string; status: number; detail?: string;
  errors?: { loc: string[]; msg: string }[] }
interface Job { job_id: string; kind: string; status: "queued"|"running"|"done"|"failed"|"cancelled";
  progress: number; message?: string; result_url?: string; created_at: string }
type Phase = "INJECTION" | "SOAK" | "PRODUCTION" | "SHUT-IN";
```

19.3 ENDPOINT CATALOGUE

System			
METHOD	PATH	PURPOSE	
GET	<code>/health</code>	Liveness + dependency status (db, redis, minio, mqtt)	
GET	<code>/meta</code>	Versions (app, physics, models), mode (<code>SIMULATION</code> / <code>FIELD</code>), field epoch, units	
GET	<code>/jobs/{job_id}</code> · DELETE	Job status · cancel	
Field			
METHOD	PATH	PURPOSE / KEY PARAMS	RESPONSE
GET	<code>/field/summary?t=</code>	Field KPIs	<code>{bopd, sor, kwh_per_bbl, co2_per_bbl, counts: {ok, watch, high, critical}}</code> (Quantities)
GET	<code>/field/map</code>	Pads, wells, generators, header geometry (GeoJSON)	<code>FeatureCollection</code>
GET	<code>/field/triage?t=&limit=5</code>	Ranked worklist	<code>{[rank, well_id, severity, reason_code, reason, value_at_risk: Quantity, deep_link]}</code>
GET	<code>/field/brief?date=</code>	Morning brief	<code>{generated_at, sections[], pdf_url}</code>
GET	<code>/field/steam-schedule</code>	Current generator plan	<code>{generators[], slots[{well_id, generator_id, start_t, end_t, tonnes}], shadow_price}</code>
POST	<code>/field/steam-schedule/solve</code>	Re-solve (job). Body: <code>{horizon_days, overrides[], generator_outages[]}</code>	<code>202 Job</code>

Wells & live state

METHOD	PATH	PURPOSE	RESPONSE (ABRIDGED)
GET	/wells	List	{well_id, pad_id, phase, severity, cycle_no}
GET	/wells/{id}	Static data: completion, rod string, pump, unit, PVT anchors	WellDetail
GET	/wells/{id}/state?t=	Posterior twin state at t	WellState (below)
GET	/wells/{id}/timeseries?vars=&from=&to=&step=	Columnar history (measured and estimated)	Series
GET	/wells/{id}/profile?t=	Depth tracks: T, P, μ , $\sigma_{\max}/\sigma_{\min}$, deposition band, fluid level	{depth_m[], tracks:{...}, markers:{...}}
GET	/wells/{id}/thermal-field?t=	r-z temperature grid for shaders	application/octet-stream float16 256×64 + X-Grid-Meta header
GET	/wells/{id}/cards?t=&n=	Last n surface/downhole cards + diagnosis	{t, surface:{x[],f[]}, downhole:{x[],f[]}, class, probs, fillage}
GET	/wells/{id}/envelope?from=&to=	Safe-SPM ceiling N_max(t) + actual + plan	Series
GET	/wells/{id}/ipr?t=	IPR curve(s) + pump line + operating point	{ipr:[...], pump:[...], op_point, limit:"RESERVOIR" "PUMP"}
GET	/wells/{id}/heat-balance?cycle=	Sankey nodes and links	{nodes[], links[]}
GET	/wells/{id}/bom	Parts list for the Blueprint	{balloon, part, spec, source}
GET	/wells/{id}/cycles	Cycle history	{cycle_no, steam_t, inj_days, soak_days, peak_bopd, oil_60d, sor, heat_util}
GET	/wells/{id}/similar?k=3	Nearest trajectories	{well_id, distance, aligned_series}
GET	/wells/{id}/forecast?h=30d&plan=baseline {plan_id}	Ensemble forecast	Series with bands
GET	/wells/{id}/explain?metric=&t=	Drivers: SHAP/physics sensitivities + causal path	{metric, value, drivers[{name, contribution, direction}], path[], evidence[]}
GET	/wells/{id}/margins?t=	Six risk margins	{kind, margin: Quantity, time_to_limit_d: Quantity, drivers[]}
GET	/wells/{id}/maintenance	Forecast + what-if	{items[{kind, due_window, risk}], whatif_url}

```
interface WellState {
  well_id: string; t: number; phase: Phase; cycle_no: number; day_in_cycle: number;
  surface: { spm: Quantity; stroke_in: Quantity; kd: Quantity; vfd_hz: Quantity;
    motor_kw: Quantity; torque_pct: Quantity };
  rods: { ppri_kn: Quantity; mpri_kn: Quantity; float_margin: Quantity; goodman_max: Quantity };
  pump: { fillage: Quantity; vol_eff: Quantity; holddown_margin: Quantity };
  wellbore:{ fluid_level_m: Quantity; submergence_m: Quantity; pip_bar: Quantity; pwf_bar: Quantity;
    wellhead_t_c: Quantity; deposition_interval_m: [number, number] | null };
  reservoir:{ sandface_t_c: Quantity; mu_cp: Quantity; heated_radius_m: Quantity;
    thermal_battery_pct: Quantity; days_to_cutoff: Quantity };
  production:{ oil_bopd: Quantity; water_cut: Quantity; liquid_bpd: Quantity };
  economy: { net_inr_per_day: Quantity; sor: Quantity; kwh_per_bbl: Quantity;
    co2_kg_per_bbl: Quantity; water_m3_per_bbl: Quantity };
  coupling_dividend: { inr_per_cycle: Quantity; oil_pct: Quantity } | null;
  health: { score: number; flags: string[] }; conditions_id: string; scenario_id: string | null;
}
```

Conditions & faults

METHOD	PATH	PURPOSE
GET	/conditions/catalog	All cards and options with parameters and provenance
GET	/conditions/presets	The six presets
POST	/wells/{id}/conditions	Apply a bundle {formation, crude, climate, completion, lift, steam_source} → returns the new conditions_id and changed parameters
POST	/faults/inject	{well_id, kind, start_t, duration_s, severity} → perturbs the generator, not the twin
GET	/faults/active · DELETE /faults/{id}	List · clear

Plans, optimisation & recommendations

METHOD	PATH	PURPOSE
POST	/wells/{id}/plans	Joint optimisation job. Body: {horizon_days, objectives:{weights}, levers: {css:bool, srp:bool, stroke_shaper:bool, hardware:bool}, constraints_override?} → 202 Job
GET	/plans/{plan_id}	Plan: CSS design, SRP schedule, expected outcomes with bands, binding constraints, gate log summary

METHOD	PATH	PURPOSE
GET	/plans/{plan_id}/candidates?status=rejected	Rejected candidates with reasons (the Safety Gate log)
POST	/optimize/pareto	Pareto front job for a well or scenario
GET	/wells/{id}/coupling-dividend?cycle=	CD + Shapley decomposition
GET	/recommendations?well_id=&status=	Open or past recommendations
GET	/recommendations/{id}	Detail incl. explain payload
POST	/recommendations/{id}/approve	{note?} → Decision
POST	/recommendations/{id}/modify	{changes:{...}, reason_code, note} → re-simulation job, then Decision
POST	/recommendations/{id}/reject	{reason_code, note} → Decision (knowledge capture)

Ledger & analytics

METHOD	PATH	PURPOSE
GET	/decisions?well_id=&from=&to=	Decision list (cursor)
GET	/decisions/{id}	ECN detail
GET	/decisions/{id}/trace	Provenance trace: plan → models (versions) → inputs → assumptions → sources
GET	/decisions/{id}/outcome	Predicted vs realised at +1/7/30 d + counterfactual band + score
GET	/ledger/verify	Recompute hash chain → {ok, length, head_hash, first_bad?}
GET	/analytics/calibration?model=	Reliability data, coverage
GET	/analytics/overrides	Override regimes

Scenarios & Arena

METHOD	PATH	PURPOSE
POST	/scenarios	Create {name, base:{well_id, t}, levers, conditions, horizon_days}
GET	/scenarios · /scenarios/{id}	List · detail
POST	/scenarios/{id}/run	Simulate (job)
POST	/scenarios/{id}/fork?t=	Branch at t
POST	/scenarios/{id}/clone	Clone
GET	/arena/strategies	Contenders with definitions and parameters
POST	/arena/runs	{strategies[], wells:"all" ...], seeds, horizon_days, conditions_preset, ablations[]} → 202 Job
GET	/arena/runs · /arena/runs/{id}	List (incl. official flag) · status and manifest
GET	/arena/runs/{id}/series?metric=oil_rate&agg=mean	Per-strategy series with seed bands
GET	/arena/runs/{id}/scoreboard	Deltas vs baseline with CIs, win rates, % oracle, violations
GET	/arena/runs/{id}/ablation · /decomposition	Ablation bars · CD decomposition
POST	/challenge/sessions · POST /challenge/sessions/{id}/actions · GET /challenge/leaderboard	Judge Challenge game

Impact

METHOD	PATH	PURPOSE
GET	/impact/summary?scope=field well illustrative&baseline=static pumpoff sequential&well_id=	12 hero numbers with bands and formulas
GET	/impact/series?metric=&scope=&baseline=	Cumulative series
GET	/impact/receipt?well_id=&cycle=	Cycle receipt (JSON + printable HTML/PDF URL)

Evidence, data & health

METHOD	PATH	PURPOSE
GET	/evidence/traceability	PS line → features → deep links
GET	/evidence/models · /evidence/models/{name}	Model cards
GET	/evidence/assumptions	Assumption registry
GET	/evidence/sources	Source registry (PUB anchors with doc refs)
GET	/evidence/provenance/summary	Share by source

METHOD	PATH	PURPOSE
POST	/data/imports	Multipart upload (CSV/Parquet/ZIP) + mapping → job (auditor)
GET	/data/imports/{id}/report	Quality report: rows, missing, duplicates, outliers, unit issues, per-column scores
GET	/data/mappings · POST	Tag → canonical field mappings
GET	/twin/health?well_id=	Sensors, calibration, freshness, OOD, disagreement

Alerts, copilot, story, export

METHOD	PATH	PURPOSE
GET	/alerts?state=active	ISA-18.2 alarms
POST	/alerts/{id}/ack · /shelve	Acknowledge · shelve {minutes, reason}
POST	/copilot/query	{question, context:{well_id,t,screen}} → {answer, numbers[{text, value, unit, tool_ref, verified}], sources[], actions[]} (SSE stream option)
GET	/story/scripts · /story/scripts/{id}	Demo beats (deep links + notes)
POST	/snapshots · GET /snapshots/{id}	Save / load a deep-link state
POST	/exports/sheet	Blueprint sheet → PDF/SVG job
POST	/exports/timelapse	Cycle time-lapse job (WebM)
GET	/exports/{id}	Download

19.4 WEBSOCKET PROTOCOL

One socket, multiplexed. Client sends {"op":"sub","ch":"well:BGW-17"} / {"op":"unsub",...}; server messages use the envelope {ch, type, t, seq, data}.

CHANNEL	TYPES	RATE
well:{id}	state (WellState delta), card (per stroke), phase_change, margin_alarm	2 Hz / per stroke
field	summary, triage_changed	0.2 Hz
alerts	raised, cleared, acked	event
job:{id}	progress {pct, message, counters}, done {result_url}, failed {problem}	event
story	beat {index} (presenter remote ↔ app)	event
challenge:{id}	tick {day, oil, cum, failures}	10 Hz

Back-pressure: the server drops intermediate state frames for slow clients (latest wins). seq gaps trigger a REST resync.

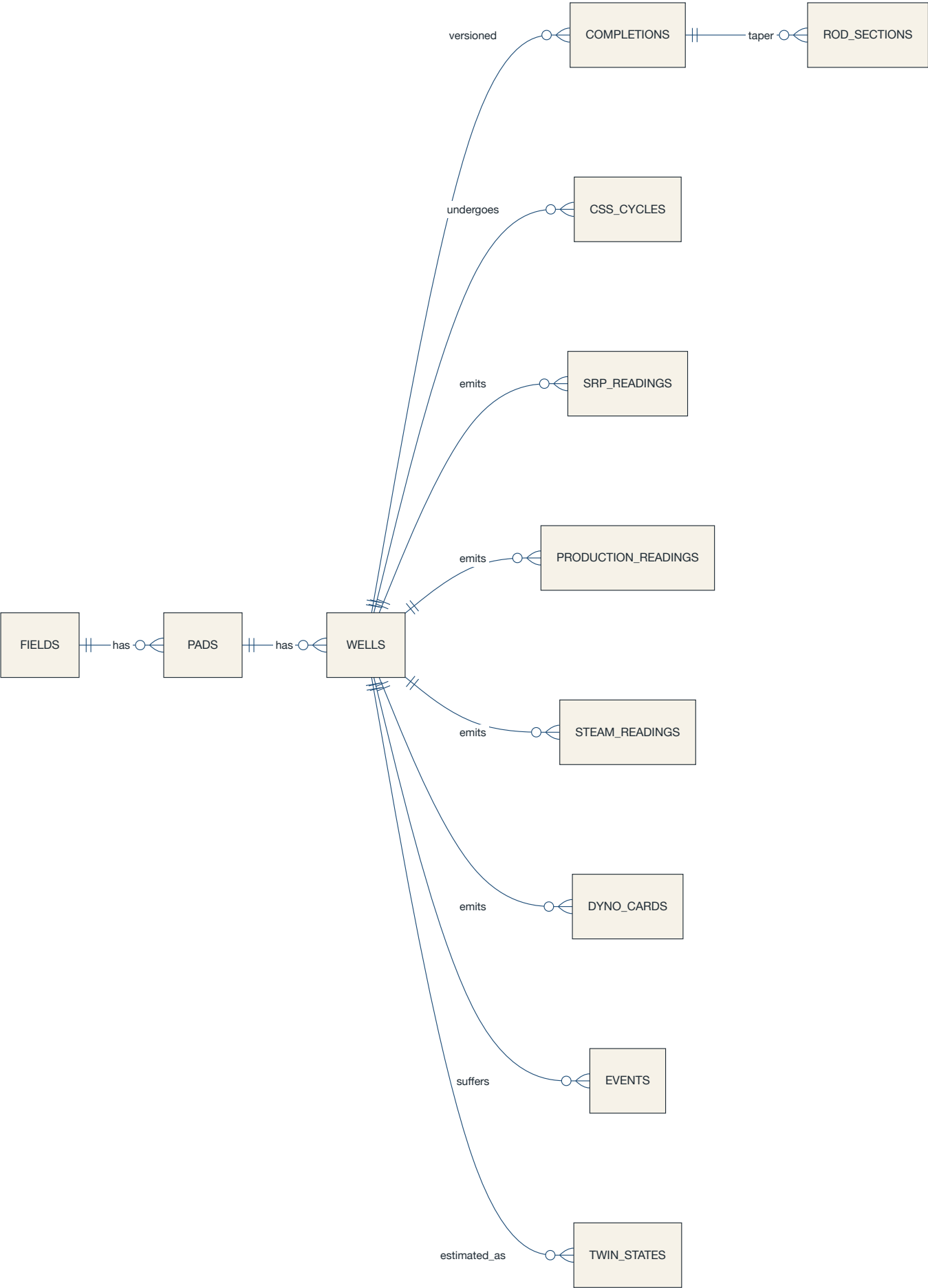
19.5 STATUS CODES

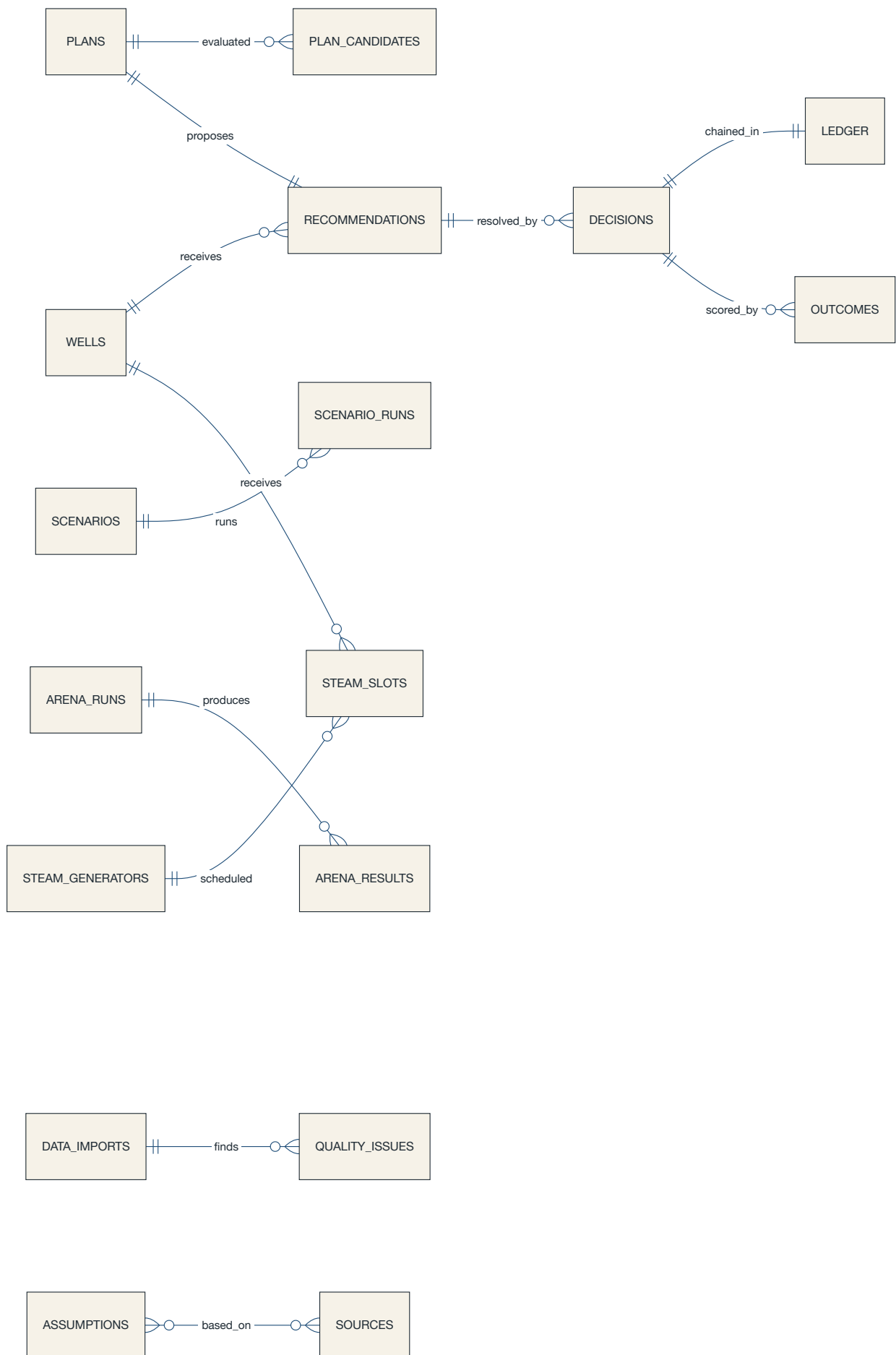
200 OK · 201 created · 202 job accepted · 400 validation (Problem with errors[]) · 404 unknown well or resource · 409 conflict (e.g. approving a closed recommendation; infeasible modify) · 422 physically invalid input (e.g. steam rate × duration ≠ volume) · 429 rate limit · 503 dependency down (the health payload names it).

20. DATABASE SCHEMA

PostgreSQL 16 + TimescaleDB. Migrations use Alembic (`packages/db/migrations`). Naming: `snake_case`, singular type names, plural tables, `id` = ULID text unless natural keys exist (`well_id`). Every time-series table has `t` (sim seconds, `bigint`) and `ts` (`timestampz`, derived from the epoch) and is a hypertable on `ts`. Every row that holds a number of uncertain origin carries `source source_kind`.

20.1 ENTITY RELATIONSHIPS (CORE)





20.2 TYPES

```
CREATE TYPE source_kind AS ENUM ('MEAS','EST','SIM','SIM_LLM','ASSUM','PUB');
CREATE TYPE phase_kind AS ENUM ('INJECTION','SOAK','PRODUCTION','SHUT_IN');
CREATE TYPE severity AS ENUM ('OK','WATCH','HIGH','CRITICAL');
CREATE TYPE rec_status AS ENUM ('OPEN','APPROVED','MODIFIED','REJECTED','EXPIRED','SUPERSEDED');
CREATE TYPE card_class AS ENUM ('NORMAL','FLUID_POUND','GAS_INTERFERENCE','ROD_FLOAT','PARTED_ROD',
                                'TV_LEAK','SV_LEAK','TAGGING','UNSETTING','OVER_TRAVEL');
CREATE TYPE event_kind AS ENUM ('ROD_FAILURE','PUMP_UNSETTING','ROD_FLOAT','FLUID_POUND','IMPACT',
                                'WORKOVER','HOT_OIL','SENSOR_FAULT','GENERATOR_OUTAGE','OTHER');
CREATE TYPE alarm_prio AS ENUM ('P1','P2','P3','P4');
```

20.3 REFERENCE DATA

```
CREATE TABLE fields (id text PRIMARY KEY, name text NOT NULL, epoch timestampz NOT NULL,
                      description text, synthetic boolean NOT NULL DEFAULT true);
CREATE TABLE pads (id text PRIMARY KEY, field_id text REFERENCES fields, name text,
                    geom geometry(Polygon,4326)); -- PostGIS optional; else jsonb
CREATE TABLE wells (well_id text PRIMARY KEY, pad_id text REFERENCES pads, spud_date date,
                    td_m real, perf_top_m real, perf_bottom_m real, formation text, api_gravity real,
                    lat real, lon real, illustrative_location boolean DEFAULT true, meta jsonb DEFAULT '{}');
CREATE TABLE completions (id text PRIMARY KEY, well_id text REFERENCES wells, valid_from bigint NOT NULL,
                           tubing_type text CHECK (tubing_type IN ('VIT','BARE')), tubing_id_mm real, casing_id_mm real,
                           casing_grade text, anchor_depth_m real, pump_depth_m real, pump_type text, plunger_in real,
                           unit_type text CHECK (unit_type IN ('BEAM','HYDRAULIC')), unit_rating jsonb, source source_kind);
CREATE TABLE rod_sections (completion_id text REFERENCES completions, idx smallint, size_in real,
                            grade text, length_m real, weight_kg_per_m real, tensile_mpa real, guides_per_rod smallint,
                            PRIMARY KEY (completion_id, idx));
CREATE TABLE pvt_samples (id text PRIMARY KEY, well_id text REFERENCES wells, temp_c real, mu_cp real,
                           density_kg_m3 real, water_cut real, asphaltene_onset_c real, source source_kind, source_ref text);
CREATE TABLE viscosity_fits (well_id text REFERENCES wells, valid_from bigint, walther_a double precision,
                             walther_b double precision, emulsion_k real, source source_kind, PRIMARY KEY (well_id, valid_from));
CREATE TABLE steam_generators (id text PRIMARY KEY, field_id text REFERENCES fields, capacity_t_per_d real,
                                efficiency real, fuel text, solar_capable boolean DEFAULT false);
```

20.4 OPERATIONAL TIME SERIES (HYPERTABLES)

```
CREATE TABLE telemetry_raw (ts timestampz NOT NULL, t bigint NOT NULL, well_id text NOT NULL,
                             tag text NOT NULL, value double precision, quality smallint DEFAULT 192, import_id text);
SELECT create_hypertable('telemetry_raw','ts');
CREATE INDEX ON telemetry_raw (well_id, tag, ts DESC);

CREATE TABLE srp_readings (ts timestampz NOT NULL, t bigint NOT NULL, well_id text NOT NULL,
                            spm real, stroke_in real, kd real, vfd_hz real, motor_kw real, motor_a real, torque_pct real,
                            ppri_kn real, mpri_kn real, fillage real, runtime_frac real, source source_kind NOT NULL,
                            PRIMARY KEY (well_id, ts));
CREATE TABLE production_readings (ts timestampz NOT NULL, t bigint NOT NULL, well_id text NOT NULL,
                                   oil_bopd real, water_bwpd real, gas_mscfd real, wellhead_p_bar real, wellhead_t_c real,
                                   is_test boolean DEFAULT false, source source_kind NOT NULL, PRIMARY KEY (well_id, ts));
CREATE TABLE steam_readings (ts timestampz NOT NULL, t bigint NOT NULL, well_id text NOT NULL,
                              rate_t_per_h real, pressure_bar real, temp_c real, quality real, generator_id text,
                              source source_kind NOT NULL, PRIMARY KEY (well_id, ts));
CREATE TABLE dyno_cards (ts timestampz NOT NULL, t bigint NOT NULL, well_id text NOT NULL,
                          surface_pos real[] NOT NULL, surface_load real[] NOT NULL, downhole_pos real[], downhole_load real[],
                          class card_class, class_probs jsonb, fillage real, area_kj real, source source_kind NOT NULL,
                          PRIMARY KEY (well_id, ts));
CREATE TABLE pressure_surveys (ts timestampz NOT NULL, well_id text NOT NULL, kind text,
                                fluid_level_m real, bhp_bar real, source source_kind, PRIMARY KEY (well_id, ts, kind));
-- all hypertables; compression after 7 days; continuous aggregates:
CREATE MATERIALIZED VIEW srp_1h WITH (timescaledb.continuous) AS
  SELECT well_id, time_bucket('1 hour', ts) AS bucket, avg(spm) spm, max(ppri_kn) ppri_max,
         min(mpri_kn) mpri_min, avg(fillage) fillage, avg(motor_kw) motor_kw
  FROM srp_readings GROUP BY 1,2; -- likewise production_1h, production_id
```

20.5 CYCLES, EVENTS, MAINTENANCE

```
CREATE TABLE css_cycles (id text PRIMARY KEY, well_id text REFERENCES wells, cycle_no int NOT NULL,
                          inj_start bigint, inj_end bigint, soak_end bigint, prod_end bigint, steam_t real, avg_inj_p_bar real,
                          avg_quality real, cutoff_reason text, oil_bbl real, sor real, heat_util real, source source_kind,
                          UNIQUE (well_id, cycle_no));
CREATE TABLE events (id text PRIMARY KEY, well_id text REFERENCES wells, t bigint NOT NULL,
                      kind event_kind NOT NULL, downtime_h real, cost_inr real, depth_m real, notes text,
                      source source_kind NOT NULL, meta jsonb);
CREATE TABLE maintenance_items (id text PRIMARY KEY, well_id text REFERENCES wells, kind text,
                                  due_from bigint, due_to bigint, risk real, status text, created_by text);
```

20.6 TWIN, CONDITIONS, FAULTS, FORECASTS

```
CREATE TABLE conditions (id text PRIMARY KEY, bundle jsonb NOT NULL, params jsonb NOT NULL,
    hash text UNIQUE NOT NULL);
-- content-addressed
CREATE TABLE well_conditions (well_id text REFERENCES wells, from_t bigint, conditions_id text
    REFERENCES conditions, PRIMARY KEY (well_id, from_t));
CREATE TABLE faults (id text PRIMARY KEY, well_id text, kind text, start_t bigint, duration_s bigint,
    severity real, active boolean DEFAULT true);
CREATE TABLE twin_states (ts timestampz NOT NULL, t bigint NOT NULL, well_id text NOT NULL,
    summary jsonb NOT NULL,
    -- WellState-shaped means + bands
    params jsonb NOT NULL,
    -- posterior parameter means/sd
    ensemble_ref text,
    -- MinIO key for the full ensemble (npz)
    innovation jsonb, health real, model_versions jsonb, PRIMARY KEY (well_id, ts));
CREATE TABLE forecasts (id text PRIMARY KEY, well_id text, issued_t bigint, horizon_s bigint,
    plan_id text, series_ref text NOT NULL, headline jsonb, model_versions jsonb);
```

20.7 PLANS, RECOMMENDATIONS, DECISIONS, LEDGER

```
CREATE TABLE plans (id text PRIMARY KEY, well_id text REFERENCES wells, created_t bigint,
    request jsonb NOT NULL, css jsonb, srp_schedule jsonb, expected jsonb, binding jsonb,
    feasible boolean NOT NULL, coupling_dividend jsonb, model_versions jsonb, job_id text);
CREATE TABLE plan_candidates (plan_id text REFERENCES plans, idx int, params jsonb, objectives jsonb,
    verified boolean, gate_pass boolean, reject_reasons text[], PRIMARY KEY (plan_id, idx));
CREATE TABLE recommendations (id text PRIMARY KEY, well_id text REFERENCES wells, plan_id text
    REFERENCES plans, created_t bigint, expires_t bigint, title text, action jsonb, deltas jsonb,
    confidence real, status rec_status NOT NULL DEFAULT 'OPEN', explain jsonb);
CREATE TABLE decisions (id text PRIMARY KEY, recommendation_id text REFERENCES recommendations,
    well_id text, t bigint, actor text NOT NULL, verdict rec_status NOT NULL, changes jsonb,
    reason_code text, note text, inputs_snapshot_ref text, model_versions jsonb);
CREATE TABLE ledger (seq bigserial PRIMARY KEY, decision_id text UNIQUE REFERENCES decisions,
    prev_hash char(64) NOT NULL, hash char(64) NOT NULL, created_at timestampz DEFAULT now());
-- trigger: hash = sha256(prev_hash || canonical_json(decision row)); UPDATE/DELETE revoked on ledger
CREATE TABLE outcomes (decision_id text REFERENCES decisions, horizon text, predicted jsonb,
    realised jsonb, counterfactual jsonb, score text, PRIMARY KEY (decision_id, horizon));
CREATE TABLE knowledge_notes (id text PRIMARY KEY, well_id text, regime jsonb, text text,
    source_decision text REFERENCES decisions, tags text[]);
```

20.8 SCENARIOS, ARENA, STEAM SCHEDULING, ALARMS

```
CREATE TABLE scenarios (id text PRIMARY KEY, name text, parent_id text REFERENCES scenarios,
    base_well text, base_t bigint, levers jsonb, conditions_id text, horizon_s bigint, created_by text);
CREATE TABLE scenario_runs (id text PRIMARY KEY, scenario_id text REFERENCES scenarios, status text,
    series_ref text, summary jsonb, code_hash text, seed bigint);
CREATE TABLE arena_runs (id text PRIMARY KEY, official boolean DEFAULT false, manifest jsonb NOT NULL,
    code_hash text, status text, started_at timestampz, finished_at timestampz);
CREATE TABLE arena_results (run_id text REFERENCES arena_runs, strategy text, well_id text, seed int,
    metrics jsonb NOT NULL, violations int DEFAULT 0, series_ref text,
    PRIMARY KEY (run_id, strategy, well_id, seed));
CREATE TABLE steam_schedule_runs (id text PRIMARY KEY, created_t bigint, horizon_s bigint,
    objective real, shadow_price real, request jsonb);
CREATE TABLE steam_slots (run_id text REFERENCES steam_schedule_runs, well_id text, generator_id text
    REFERENCES steam_generators, start_t bigint, end_t bigint, tonnes real, locked boolean DEFAULT false);
CREATE TABLE alarms (id text PRIMARY KEY, well_id text, kind text, prio alarm_prio, raised_t bigint,
    cleared_t bigint, acked_by text, shelved_until bigint, group_id text, message text);
```

20.9 EVIDENCE, DATA QUALITY, COPILOT, MISC

```
CREATE TABLE sources (id text PRIMARY KEY, title text, publisher text, year int, url text,
    file_ref text, pages text, verified boolean DEFAULT false);
CREATE TABLE assumptions (key text PRIMARY KEY, value jsonb, unit text, basis text,
    replace_with text, status text CHECK (status IN ('ASSUMED', 'CALIBRATED', 'MEASURED')),
    source_ids text[]);
CREATE TABLE model_registry (name text, version text, kind text, metrics jsonb, dataset_tag text,
    card_md text, artifact_ref text, PRIMARY KEY (name, version));
CREATE TABLE data_imports (id text PRIMARY KEY, filename text, uploaded_at timestampz, mapping jsonb,
    rows int, valid_rows int, score real, status text);
CREATE TABLE quality_issues (import_id text REFERENCES data_imports, row_idx int, column_name text,
    kind text, detail text);
CREATE TABLE copilot_messages (id text PRIMARY KEY, session_id text, role text, content text,
    tool_calls jsonb, verified boolean, created_at timestampz DEFAULT now());
CREATE TABLE snapshots (id text PRIMARY KEY, state jsonb NOT NULL, created_at timestampz DEFAULT now());
CREATE TABLE challenge_sessions (id text PRIMARY KEY, player text, score real, result jsonb,
    created_at timestampz DEFAULT now());
```

20.10 DATA LIFECYCLE

DATA	RETENTION (DEMO)	NOTES
telemetry_raw	30 d raw, then compressed	Canonical tables keep full resolution for the sim span
Canonical time series	Full span (3 y sim)	Compressed after 7 d; continuous aggregates for 1 h / 1 d
twin_states	Hourly for 90 d, daily after	Ensembles in MinIO, pruned to daily
Ledger	Forever, append-only	UPDATE/DELETE revoked at role level
Arena	Official runs forever; ad-hoc 30 d	Series in MinIO (Parquet)

Seeding: seed (a one-shot container) loads the Baghewala-S snapshot (Parquet, $\approx$ 300 MB for 30 wells $\times$ 3 y at hourly resolution; cards subsampled), the official Arena run, sources, assumptions and model cards. The first run takes about 2 minutes; after that the stack starts in seconds.

21. REPOSITORY LAYOUT AND DEVELOPMENT WORKFLOW

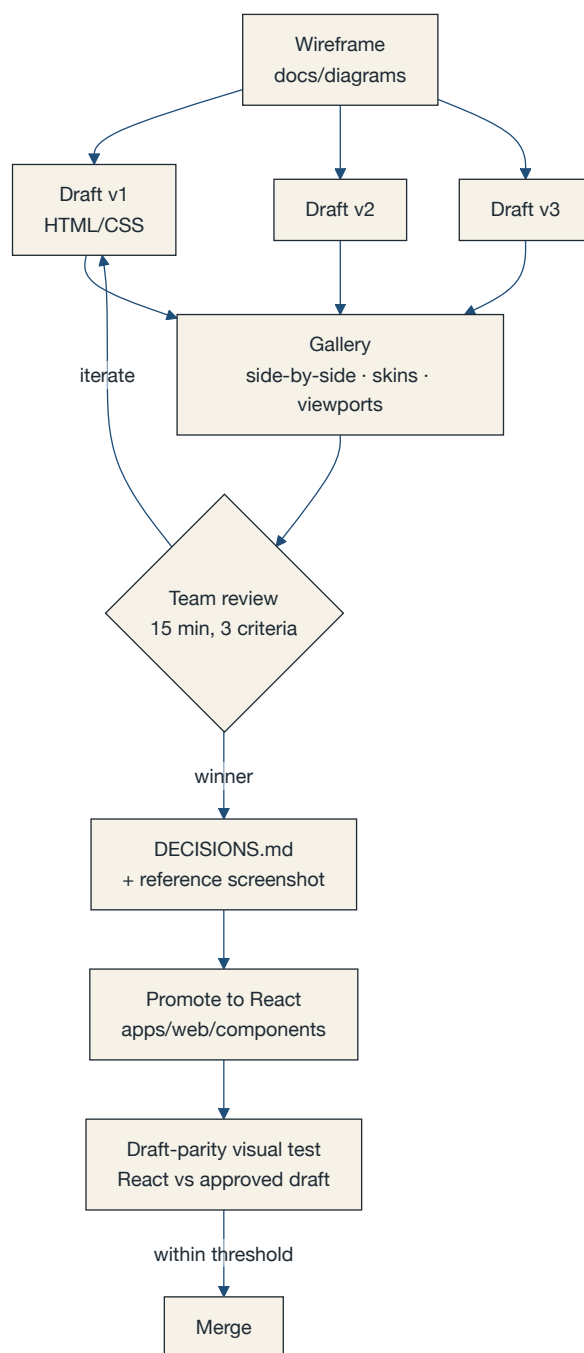
21.1 MONOREPO LAYOUT

mantle/	
├ README.md	plan.md · plan.pdf
├ compose.yaml	the single-command stack (profiles: default, dev, test, drafts, ml, obs)
├ .env.example	all configuration (Appendix E)
├ Makefile	thin aliases (make up / test / drafts / bench / pdf)
├ pyproject.toml · uv.lock	uv workspace root (Python 3.12)
├ package.json · pnpm-workspace.yaml · pnpm-lock.yaml	
├ apps/	
│ └ web/	Next.js app (App Router)
│ │ └ app/	routes (§18.4)
│ │ │ └ components/	ui/ (design system) · twin/ · blueprint/ · charts/ · overlays/
│ │ │ └ three/	scene graph, materials, shaders (glsl), kinematics worker
│ │ │ └ lib/	api client (generated), ws, stores, deep links, units
│ │ │ └ tests/	vitest units · playwright e2e & visual
│ └ gateway/	FastAPI app (routers/, ws/, deps/, tests/)
├ worker/	Arq tasks
├ scada/	synthetic SCADA publisher, replay, faults
├ ingest/	MQTT/OPC-UA/CSV adapters, auditor
├ packages/	
│ └ tokens/	tokens.json → tokens.css (drafts + web) · tokens.ts
│ └ core/	(py) schemas, units, provenance, ids
│ └ physics/ estimator/ optimize/ ml/ rl/ synth/ arena/ copilot/ db/	(py packages)
│ └ api-types/	(ts) generated OpenAPI types
├ drafts/	HTML/CSS drafts (§21.3)
│ └ index.html	gallery: variants side by side
│ └ screens/<screen>/v1.html v2.html v3.html ...	
│ └ components/<component>/v1.html ...	
│ └ fixtures/	JSON mirrors of API responses
├ DECISIONS.md	which variant won and why
├ assets/3d/	mantle_well.glb, variants, depth_map.json, kinematics.json, anchors.json
├ data/	schema/ (JSON Schema, dictionary) · seeds/ (manifests; data via DVC or releases)
├ bench/	arena configs, official results, RESULTS.md (generated)
├ evidence/	sources/ · assumptions.yaml · model_cards/
├ docs/	adr/ · diagrams/ (wireframes, rendered) · demo/ (scripts) · api/ (generated)
├ infra/	docker/ (Dockerfiles) · caddy/ · mosquito/ · grafana/
├ .github/workflows/	ci.yml · nightly.yml · release.yml

21.2 TOOLCHAIN

AREA	TOOL
Python env & deps	uv (workspace, uv.lock, uv run, uv sync --frozen in Docker/CI)
Python quality	ruff (lint + format), pyright (strict on packages), pytest, hypothesis
JS/TS	pnpm workspaces; Next.js; TypeScript strict; Biome (lint + format); Vitest; Playwright
Git hooks	lefthook: ruff, biome, typecheck on staged files; conventional-commit message check
Docs	Markdown + Mermaid; make pdf renders plan.pdf (Mermaid → SVG → Chrome headless)
Large data	Git LFS for GLB; seed snapshots as release artefacts (or DVC)

21.3 DRAFT-FIRST UI WORKFLOW (HTML/CSS → NEXT.JS)



- **Rules for drafts:** plain HTML + CSS, no framework, no build step. They import `packages/tokens/tokens.css` (the same tokens the app uses) and read `drafts/fixtures/*.json` via a tiny inline `fetch` for realistic numbers. 3D is represented by a still render (or a minimal three.js CDN embed when evaluating glass-over-scene).
- **Variants:** at least 2 for every screen and every non-trivial component (callout, recommendation card, plate, peek card, timeline, condition card). Each variant targets a clear hypothesis ("denser right stack", "callouts as dots until hover").
- **Gallery:** `drafts/index.html` shows variants side by side in iframes at 1440×900 and 1920×1080, with a Glass/Paper toggle, an overlay "onion-skin" comparer and a grayscale check (colour carries meaning only).
- **Review criteria:** (1) does the screen answer its one question in 3 seconds, (2) ≤ 7 visible numbers per region and no paragraphs, (3) legible on a projector (tested at 50% brightness).
- **Promotion checklist:** tokens only (no raw colours) · semantic HTML preserved · keyboard path defined · states covered (loading/empty/error/stale) · Storybook-free: the draft is the reference, and the **draft-parity test** (Playwright screenshot of the React component with fixture data vs the approved draft screenshot, perceptual diff $\leq 2\%$) guards drift.

21.4 ENGINEERING CONVENTIONS

- **Trunk-based development**, short-lived branches `feat/...`, `fix/...`; PRs need green CI + 1 review; squash merge.
- **Conventional Commits.** The changelog is generated.
- **ADRs** in `docs/adr/NNNN-title.md` for every non-obvious decision (Appendix F).

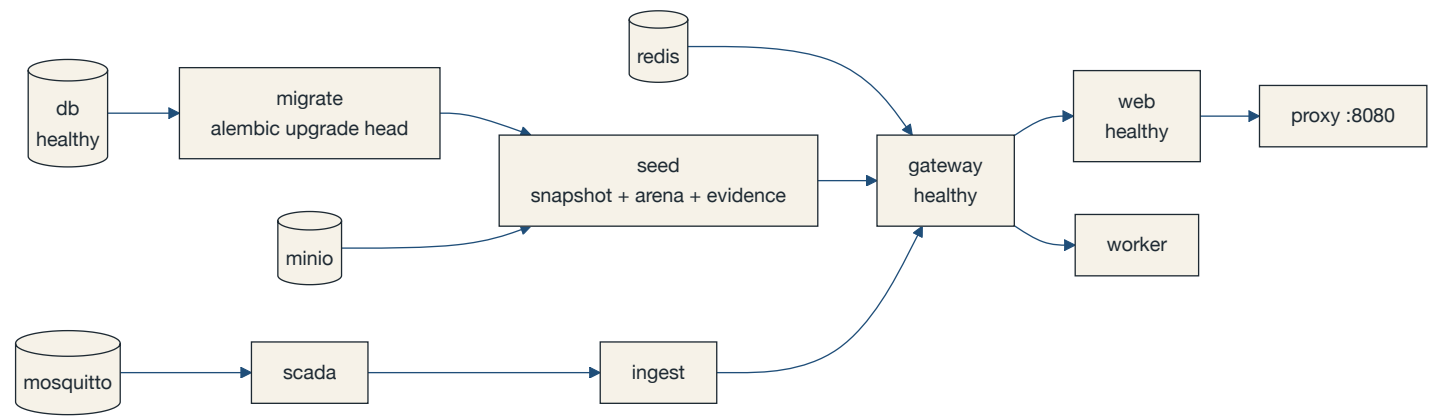
- **Definition of Done:** code + tests (§23) + docs updated + provenance tags on new numbers + demo deep link where visible.

22. DOCKER: ONE COMMAND TO RUN EVERYTHING

22.1 THE COMMAND

```
docker compose up
```

That's it. On first run it builds images, starts infrastructure, runs migrations, seeds Baghewala-S and the official Arena run, then serves the app at **http://localhost:8080**. Subsequent runs start in seconds. `make up` is an alias.



22.2 SERVICES

SERVICE	BUILD / IMAGE	PORT (HOST)	DEPENDS ON	HEALTHCHECK
proxy	caddy:2	8080	web, gateway healthy	GET /healthz
web	infra/docker/web.Dockerfile (Next standalone)	—	gateway	GET /api/health
gateway	infra/docker/python.Dockerfile target gateway	— (8000 internal)	db, redis, minio, seed done	GET /api/v1/health
worker	same image, target worker	—	db, redis, minio	arq health key
scada	same image, target scada	—	mosquitto, seed	heartbeat topic
ingest	same image, target ingest	—	mosquitto, redis, db	heartbeat
migrate	same image, one-shot	—	db healthy	exit 0
seed	same image, one-shot, idempotent	—	migrate, minio	exit 0
db	timescale/timescaledb:latest-pg16	5432 (dev profile only)	—	pg_isready
redis	redis:7	—	—	redis-cli ping
minio	minio/minio	9001 console (dev)	—	/minio/health/live
mosquitto	eclipse-mosquitto:2	1883 (dev)	—	\$SYS topic

22.3 PROFILES

COMMAND	WHAT YOU GET
docker compose up	Full demo stack (production builds, seeded, offline-capable)
docker compose --profile dev watch	Hot reload: Next.js dev server + uvicorn --reload, source synced via Compose Watch; DB/MinIO ports exposed
docker compose --profile test run --rm tests	Entire test suite in containers (§23), exits non-zero on failure
docker compose --profile drafts up drafts	Draft gallery at http://localhost:8090
docker compose --profile ml up mlflow	MLflow UI + training jobs (make train)
docker compose --profile obs up	Prometheus + Grafana dashboards
docker compose --profile bench run --rm bench	Official Arena run → bench/

22.4 IMAGE DESIGN

- **Python image** (multi-stage, one Dockerfile, multiple targets): FROM python:3.12-slim, copy uv from ghcr.io/astral-sh/uv, uv sync --frozen --no-dev --package <app> with cache mounts, UV_COMPILE_BYTECODE=1, non-root user. Numba cache is warmed at build time for the physics kernels.

- **Web image:** `node:24-slim` build stage with `pnpm`, then `next build` with `output: "standalone"`; the runtime stage copies `.next/standalone + static + public`; non-root.
- **Sizes (targets):** python $\leq$ 1.2 GB (PyTorch CPU wheels), web $\leq$ 250 MB.
- **Offline demo bundle:** `make bundle` runs `docker save` on all images plus the seed snapshot into a single tarball for the demo laptop; `make load` restores it with no network needed.

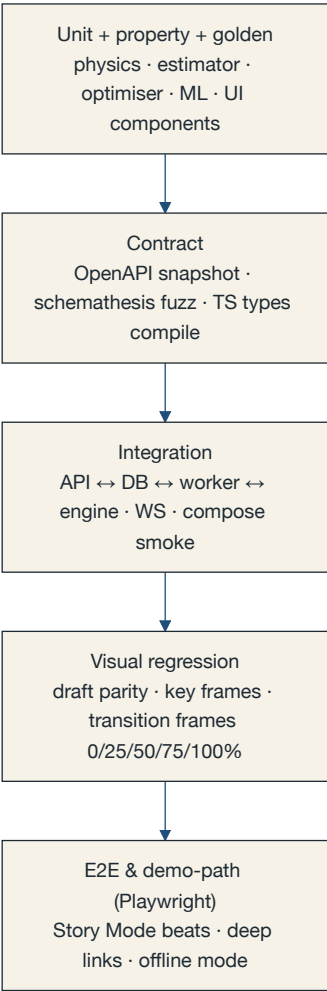
22.5 RESOURCE NEEDS

Demo laptop: 4+ cores, **16 GB RAM** recommended (8 GB minimum with `MANTLE_WORKERS=1`), ~10 GB disk. GPU not required (RL training and big Arena runs are done ahead of time; the official results ship in the seed).

23. TESTING STRATEGY: TESTS FOR EVERYTHING

Principle: every claim Mantle makes on screen is backed by a test that would fail if the claim became false. The physics is tested against textbooks, the numbers against the physics, the screens against the drafts, and the demo against a script.

23.1 TEST PYRAMID



23.2 WHAT IS TESTED, HOW, AND THE GATE

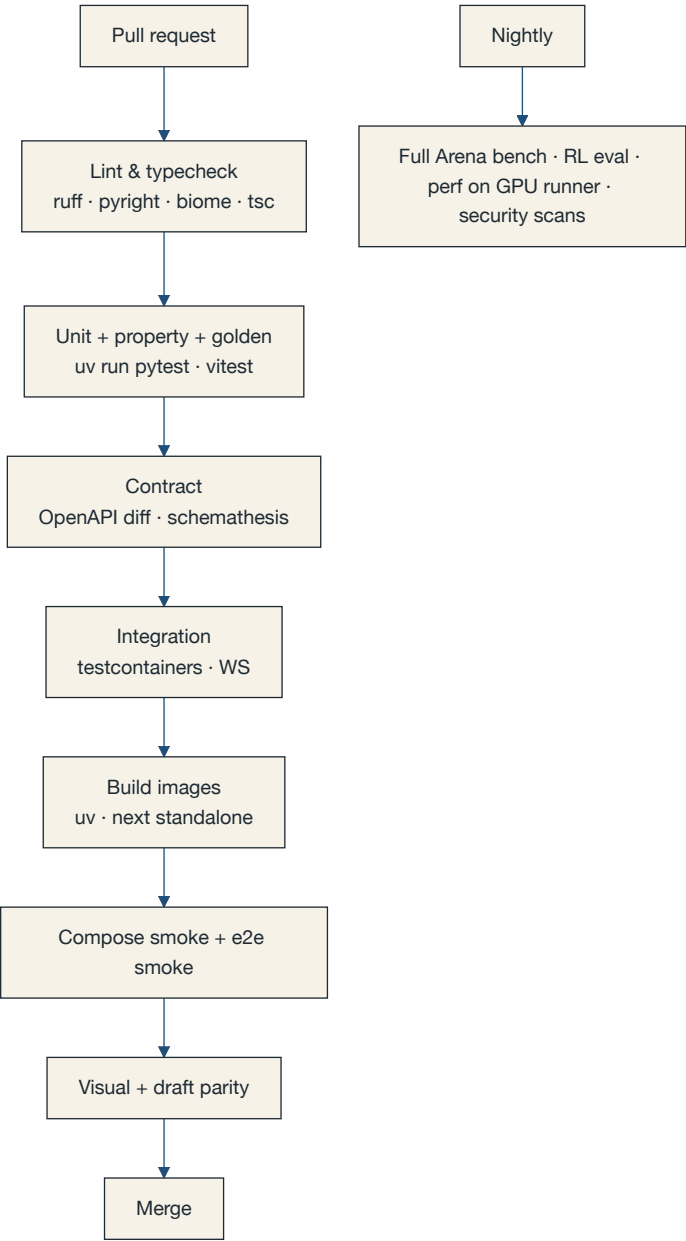
LAYER	TOOL	WHAT	GATE
Physics golden	pytest	API RP 11L worked examples (PPRL/MPRL/PD within 2%); Marx–Langenheim $G(t_D)$ table values; IAPWS-IF97 check values; Walther fit round-trip; Boberg–Lantz ratio limits; Ramey vs published example; Goodman allowable; four-bar kinematics vs API 11E geometry	100% pass
Physics invariants	hypothesis	μ strictly decreasing in T; $PD \geq 0$; energy balance closes $\leq 1\%$; soak never heats without a source; fluid-level mass balance; float margin decreases with SPM at fixed μ	100% pass
Numerics	pytest	Wave-equation CFL stability; grid-refinement convergence (order ≈ 2); thermal-grid conservation; deterministic outputs for fixed seeds	pass
Estimator	pytest	Twin experiments: truth run + noisy obs $\rightarrow$ EnKF recovers parameters inside P10–P90 in $\geq 75\%$ of trials; coverage $\approx$ nominal; innovation test flags injected stuck sensors within N hours	thresholds
Optimiser	pytest	Safety Gate rejects each seeded bad plan (one per constraint); INFEASIBLE returned (never a fallback) on impossible requests; Pareto set non-dominated; surrogate top-k re-verified with physics	100% pass
ML	pytest + pandera	Data contracts on training frames; metric floors (card classifier macro-F1 ≥ 0.9 on synthetic hold-out; ECE ≤ 0.05); determinism with seeds; slice metrics per class and condition; model cards generated	thresholds
RL	pytest	Env API compliance (Gymnasium checker); reward sanity; shielded policy has 0 unsafe executed actions on the eval set	pass
Arena	pytest (small)	CRN reproducibility (same seed $\rightarrow$ identical metrics); stats code vs known distributions; CI mini-arena: Mantle – Static > tolerance band	band
Synthetic data	pytest	Generator outputs match schema; distributions within calibration anchors; faults appear where injected	pass
API	pytest + httpx	Every endpoint: happy path, validation errors (Problem shape), 404/409/422; auth persona handling	100% endpoints

LAYER	TOOL	WHAT	GATE
API contract	schemathesis	Property-based fuzzing from OpenAPI (no 5xx on valid input); OpenAPI snapshot diff reviewed in PRs	no 5xx
DB	pytest + testcontainers	Alembic up/down round-trip; constraints; ledger hash-chain trigger + tamper detection; hypertable/aggregate refresh	pass
WebSocket	pytest	Subscribe/unsubscribe, seq ordering, back-pressure drop, resync	pass
Ingest	pytest	MQTT/CSV adapters; unit conversions; auditor catches seeded issues (gaps, duplicates, impossible values, unit mismatch)	100% of seeded issues
Copilot	pytest (LLM mocked + recorded)	Tool routing; numeric verifier blocks invented numbers (adversarial set); offline template fallback	100% on adversarial set
Web units	Vitest + Testing Library	Stores, deep-link encode/decode, unit conversion, formatters, WS manager, callout layout solver (no overlaps)	≥ 85% lines in lib/
Components	Vitest + Playwright CT	Each component in all states (loading/empty/error/stale/alarm) with fixtures	all states
Draft parity	Playwright screenshots	React component vs approved draft screenshot (perceptual diff ≤ 2%)	threshold
Visual regression	Playwright	30 key frames incl. transition at 0/25/50/75/100% (time-seeked GSAP timeline, fixed seed, fixed camera)	diff ≤ 0.5%
3D asset	glTF-Validator + custom script	GLB validates; all required nodes, anchors and pivots present by name; triangle and draw-call budgets; line-art registration ≤ 1 px vs ortho render	pass
Shaders	Playwright (headless GPU)	Compile all materials at load; no WebGL errors in console	0 errors
Performance	Playwright trace + Lighthouse CI	Twin ≥ 55 fps median on a CI GPU runner (nightly; demo-laptop check manual); transition 0 dropped frames; LCP < 2.5 s	budgets §24
Accessibility	axe-core	Paper screens: 0 serious violations; keyboard path through every interaction	0 serious
E2E demo path	Playwright	Story Mode script start to finish; every Proof "demo" link resolves; offline mode (network disabled)	pass
Compose smoke	CI job	docker compose up → health green in < 5 min → e2e smoke	pass
Security	pip-audit, pnpm audit, Trivy, gitleaks	No high CVEs; no secrets	0 high

23.3 GOLDEN SCENARIOS

Six seeded scenarios (the Condition Deck presets) run end-to-end on every CI build. Their headline numbers (oil, SOR, float hours, the recommendation) and screenshots are snapshot-tested. Any change must be deliberate, reviewed and re-baselined with a reason in the PR.

23.4 CI PIPELINE



Coverage targets: physics/estimator/optimise ≥ 90% lines + branches on the Safety Gate = 100%; gateway ≥ 85%; web lib/ ≥ 85%. Mutation testing (mutmut) runs weekly on mantle-physics and the Safety Gate; surviving mutants are triaged.

23.5 DEMO-DAY TESTS (MANUAL CHECKLIST, AUTOMATED WHERE POSSIBLE)

Projector at 1080p and 4K · offline (Wi-Fi off) full run · battery-saver mode fps · second-screen presenter notes · phone remote pairing · Judge Challenge kiosk loop for 2 hours without leaks (memory profile) · recovery from browser refresh mid-transition.

24. PERFORMANCE AND ACCESSIBILITY BUDGETS

24.1 PERFORMANCE BUDGETS (DEMO LAPTOP CLASS: APPLE M1/M2 OR RTX LAPTOP)

ITEM	BUDGET
Twin frame time	≤ 16.6 ms (60 fps) at 1440p; floor 30 fps on integrated Intel Iris Xe at 1080p (auto-quality)
Transition	0 dropped frames over 2.2 s (pre-warm shaders; compile all materials at load)
First meaningful paint	< 2.5 s (Twin shell); GLB streamed with a progressive placeholder silhouette
GLB	≤ 25 MB meshopt; KTX2 textures
Blurred glass area	≤ 30% of viewport, ≤ 5 surfaces
WS state update	2 Hz state, per-stroke cards; < 50 ms end-to-end locally
Optimization (per well, joint)	< 8 s to the first Pareto front, < 30 s to the verified plan
Scrub latency	< 50 ms per frame (precomputed dense trajectories and client-side interpolation of the returned arrays, not physics)

Auto-quality ladder: particles → SSAO → shadows → post-FX → DPR, stepped down by measured frame time.

24.2 TESTING

See §23.

24.3 ACCESSIBILITY

Keyboard navigation for everything, including 3D part selection (Tab cycles through anchors). Colour-blind-safe thermal ramp. Text contrast AA on both skins. `prefers-reduced-motion` honoured. Every chart has a data-table fallback.

25. THE THREE IMPACTS: ECONOMIC, ENVIRONMENTAL, SOCIAL

Method rule: the numbers come from the Strategy Arena (§16). every impact number is **Mantle policy vs baseline policies on identical simulated wells** (Baghewala-S, N seeds, mean ± sd), labelled SIM. Where we extrapolate to the field ("if applied to 33 wells..."), we label it **ILLUSTRATIVE** and show the arithmetic.

25.1 ECONOMIC

METRIC	DEFINITION	WHERE
Incremental oil	Δ cumulative oil vs baseline, per cycle and per 180 d	Impact, Battle
Coupling Dividend	§15.6	Twin, Impact
Lifting cost per bbl	(steam + power + water + maintenance + chemicals) / oil	Impact
Steam cost per bbl, marginal steam ROI	§13.3	Cycle
Avoided failure cost	Δ expected failures × (workover + deferred oil)	Impact, Lift
Deferred-production avoided	Δ downtime × rate	Impact
Equipment life	Δ Goodman-loading cycles consumed per bbl	Lift
Energy security (national frame)	Barrels of domestic heavy oil recovered more efficiently. India imports most of its crude (cite the latest PPAC figure).	Impact narrative

25.2 ENVIRONMENTAL

METRIC	DEFINITION
SOR (CWE)	Steam per barrel. Lower means less fuel and water per barrel.
Steam avoided (t)	Δ steam vs baseline for equal or greater oil
CO₂e per bbl	Steam-generator fuel × emission factor + grid power × grid factor (CEA baseline database; verify the current value)
Solar-steam share	Fraction of heat from solar thermal (hybrid condition)
Water intensity	m ³ fresh water per bbl (steam feed – recycled produced water). Framed around the Thar desert: water is the scarcest resource there.
Heat utilization	Useful reservoir heat / fuel energy (from the heat balance)
Wasted thermal energy	Surface + wellbore + caprock losses avoided (VIT, timing)
Workover footprint	Δ interventions (each is rig mobilization, diesel and waste)

25.3 SOCIAL (GROUNDED, NOT DECORATIVE)

METRIC	DEFINITION / RATIONALE
Worker safety: high-risk states avoided	Δ hours in float/impact/overload states (these precede failures and emergency interventions)
Heat-stress exposure avoided	Δ unplanned site visits × heat index for the date (Thar summers exceed 45 °C). Remote diagnosis means fewer people on hot pads.
Emergency interventions avoided	Δ unplanned workovers
Knowledge retention	Captured override reasons and "known well behaviours". Senior operators' judgement becomes institutional memory.
Skilling	Challenge mode as a training simulator for young engineers; hours trained, score improvement
Decision transparency	% of actions with a full provenance trail
Regional economy (narrative)	Sustained production in Rajasthan supports local employment and state revenue. Qualitative only; don't quantify without sources.

25.4 IMPACT SCREEN LAYOUT (SEE ALSO §6.8)

Three paper columns (Economic · Environmental · Social), each with 4 hero numbers (± band, SIM tag), a cumulative chart vs both baselines (historical practice, pump-off controller) and a "how we computed this" fold-out showing the formula and data. A top toggle switches between **per well, field (simulated 30 wells)** and **illustrative extrapolation**.

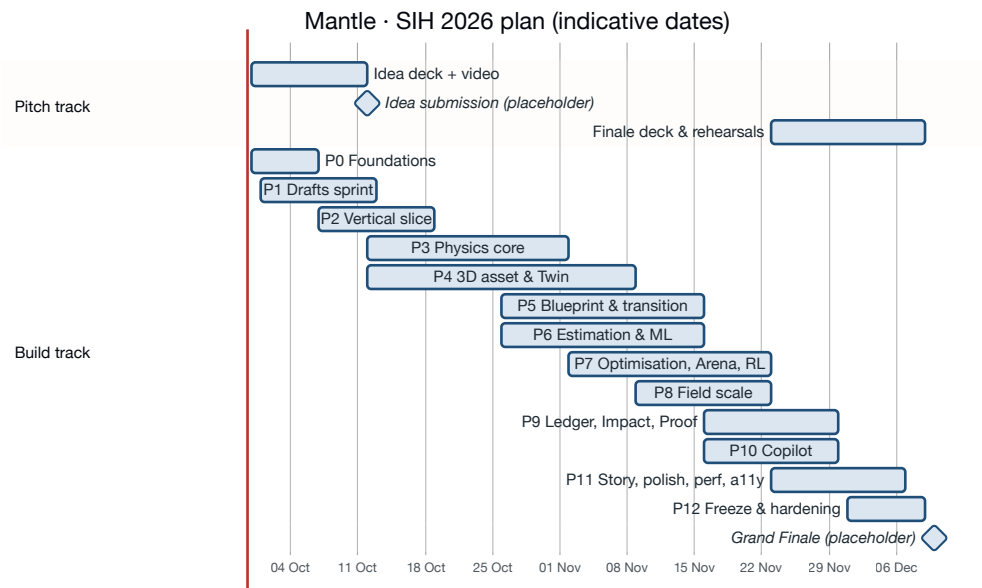
26. HACKATHON PLAN

Dates below are indicative. Replace them with the official SIH 2026 calendar from the portal and the institute SPOC. The sequence and durations are what matter.

26.1 SIH STAGES AND WHAT WE DELIVER AT EACH

STAGE	WHAT JUDGES OR EVALUATORS SEE	OUR DELIVERABLE	OWNER
Internal hackathon (institute)	Idea pitch + early prototype	5-min pitch, the vertical slice (Phase 2), a 60-s video	Product lead
Idea submission (SIH portal, via SPOC)	The SIH idea PPT template + optional video	Deck per §29.2; 2-min demo video recorded from Story Mode	Product lead + 3D lead
Shortlisting	The deck	— (keep building)	—
Grand Finale (multi-round evaluation, typically ~36 h, on site)	Live product, progress between rounds, Q&A	Frozen demo build + planned on-site increments (§26.4)	Whole team

26.2 TIMELINE



26.3 WEEKLY RHYTHM

- Monday: plan (30 min), pick the week's golden-scenario targets.
- Daily: 10-min stand-up; the demo build must stay green (main always demoable).
- Wednesday: draft review (15 min per screen, three criteria).
- Friday: full Story Mode run-through, recorded. One "judge" teammate asks 5 hostile questions and we log the answers (feeds §30).
- Every 2 weeks: mentor/faculty review with the Proof screen.

26.4 GRAND FINALE PLAYBOOK (≈ 36 H, ADAPT TO THE OFFICIAL FORMAT)

Tactic: arrive with a finished, frozen product. Reserve 3 small, pre-scoped, visible increments to build on site, so each evaluation round sees real progress. Also keep a slot for mentor-requested changes.

PRE-SCOPED ON-SITE INCREMENTS (EACH ≤ 4 H, PRE-RESEARCHED)	
1.	A new Condition Deck preset from mentor input (e.g. a specific well pattern or climate) plus an Arena re-run
2.	An OIL-requested metric or unit added to the Blueprint and Impact
3.	A new card-class or fault type, with the injector and detection shown live

HOURS	ACTIVITY
0–2	Set up the booth: demo laptop + external display + phone remote; offline check; Judge Challenge kiosk on a second laptop
2–6	Mentor round: listen, log requests, triage to increments
6–12	Build increment 1; rehearse the 7-min script twice
Round 1	Full demo (§29.3) + Q&A (§30)

HOURS	ACTIVITY
12–20	Increment 2; sleep rota (two people always awake; the physics owner sleeps before Round 2)
Round 2	Demo with "what changed since Round 1" opener (show increment 1 and 2)
20–30	Increment 3; final Arena run with any new preset; freeze by hour 30
Final round	7-min pitch, clean, rehearsed; Q&A; leave a printed Blueprint well card with judges
Always	One person owns the demo laptop; nobody pulls or updates it after the freeze

26.5 BOOTH AND KIT CHECKLIST

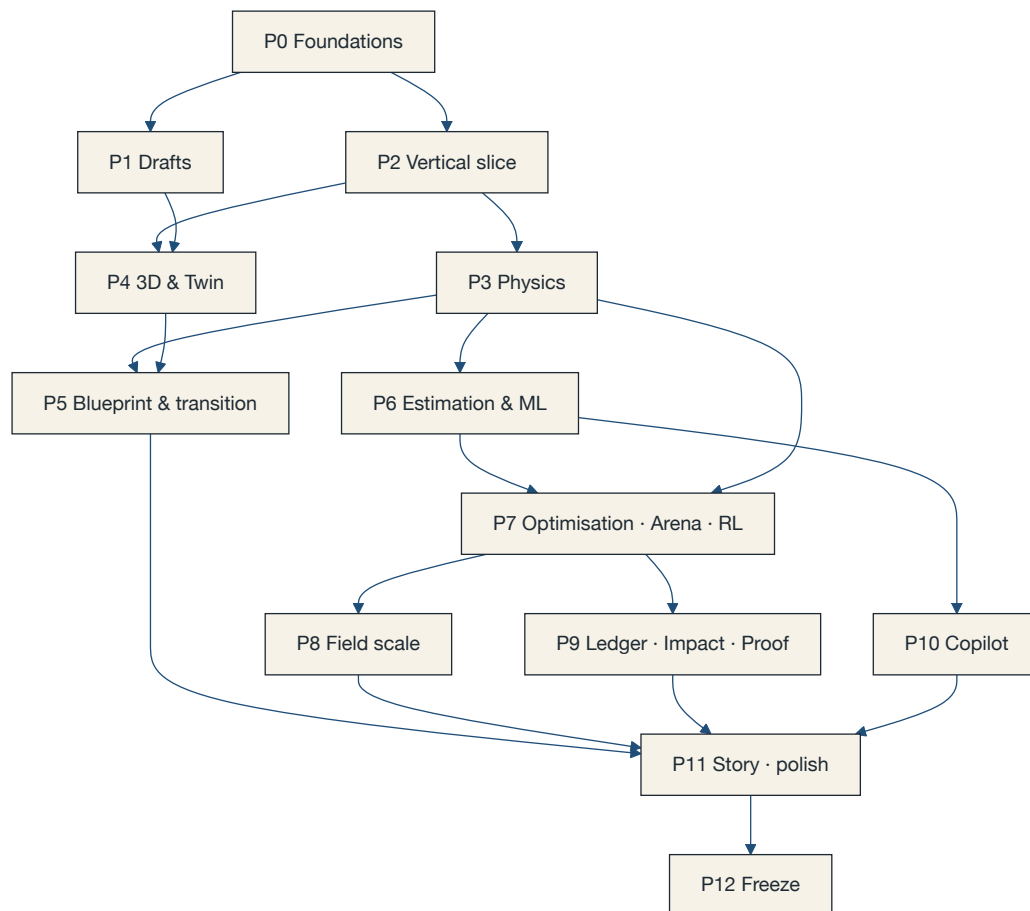
Demo laptop (charged, performance mode, notifications off, `make load` bundle already restored) · spare laptop with the same bundle · HDMI/USB-C adapters · clicker + phone remote · printed A3 Blueprint well cards (×20) · printed cycle receipts · QR standee to `/m/BGW-17` and `/play` · backup video of the full demo on a USB drive and on both laptops · extension board.

26.6 TEAM ROLES DURING THE EVENT

ROLE	DURING DEMO	DURING Q&A
Product/domain lead	Narrator	Impact, PS mapping, OIL deployment
3D lead	Drives the laptop	Visuals, performance
Physics lead	On call	Physics, rod float, CSS
ML/estimation lead	On call	EnKF, models, RL, uncertainty
Backend/optimisation lead	Arena run	Architecture, Arena, safety
Frontend lead	Kiosk + phone remote	UX, accessibility, drafts

27. TECHNICAL IMPLEMENTATION PLAN

Each phase has a **goal**, **tasks by role**, **deliverables**, **tests** and **exit criteria**. Phases overlap (see the Gantt). `main` must always be demoable.



P0: FOUNDATIONS (WEEK 1)

- **Goal:** the empty shell runs with one command, and CI is green.
- **Tasks:**
 - *Backend:* uv workspace skeleton (`packages/*`, `apps/*`); FastAPI `/health`, `/meta`; Alembic baseline; Compose with db, redis, minio, mosquito, migrate, seed (no-op); Caddy.
 - *Frontend:* Next.js app with App Router shell, tokens pipeline (`tokens.json` → CSS), persistent `SceneHost` placeholder, routes for the 6 screens.
 - *Physics:* package skeleton + first golden test (API RP 11L example).
 - *Product:* source-truth folder; SIH traceability table; ADR-0001...0005.
 - *All:* lefthook, CI (lint, typecheck, unit, compose smoke).
- **Deliverables:** `docker compose up` shows 6 empty screens behind Caddy; CI green; README accurate.
- **Exit:** a fresh clone runs in ≤ 10 min on a teammate's machine.

P1: DRAFTS SPRINT (WEEKS 1–2, PARALLEL)

- **Goal:** approved HTML/CSS drafts for every screen and core component.
- **Tasks:** fixtures from the §19 schemas; ≥ 2 variants for 6 screens × 2 skins + 12 components (callout, peek, sheet, dialog, rec card, risk bars, timeline, plate, title block, condition card, scoreboard, receipt); gallery; reviews; DECISIONS.md.
- **Exit:** every screen has a winning draft with a reference screenshot; tokens frozen v1.

P2: VERTICAL SLICE (WEEKS 2–3)

- **Goal:** one well end-to-end: lumped physics → state → Twin (placeholder GLB) → Blueprint Plate 3 → Timeline.
- **Tasks:** lumped thermal + Walther + RP 11L + float margin; `/wells/{id}/state`, `/envelope`, `/timeseries`; WS state; placeholder 3D well (primitives) with 3 callouts; Plate 3 SVG; timeline scrub.
- **Exit:** scrub 60 days and watch the SPM envelope cross actual SPM in both views; demo-able to mentors.

P3: PHYSICS CORE (WEEKS 3–5)

- **Goal:** the full ~~mantle~~-physics of §14, tested.
- **Tasks:** IAPWS steam + surface loss; Ramey/VIT wellbore; Marx–Langenheim + r–z thermal grid; Boberg–Lantz inflow; annulus material balance; Gibbs wave equation (Numba) + card generation; four-bar kinematics; Goodman, torque, hold-down, casing stress, deposition; economics, CO₂, water; cut-off rule.
- **Tests:** all golden + property + convergence tests (§23.2).
- **Exit:** physics coverage ≥ 90%; cards look right to a domain review; ≤ 50 ms per sim-day per well.

P4: 3D ASSET & TWIN (WEEKS 3–6)

- **Goal:** the final GLB integrated; the Twin looks finished.
- **Tasks:**
 - *3D agent*: the §8 brief.
 - *3D lead*: asset validation script; materials; thermal-field shader from ~~/thermal-field~~; lenses; particles (flow); rod segments driven by u(z,t); callout layout solver; phase behaviours; Condition Deck visual responses; performance ladder.
- **Exit:** 60 fps on the demo laptop; all six lenses; all special states (§8.10) triggerable.

P5: BLUEPRINT & TRANSITION (WEEKS 5–7)

- **Goal:** the living drawing and the signature transition.
- **Tasks:** Plates 1–6 + tracks from the API; SVG line-art projection from ~~LINE_*~~; hatch dissolve shader; paper wipe; the GSAP master timeline authored in Theatre; FLIP of numbers into the BOM; reverse; reduced motion; export sheet.
- **Exit:** transition visual tests at 0/25/50/75/100%; 0 dropped frames; the Twin and Blueprint numbers are identical at the same ~~t~~.

P6: ESTIMATION & ML (WEEKS 5–7)

- **Goal:** the twin tracks truth with honest uncertainty; perception models work.
- **Tasks:** Baghewala-S generator v1 (30 wells × 3 y, faults, noise); EnKF (M1); card classifier (M4); hazard (M5); anomaly (M6) + Petrobras 3W validation; residual learner (M3); conformal wrappers; SHAP; model cards; the data decision ADR (§17.7).
- **Exit:** EnKF twin-experiment tests pass; classifier and hazard metrics meet floors; Proof shows model cards.

P7: OPTIMISATION, ARENA, RL (WEEKS 6–8)

- **Goal:** the evidence engine.
- **Tasks:** cycle surrogate (M2); NSGA-III + physics verification + Safety Gate + refinement; receding horizon; Stroke Shaper; Coupling Dividend + Shapley; strategies S0–S8; the Gymnasium env + shielded PPO training (M8); Arena runner with CRN + stats; the official run; the Arena screen.
- **Exit:** §16.5 criteria evaluated on the official run; the Arena screen loads it in < 1 s; the live mini-run finishes in < 30 s.

P8: FIELD SCALE (WEEKS 7–8)

- **Tasks:** field map (MapLibre cream style, offline tiles); triage ranking; CP-SAT steam scheduler + dialog; similarity search; morning brief; fleet table; field time-lapse.
- **Exit:** the Field screen matches its draft; the scheduler re-solves in < 5 s for 30 wells × 180 d.

P9: LEDGER, IMPACT, PROOF (WEEKS 8–9)

- **Tasks:** recommendations → decisions → hash-chained ledger (trigger + verify); outcomes and counterfactual scoring; calibration + override analytics; Impact summary/series/receipt; Proof cards, CSV import + auditor, assumption and source registries, twin health.
- **Exit:** all Proof buttons work live; ledger tamper test passes.

P10: COPILOT (WEEKS 8–9)

- **Tasks:** tool layer over the engine; LLM integration; numeric verifier + adversarial test set; offline templates; answer cards with "Show me"; command palette actions.
- **Exit:** 100% of the adversarial set is blocked or corrected; offline mode works.

P11: STORY, POLISH, PERFORMANCE, ACCESSIBILITY (WEEKS 9–10)

- **Tasks:** Story Mode script + presenter remote; intro sequence; Judge Challenge kiosk; sound (optional); units toggle; performance pass; axe fixes; the pitch video from the time-lapse export; printed well cards.
- **Exit:** three clean end-to-end rehearsals; e2e demo-path test green.

P12: FREEZE & HARDENING (WEEK 10)

- **Tasks:** bug bash; offline bundle; demo-laptop checklist (§23.5); final Arena run; plan.pdf Rev C; the backup video.
- **Exit:** tagged release ~~v1.0-final~~; bundle verified on both laptops.

28. VISUAL EVIDENCE MAP: WHAT JUDGES DON'T SEE DOESN'T EXIST

Every claim needs a **visual**, a **place** (deep link), **data** behind it and a **test** that keeps it true.

#	CLAIM	THE VISUAL THAT PROVES IT	WHERE	BACKED BY	TEST
1	We model the whole well-to-surface chain	Lens switching on one 3D model; the Blueprint depth tracks aligned on one depth axis	Well · Twin/Blueprint	\$14 physics	golden + parity tests
2	Heat decays and viscosity rises	Thermal lens fading over a timeline scrub; the Walther chart marker moving	Well	thermal grid, Walther	visual frames
3	That makes rods float	Bridle slack + impact flash in 3D; float margin bar crossing zero; Plate 3 envelope crossing	Well	wave eq., margin	special-state tests
4	We detect it	Card shape changes; the classifier label on Plate 2; P2 alarm	Well, Alerts	M4	classifier floors
5	We optimise CSS + SRP jointly	Optimiser progress stream; the recommendation's binding constraint	Well	\$15.4	optimiser tests
6	Joint beats separate	Coupling Dividend decomposition bars	Arena, Twin chip	S7 – S5	Arena CI band
7	Mantle beats every alternative	Production-output race , scoreboard with CIs, % of oracle	Arena	official run	\$16.5 criteria
8	Not just tuning, not just RL	Tuned-static and RL lines on the same chart	Arena	S1, S6	Arena
9	Safe by construction	"1,120 rejected by Safety Gate" counter; the ablation "no Safety Gate" disqualified	Well, Arena	Safety Gate log	100% gate tests
10	It knows when it's unsure	P10–P90 fans; OOD and disagreement badges; the sensor-fault pause	Well, Proof	EnKF, M6	twin experiments
11	It learns from outcomes	Predicted-vs-realised chart; calibration scatter improving by cycle	Ledger	outcomes	ledger tests
12	Humans stay in charge	Approve / Modify / Reject dialog; override analytics	Well, Ledger	decisions	e2e
13	Tamper-evident audit	"CHAIN VERIFIED" seal; live verify	Ledger, Proof	hash chain	tamper test
14	Field-scale value	Steam scheduler Gantt; shadow price of a generator-day	Field	CP-SAT	solver tests
15	Environmental benefit	Impact column; cycle receipt ; solar-steam preset flipping CO ₂ /bbl	Impact	economics	golden scenarios
16	Social benefit	High-risk hours avoided; heat-stress-weighted visits avoided; knowledge notes count	Impact	\$25.3	golden scenarios
17	Real-data ready	Live CSV import with the auditor report; SCADA readiness diagram; Petrobras 3W result	Proof	ingest	auditor tests
18	Honest	Provenance mode tinting every number; assumption registry	any, Proof	provenance	contract tests
19	Runs anywhere	docker compose up on the judge's request (if asked)	—	Compose	smoke test
20	Answers the PS line by line	Traceability card with demo links	Proof	Appendix A	e2e link test

29. SCRIPTS: HOW WE TALK ABOUT MANTLE

29.1 THE 30-SECOND PITCH

"Baghewala's oil is as thick as honey. Oil India heats it with steam and lifts it with a 1.1-kilometre steel rod pump. But as the heat fades, the oil thickens, and the pump speed that was safe last week starts breaking rods. Today steam and pump are tuned separately, by experience. **Mantle** is a digital twin that sees the whole chain, from steam to rock to rod to rupees. It plans steam and pump as one decision, and it proves it: on identical simulated wells, it beats every alternative we could build, from today's practice to a reinforcement-learning agent. And it tells you when not to trust it."

29.2 IDEA-SUBMISSION DECK (SIH TEMPLATE)

SLIDE	TITLE	CONTENT	VISUAL
1	Title	Mantle · SIH26120 · team, institute	Cover drawing (plan.pdf page 1 style)
2	Problem & idea	The thermal-battery-on-a-steel-spring metaphor; "optimised separately" is the problem	Causal chain diagram (§1)
3	Proposed solution	Twin ⇌ Blueprint; the 6 screens; Coupling Dividend	Twin and Blueprint wireframes
4	Technical approach	Physics + EnKF + ML + joint MPC + Safety Gate; Next.js/R3F, FastAPI/uv, Docker	Architecture diagram (§18.1)
5	Feasibility & viability	Synthetic-first but real-data-ready; risks and mitigations; the prototype already running (screenshots)	Readiness diagram; Arena chart
6	Impact & benefits	Economic · environmental · social with the method	Impact wireframe
7	Research & references	Physics standards, public sources, competitor review	Reference list

Video (2 min): Story Mode auto-play with voice-over of §29.3 beats 1–6.

29.3 THE 7-MINUTE FINALE DEMO

TIME	BEAT	SCREEN & ACTION	SPOKEN LINE (VERBATIM GUIDE)
0:00	Hook	Intro sequence: the drawing comes alive into the 3D Twin at dusk	"This is a heavy-oil well in the Thar. Twelve hundred metres down, oil as thick as honey."
0:25	The machine	Slow orbit; callouts breathe; hover the polished rod	"We keep it moving with steam, and with a steel rod over a kilometre long that we push up and down five times a minute."
0:50	The coupling	Thermal lens; scrub the timeline +30 days; the heated zone fades; the viscosity callout climbs; the Mechanical lens turns the rods red	"As the heat fades, the oil thickens. The same pump speed that was safe last week now makes the rods float, and floating rods break."
1:40	The failure, visible	The float margin hits amber; bridle slack + impact flash	"That flash is impact loading. Oil India lists it in the problem statement. It's how rods fail."
2:00	Engineering view	Press B : the transition; Plate 3	"Here's the engineering. The red hatch is where rods float. The safe speed shrinks as the well cools, and today's fixed setting walks straight into it."
2:40	Why	Click <i>Why?</i> → waterfall → causal trace	"Sixty-two percent of this risk is viscosity, and the viscosity was decided forty days ago by the steam plan. That's why steam and pump have to be one decision."
3:10	Decide	Back to Twin → Plan; the progress stream; the recommendation	"Mantle searched about five thousand joint plans, threw out eleven hundred unsafe ones, and recommends this: slow the downstroke, keep the upstroke fast, and advance the next steam cycle four days."
3:50	Prove (3)	Arena: the production race plays	"Is it actually better? Same thirty wells, twenty random seeds, three years. Grey is today's practice, orange the industry pump-off controller, red the greedy strategy that burns out its rods, purple a reinforcement-learning agent. Blue is Mantle, at about ninety-four percent of the theoretical best."
4:40	The number	Coupling Dividend decomposition	"This bar is the value of optimising together instead of separately. It's the problem statement, as a number."
5:10	Trust	Inject "stuck sensor in a sandstorm"; the pause; ⌘K "why did you stop?"	"It also knows when not to trust itself. A sensor froze in a sandstorm, so Mantle paused and told us why. It won't optimise on bad data."
5:45	Human in charge (4)	Modify → approve → Ledger seal; outcome chart	"The engineer stays in charge. Every decision is recorded, tamper-evident, and scored later against what really happened."
6:15	Impact (5)	Three columns; the solar preset; the cycle receipt	"More oil from less steam, less water in a desert, and fewer people on hot pads at forty-eight degrees."
6:40	Close (6)	Proof → live CSV import; back to the Twin	"Every number here is labelled simulated or measured, and it plugs into Oil India's SCADA. A flight simulator and autopilot for heavy oil."

Fallbacks: if the 3D stutters, press **Q** (quality low) and continue. If the live Arena run is slow, show the official run ("this is the pre-computed official run; the live one is still going in the corner"). If the network is down, nothing changes, since everything is local. If the laptop fails, the spare laptop has the same bundle, and the backup video is on USB.

29.4 ARCHITECTURE WALKTHROUGH (90 SECONDS, WITH §18.1 ON SCREEN)

"Three layers. **At the edge**, a synthetic SCADA publishes real protocols: MQTT today, OPC-UA ready. It's the same path Oil India's data would take. **In the middle**, a Python engine managed with uv. One physics library, `mantle-physics`, is the only place any physics is computed. An Ensemble Kalman Filter keeps the twin synchronised with the well and quantifies uncertainty. The optimiser screens thousands of joint plans with a learned surrogate, verifies the best with full physics, and a deterministic Safety Gate has the final say. **At the top**, a Next.js app with a persistent 3D scene and a living SVG blueprint. It's all in Docker: one command, fully offline. And all of it is tested, from textbook physics to the pixels of the transition."

29.5 ONE-LINERS PER SCREEN (FOR THE BOOTH)

SCREEN	ONE-LINER
Well · Twin	"The well, live. Every colour is a calculation."
Well · Blueprint	"The same well, as an engineer draws it, and it's still moving."
Field	"Thirty wells, ranked by value at risk, and a steam schedule for the whole field."
Arena	"Nine strategies, the same wells, one winner, with confidence intervals."
Ledger	"Every decision, who made it, why, and whether it worked."
Impact	"What it means for money, the environment and people, and how we computed it."
Proof	"Press any button: the evidence runs live."

29.6 STAGE ROLES AND HAND-OFFS

Narrator (product lead) speaks. The 3D lead drives. For questions, the narrator routes by topic (§26.6) with "X *built that*. X?". **Never** say a simulated number is a field result. Say "in our simulated field...".

30. JUDGES' Q&A

30.1 IMPACTS AND BENEFITS: OUR ARGUMENTS

ECONOMIC

Claim	Mantle raises oil recovered per cycle and lowers lifting cost per barrel, mainly by avoiding rod failures and by timing steam better.
Mechanism	(1) The safe-SPM envelope stops over-speed, so fewer float and impact events, fewer rod failures and workovers, less deferred production. (2) Marginal-value cut-off and a joint steam design mean steam is spent where it returns the most oil. (3) The Stroke Shaper recovers production without extra steam. (4) The field steam scheduler allocates a scarce generator to the highest-value wells.
Evidence shown	Arena scoreboard (net value Δ with CI, failures Δ); Coupling Dividend; steam scheduler shadow price; Impact economic column.
Method	Policy vs policy on identical simulated wells; costs from the assumption registry (steam ₹/t, power ₹/kWh, workover ₹); Monte-Carlo over seeds.
Caveat, said proactively	"Absolute rupees depend on Oil India's cost data, which we don't have. The <i>relative</i> ranking and the mechanisms are what we show. The EnKF is how we'd calibrate to their numbers."
Why OIL cares	OIL's Rajasthan strategy emphasises production optimisation and competitive lifting cost.

ENVIRONMENTAL

Claim	Fewer tonnes of steam, less fuel gas, less water and lower CO ₂ e per barrel.
Mechanism	Lower SOR (less steam per barrel) → less fuel burned → less CO ₂ . Less steam also means less boiler feed water, which matters in the Thar desert. VIT and timing reduce wasted heat. The solar-thermal hybrid option shifts steam to daytime solar. Fewer workovers mean fewer rig moves and less diesel and waste.
Evidence shown	Impact environmental column; cycle receipt; the "Sun-powered steam" preset; the heat-balance Sankey (where the heat goes).
Method	Fuel = steam heat / generator efficiency × emission factor (IPCC default for natural gas, flagged ASSUM); grid power × CEA emission factor; water = steam feed – recycled produced water.
Caveat	"We don't claim oil production is green. We make an existing thermal process less wasteful per barrel, and we show how solar steam could change the picture."

SOCIAL

Claim	Safer operations, less heat-stress exposure, retained know-how, faster skilling.
Mechanism	(1) Fewer high-risk mechanical states means fewer emergency interventions, and rod failures and workovers are hazardous work. (2) Remote diagnosis means fewer unplanned site visits in 45–48 °C summers. (3) Knowledge capture on every override keeps senior operators' judgement as institutional memory. (4) The Challenge/Sandbox mode trains young engineers on a safe simulator. (5) Transparent decisions (ledger) build trust between field and office.
Evidence shown	Impact social column (high-risk hours, heat-stress-weighted visits avoided, knowledge notes, decisions with a full trail); the Judge Challenge itself.
Method	Counts from the simulator's event logs (policy vs policy); heat index by date from the climate condition.
Caveat	"We don't invent social metrics. Everything here is safety, exposure or knowledge, measured in the simulation."

Also relevant nationally (narrative, sourced): India imports most of its crude (cite the latest PPAC figure). Making domestic heavy-oil production more efficient supports energy security.

30.2 THE PS'S EXPECTED BENEFITS → OUR PROOF

EXPECTED BENEFIT (PS)	PROOF SHOWN	METRIC
Increased oil production and recovery	Arena race; Impact	Δ cumulative oil, % of oracle
Reduced SOR	Arena scoreboard; receipt	Δ SOR (CWE)
Lower energy per barrel	Arena; economy strip	Δ kWh/bbl
Reduced rod failures and pump unsettling	Arena failures; Risk margins	Δ failure counts, float hours
Improved equipment life and reliability	Goodman damage; maintenance forecast	Δ fatigue consumed per bbl
Data-driven, predictive decisions	Ledger calibration; forecasts with bands	coverage, outcome scores

30.3 HARD QUESTIONS

QUESTION	ANSWER (SHORT)	SHOW
Is this real data?	No. It's physics-based synthetic data calibrated to public figures, labelled on every screen. The anomaly pipeline was also validated on Petrobras 3W real well data.	Provenance mode; Proof
How accurate is it?	Against the simulator: model cards with held-out wells and time splits, and interval coverage. Against the field: unknown until calibrated. That's the EnKF's job, and we built the ingestion for it.	Proof → Model cards

QUESTION	ANSWER (SHORT)	SHOW
Why not just machine learning?	Physics extrapolates to states you haven't seen; ML doesn't. We use ML where it helps (speed, residuals, perception) and alarm when the two disagree.	Twin health "physics≈ML"
Do you use reinforcement learning?	Yes, as a shielded contender in the Arena. MPC wins on safety, explainability and data efficiency, and the Arena shows the numbers. If RL ever wins, it becomes a proposal generator behind the same Safety Gate.	Arena purple line
What's new compared with other teams?	The coupled physics-driven 3D/2D twin, the Coupling Dividend, the Stroke Shaper, the Arena with an oracle bound, and the honest learning loop.	§31
How would Oil India deploy it?	Read-only first: SCADA → MQTT/OPC-UA adapter → Advise mode. Their control system executes. On-prem Docker. Calibrate per well via EnKF.	Proof → Readiness
What if the optimiser is wrong?	A deterministic Safety Gate, INFEASIBLE instead of a guess, human approval, outcome scoring and override analytics.	Ledger
Does it work in real time?	Twin state at 2 Hz and a card per stroke. The EnKF updates hourly and daily, which matches CSS time scales. Plans take seconds to minutes.	Well live
Scalability?	Stateless gateway, sharded estimators, a job queue; 30 wells on a laptop, so a field's ~50 wells on a small server.	Architecture
Cyber-security?	Read-only integration, no write-back to PLCs, on-prem, OIDC/RBAC, audited dependencies, tamper-evident ledger.	§18.6
Cost to run?	Open-source stack on one server; LLM optional (templated offline mode).	—
Isn't the 3D just decoration?	Every visual is computed: thermal colour = $T(r,z)$, particle speed = Darcy velocity, rod colour = wave-equation stress. The Blueprint shows the same numbers as an engineer's drawing.	Lens toggle
What's done versus planned?	Answer honestly by feature tier (§13). The Proof page lists model status.	Proof
What's the business case?	"In our simulated field, X% more oil and Y% fewer failures." Scale the arithmetic in the <i>Illustrative</i> ×33 view, labelled as illustrative.	Impact

31. COMPETITIVE ANALYSIS: 25 PUBLIC REPOS

GitHub search on 29 Sep 2026 for `SIH26120`, `SIH 26120` and `baghewaLa` found **25 repositories**. 21 have substantive content. Private repos and teams without public code are invisible to us, so treat this as a floor, not a census.

31.1 WHO HAS WHAT

REPO	STACK	STAND-OUT FEATURES	WEAK SPOTS WE EXPLOIT
XploY04/Aavartan	Next.js + Three.js, FastAPI, SQLite	3D twin , IAPWS-IF97, Marx–Langenheim, Walther, Boberg–Lantz, API RP 11L, Couette drag, Goodman; equipment-limit gate; field schedule calendar; best-documented formulas; conformal and LightGBM <i>designed</i>	Minimal light UI; 3D is basic; ML layer not built; 4-well demo
Sharon-codes/SIH-120 (Urja Twin)	React + FastAPI, Vercel	SVG animated cutaway; 730-plan search with infeasibility; assistant with numeric verification (Nemotron); 43 tests + CI; decision export; research docs	2D SVG only; screening proxies; illustrative dynocard
Rohit-girish-Belagali (Inferno)	Vanilla JS + stdlib server	EnKF P10–P90; Gibbs wave-equation cards ; Ramey; marginal-value cut-off ; rod-float envelope with downstroke share ; 3 policies incl. pump-off baseline; ISA-style HMI; Day/Night	Plain UI; steam MILP and multi-taper not built
Dhayananth1511/Polaris (MIT)	React + FastAPI + Postgres	7-table dataset incl. 38,400 card points; SPM_crit; telemetry replay 1–50×; Isolation Forest; 12 pages; approve/reject	Page sprawl; lumped physics
maddy-bit/PetroTwin-AI	React + FastAPI + Spring Boot	180-day time machine; conformal 95% CI; SHAP waterfall; agentic copilot endpoint ; WebSocket; parted-rod classifier; tiered data honesty	Three backends; results table looks optimistic (e.g. "0.0% unsafe")
nishanth24vv	React + FastAPI + SQLite	Wellbore hydraulics (Pwf, PIP, fluid level); anomaly injector ; 11-step judging demo; 15 wells	A competitor review found fallback results substituted when infeasible
sandhya-parekar/WellTwin-AI	React + FastAPI	30 modules : RUL, health gauge, root cause, dossier generator, SIH requirement mapping , judge presentation mode , RBAC	Breadth over depth; "R ² 0.9997"-style claims
Devansh-567	Web	Calibration table with sources ; OSM map with real pads; sensor-integrity page; Everitt–Jennings; Arps	Small
bharatmalik-cs	React + FastAPI	Gibbs inversion; card fault classes; Beggs & Brill; canvas pseudo-3D pump; GIS map; WS telemetry	Rule-based assistant
Apaar-Gupta/SIH-2026	FastAPI + static HTML	Steam-driven hourly 720-h forecast ; steam quality; casing risk; cut-off hour; SPM schedule; honest assumptions	Single-page UI; ML regresses grid-search labels
dasher06 (MIT)	Python	SRP physics + GBM surrogate; trained on real "NK" SCADA (7 wells)	Component only
naveed1756	Python + web	9-class dynocard library ; figures for the PPT	PoC only
varshini2419	React + FastAPI	RAG over OIL public documents with a source index; soft-sensor nowcast; 7-day alarm classifier; constraints registry	Unstructured repo
Bhuvanesh0097/WellWise	React + FastAPI	Approve / Modify / Reject; 24 h prediction	Narrow
DevanshS1ngh/WELL-NEX	React + FastAPI	Cream / desert palette , Leaflet map, CSS/SRP simulate/apply	Similar palette to ours, so our differentiation must be the <i>living blueprint</i> concept, not the colour
ZotacMaster	React + FastAPI + TimescaleDB + Redis	Validation benchmark; in-memory fallback	5 pages
Bis28	Streamlit + PyCaret + Ollama Llama 3.2	Offline local copilot with deterministic safety overrides	Streamlit look
xarjunpatil	Static HTML + FastAPI	Glassmorphic dark dashboard , cryptographic audit logs, PostGIS	Mostly a whitepaper
Others (Rajatgupta2760, viveky1621, n230837, SRILEKHA-910, kr7561846, aniketcoder15, Vvl1232)	various	Concept READMEs, ML vs physics comparison (viveky), large synthetic SRP set (n230837)	Early-stage

31.2 WHAT WE ADOPT (AND CREDIT AS PRIOR ART IN OUR RESEARCH DOC)

Wellbore hydraulics · fluid level · PPRL/MPRL · card metrics · fluid pound as a separate mode · casing thermal risk · steam quality · heat-delivery efficiency · SPM_crit / float envelope with downstroke share · dynamic SPM schedule · cut-off and re-steam timing (marginal value) · 180-d horizon · conformal intervals · anomaly injection · telemetry replay · multi-well fleet · SHAP · safety gate · approve/modify/reject · audit trail · data-quality auditor · physics-vs-ML · OOD · tool-using copilot with numeric verification · knowledge capture · scenario library · requirement mapping page · judge presentation mode · pump-off baseline · calibration-with-sources table · RAG-style source index (as provenance, not chat) · offline copilot fallback.

All are **re-implemented by us** from the published physics and standards. We copy no code from unlicensed repos.

31.3 WHAT NOBODY HAS (OUR MOAT, AS OF THE REVIEW DATE)

- 1. **The living Blueprint** and the **in-place 3D → 2D transition** with continued animation in line art.
- 2. **Physics-driven visuals:** rendering the actual T(r,z) field, Darcy-speed particles and wave-equation stress bands in 3D.
- 3. **Coupling Dividend** with a Shapley decomposition. It quantifies the PS thesis.
- 4. **Stroke Shaper as an optimizer lever**, visualized.
- 5. **Condition Deck** covering formation, crude, climate, completion, **solar-thermal steam** and faults, changing physics and scene together.
- 6. **Field Steam Scheduler** (CP-SAT) *built* (one competitor planned it).
- 7. **Asphaltene/wax deposition band** from the wellbore temperature profile (the PS mentions asphaltenes and no repo models it).
- 8. **Hash-chained ledger + counterfactual scoring + override analytics** as one learning loop.
- 9. **Provenance mode** on every number.
- L0. **Safe Calibration Probe** (active learning with a safety constraint).
- L1. **Strategy Arena with an oracle bound and an RL contender**, all on common random numbers.
- L2. **Real-data method validation** on Petrobras 3W for the anomaly pipeline.

32. HONESTY RULES AND RISKS

32.1 HONESTY RULES (NON-NEGOTIABLE)

- 1. Never present synthetic results as Baghewala field results. Every impact number is tagged SIM.
- 2. Never report a metric without its dataset and split.
- 3. Every constant not from a source lives in the Assumption Registry, tagged ASSUM, with "replace with...".
- 4. The optimizer returns INFEASIBLE when it is infeasible.
- 5. The copilot's numbers must pass the verifier.
- 6. No real company branding or logos in the 3D scene or the UI. Mantle is an independent prototype for the PS, not an Oil India product.
- 7. Credit competitors' ideas as prior art in our research document; copy no unlicensed code or data.

32.2 RISKS

RISK	LIKELIHOOD	MITIGATION
3D performance on the venue machine	Med	Own laptop; auto-quality ladder; pre-rendered video fallback of the transition
Venue network down	High	Fully offline demo; copilot offline mode
Physics too slow for interactive scrubbing	Med	Precompute dense trajectories per scenario; surrogates in the optimizer inner loop
Judges challenge synthetic data	High	Provenance mode, source table, Petrobras 3W real-data validation, readiness story
Scope sprawl, nothing polished	High	Phases with demoable exits; P0 first; golden scenarios
Transition looks "cheap" if mistimed	Med	Author in Theatre.js, review frame by frame, visual regression on transition frames
Domain errors (wrong petroleum terms)	Med	Domain lead review; ask the OIL mentor; the source truth folder
RL training time or instability	Med	Train offline on nightly runs; the Arena reports RL honestly even if it's weak; MPC doesn't depend on it
Next.js + WebGL hydration and route pitfalls	Med	Persistent client-only canvas in the root layout; e2e tests on route changes
Data decision drags on	Med	Adapters make sources swappable; deadline end of P2 (ADR-0007)
Drafts multiply and never converge	Med	Time-box: 2 variants, 15-minute review, decision logged
Competitor convergence (others add 3D)	Med	Our moat is the <i>coupled physics-driven visuals</i> and the Coupling Dividend, not 3D alone

APPENDICES

APPENDIX A: SIH26120 TRACEABILITY MATRIX

PS ITEM	MANTLE FEATURE(S)	EVIDENCE IN APP
Integrated reservoir + wellbore + surface twin	Twin, physics chain §14, EnKF	Twin, Blueprint depth tracks
Real-time monitoring	Synthetic SCADA → WS → Twin; Alerts (ISA-18.2)	Live badge, Alerts drawer
Prediction	30/90/180-d forecasts with P10–P90	Timeline fan, Reservoir sheet
Optimize CSS (steam volume, injection pressure, soak time, production cut-off)	Joint optimizer, cut-off rule, marginal steam, Scheduler	Well → Reservoir sheet, Recommendation card
Predict heating, cooling, production	Thermal grid + EnKF	Thermal lens, Plate isotherms
Continuously optimize SRP (stroke speed, SPM)	Receding-horizon schedule, Stroke Shaper	Well → Lift sheet, Plate 3
Detect rod floating, minimize impact loading	Float margin, wave-equation impact, card classifier	Risk Stack, Plate 2
Pump efficiency, equipment reliability	Fillage, Goodman, hazard models, maintenance forecast	Well → Lift sheet
Optimize steam and energy, reduce cost	Economics, SOR, kWh/bbl, steam ROI	Impact, Battle
Rod failures, pump unsetting ↓	Hazard + margins + Safety Gate	Risk Stack, Impact
Data-driven, predictive decisions	Decision Ledger, outcome scoring	Ledger
Uses the listed data categories	Canonical schema §17.3	Proof → Real-data readiness

APPENDIX B: METRIC DICTIONARY (CORE)

METRIC	UNIT	FORMULA / SOURCE	SHOWN IN
SPM	1/min	measured / plan	Twin callout 2, Lift
Downstroke share k_d	–	VFD profile	Twin callout 2, Plate (Stroke Shaper)
PPRL / MPRL	kN	wave eq.; RP 11L fast form	Twin callout 1, Plate 2
Rod-float margin	%	$1 - (F_{drag} + F_{fric})/W_b$	Risk Stack, callout 1
N_max (safe SPM)	1/min	$120 \cdot k_d \cdot v_{max} / (\pi \cdot S)$, corrected	Plate 3
Pump fillage	%	pump model / card	Callout 6
Volumetric efficiency	%	q_{liquid} / PD	Callout 6
Hold-down margin	×	rating / (friction + transient)	Risk Stack
Goodman loading	–	$S_{range} / S_{a,allow}$	Plate 6, Risk Stack
Gearbox torque	% rating	torque factor × net load	Callout 2
Fluid level / submergence	m	annulus balance	Callout 5
Pwf, PIP	bar	hydraulics	Pressure track
Sandface temperature	°C	thermal grid / EnKF	Callout 7
Viscosity	cP	Walther + emulsion	Callout 7, μ track
Heated radius	m	Marx–Langenheim / grid	Callout 7
J_hot/J_cold	–	Boberg–Lantz	Plate 4
Oil rate	BOPD	min(inflow, pump)	Callout 4
SOR (CWE)	bbl/bbl	$steam\ t \times 6.2898 / oil\ bbl$	Economy strip
Energy / bbl	kWh/bbl	(motor + steam-gen fuel)/oil	Economy strip
CO ₂ e / bbl	kg/bbl	fuel EF + grid EF	Economy strip, Impact
Water / bbl	m³/bbl	(feed – recycle)/oil	Economy strip, Impact
Casing thermal stress	% yield	$E \cdot \alpha \cdot \Delta T / yield$	Callout (injection), Risk
Deposition interval	m	$T(z) < T_{onset}$	Temperature track
Coupling Dividend	₹/cycle, bbl, ΔSOR	§15.6	Twin chip, Impact
Twin health	%	weighted sensor integrity, calibration, freshness, OOD	Twin ring, Proof

APPENDIX C: GLOSSARY

API gravity: density scale for crude; lower means heavier. **BOPD:** barrels of oil per day. **CSS:** cyclic steam stimulation (inject → soak → produce). **CWE:** cold-water-equivalent steam volume. **Dynamometer card:** load vs position loop of one pump stroke. **EnKF:** Ensemble Kalman Filter. **Fillage:** fraction of the pump barrel filled with liquid per stroke. **Fluid pound:** the plunger striking the liquid in a partially

filled barrel. **Goodman diagram**: fatigue allowable-stress chart for rods. **IPR**: inflow performance relationship. **MPRL / PPRL**: minimum / peak polished-rod load. **Pump unsetting**: the insert pump pulled off its seating nipple. **Rod float**: the rod string failing to follow the surface unit on the downstroke because of viscous drag. **SOR**: steam-oil ratio. **SPM**: strokes per minute. **VFD**: variable-frequency drive. **VIT**: vacuum-insulated tubing.

APPENDIX D: REFERENCES (PRIMARY; OBTAIN AND VERIFY)

- Oil India Limited, *Rajasthan Fields* (field description, CSS/SRP, VIT, viscosity).
- SIH 2026 Problem Statement **SIH26120** (Oil India Limited).
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- ISA-101 (HMI design) · ISA-18.2 (alarm management) · ISO 128 (technical drawing lines) · ISO/PAS 12835 (thermal well casing connections).
- ACT SHARP consortium, D4.1 (2022). Geological characterization including Baghewala (as indexed by a competitor; obtain the original).
- Competitor repositories listed in §31 (accessed 29 Sep 2026).

End of plan. Revision A. The next revision should add: confirmed source numbers, the final asset review, benchmark results, and the rehearsed demo timings.

APPENDIX E: CONFIGURATION (ENVIRONMENT VARIABLES)

VARIABLE	DEFAULT	USED BY	MEANING
MANTLE_MODE	SIMULATION	all	SIMULATION or FIELD (locked in the demo)
MANTLE_PROFILE	demo	all	demo (seeded, deterministic) · dev · bench
MANTLE_SEED	2026	synth, arena	Global seed for the demo field
MANTLE_WELLS	30	synth	Wells in Baghewala-S
MANTLE_WORKERS	2	worker	Concurrent job workers
DATABASE_URL	postgresql+asyncpg://mantle:mantle@db:5432/mantle	py	DB
REDIS_URL	redis://redis:6379/0	py	Streams, queue
S3_ENDPOINT / S3_BUCKET / S3_ACCESS_KEY / S3_SECRET_KEY	http://minio:9000 / mantle / - / -	py	Object store
MQTT_URL	mqtt://mosquitto:1883	scada, ingest	Broker
OPCUA_ENDPOINT	empty	ingest	Optional OPC-UA source
LLM_PROVIDER / LLM_API_KEY / LLM_MODEL	anthropic / empty / e.g. claude-sonnet-5-5	copilot	Empty key means offline templated mode
NEXT_PUBLIC_WS_PATH	/ws	web	WebSocket path behind the proxy
GATEWAY_INTERNAL_URL	http://gateway:8000	web (server components)	Internal API base
WEB_QUALITY	auto	web	auto · high · low
TZ	Asia/Kolkata	all	Display time zone

APPENDIX F: DECISION RECORDS TO WRITE (ADRS)

ADR	TITLE	STATUS
0001	Monorepo with pnpm + uv workspaces	Accepted (Rev B)
0002	Next.js App Router with a persistent R3F canvas	Accepted
0003	FastAPI gateway; one physics library as the only calculation path	Accepted
0004	Docker Compose single command; Caddy as the single entry point	Accepted
0005	HTML/CSS draft-first UI with draft-parity tests	Accepted
0006	EnKF for state estimation (vs particle filter / UKF)	Proposed
0007	Data sources: synthetic / competitor / LLM mix	Open (decide end of P2)
0008	MPC primary, shielded RL as contender	Accepted
0009	Ledger hash chain in Postgres (vs external)	Proposed
0010	LLM provider and offline fallback	Proposed
0011	Map stack (MapLibre + offline tiles)	Proposed

End of plan, Revision B. Rev C should add: confirmed source numbers, the approved drafts, the final asset review, official Arena results and rehearsed demo timings.